

## 35 USB on-the-go high-speed (OTG\_HS)

This section applies to the whole STM32F4xx family, unless otherwise specified.

### 35.1 OTG\_HS introduction

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This section presents the architecture and the programming model of the OTG\_HS controller.

The following acronyms are used throughout the section:

|      |                                     |
|------|-------------------------------------|
| FS   | full-speed                          |
| HS   | High-speed                          |
| LS   | Low-speed                           |
| USB  | Universal serial bus                |
| OTG  | On-the-go                           |
| PHY  | Physical layer                      |
| MAC  | Media access controller             |
| PFC  | Packet FIFO controller              |
| UTMI | USB Transceiver Macrocell Interface |
| ULPI | UTMI+ Low Pin Interface             |

References are made to the following documents:

- USB On-The-Go Supplement, Revision 1.3
- Universal Serial Bus Revision 2.0 Specification

The OTG\_HS is a dual-role device (DRD) controller that supports both peripheral and host functions and is fully compliant with the *On-The-Go Supplement to the USB 2.0 Specification*. It can also be configured as a host-only or peripheral-only controller, fully compliant with the *USB 2.0 Specification*. In host mode, the OTG\_HS supports high-speed (HS, 480 Mbits/s), full-speed (FS, 12 Mbits/s) and low-speed (LS, 1.5 Mbits/s) transfers whereas in peripheral mode, it only supports high-speed (HS, 480Mbits/s) and full-speed (FS, 12 Mbits/s) transfers. The OTG\_HS supports both HNP and SRP. The only external device required is a charge pump for VBUS in OTG mode.

### 35.2 OTG\_HS main features

The main features can be divided into three categories: general, host-mode and peripheral-mode features.

### 35.2.1 General features

The OTG\_HS interface main features are the following:

- It is USB-IF certified in compliance with the Universal Serial Bus Revision 2.0 Specification
- It supports 3 PHY interfaces
  - An on-chip full-speed PHY
  - An ULPI interface for external high-speed PHY.
- It supports the host negotiation protocol (HNP) and the session request protocol (SRP)
- It allows the host to turn  $V_{BUS}$  off to save power in OTG applications, with no need for external components
- It allows to monitor  $V_{BUS}$  levels using internal comparators
- It supports dynamic host-peripheral role switching
- It is software-configurable to operate as:
  - An SRP-capable USB HS/FS peripheral (B-device)
  - An SRP-capable USB HS/FS/low-speed host (A-device)
  - An USB OTG FS dual-role device
- It supports HS/FS SOFs as well as low-speed (LS) keep-alive tokens with:
  - SOF pulse PAD output capability
  - SOF pulse internal connection to timer 2 (TIM2)
  - Configurable framing period
  - Configurable end-of-frame interrupt
- It embeds an internal DMA with shareholding support and software selectable AHB burst type in DMA mode
- It has power saving features such as system clock stop during USB suspend, switching off of the digital core internal clock domains, PHY and DFIFO power management
- It features a dedicated 4-Kbyte data RAM with advanced FIFO management:
  - The memory partition can be configured into different FIFOs to allow flexible and efficient use of RAM
  - Each FIFO can contain multiple packets
  - Memory allocation is performed dynamically
  - The FIFO size can be configured to values that are not powers of 2 to allow the use of contiguous memory locations
- It ensures a maximum USB bandwidth of up to one frame without application intervention

### 35.2.2 Host-mode features

The OTG\_HS interface features in host mode are the following:

- It requires an external charge pump to generate  $V_{BUS}$
- It has up to 12 host channels (pipes), each channel being dynamically reconfigurable to support any kind of USB transfer
- It features a built-in hardware scheduler holding:
  - Up to 8 interrupt plus isochronous transfer requests in the periodic hardware queue
  - Up to 8 control plus bulk transfer requests in the nonperiodic hardware queue
- It manages a shared RX FIFO, a periodic TX FIFO, and a nonperiodic TX FIFO for efficient usage of the USB data RAM
- It features dynamic trimming capability of SOF framing period in host mode.

### 35.2.3 Peripheral-mode features

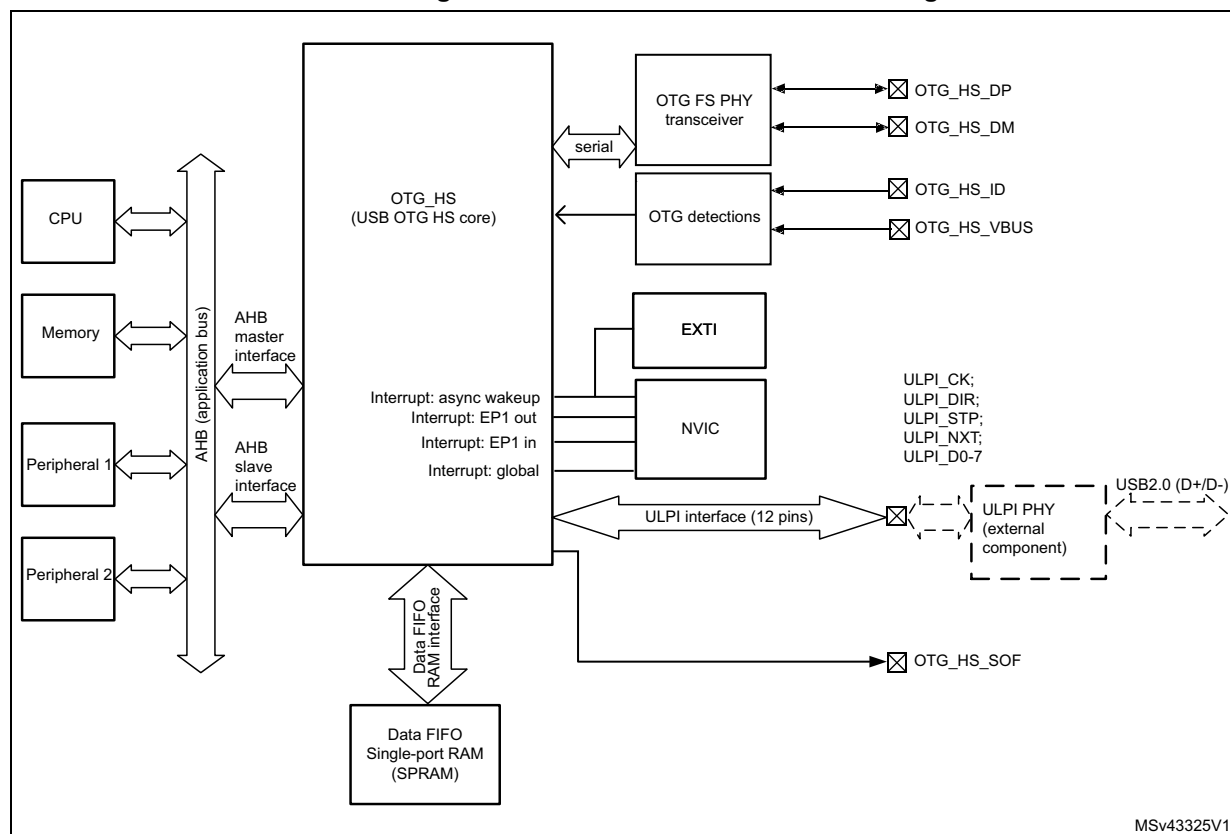
The OTG\_HS interface main features in peripheral mode are the following:

- It has 1 bidirectional control endpoint 0
- It has 5 IN endpoints (EP) configurable to support bulk, interrupt or isochronous transfers
- It has 5 OUT endpoints configurable to support bulk, interrupt or isochronous transfers
- It manages a shared Rx FIFO and a Tx-OUT FIFO for efficient usage of the USB data RAM
- It manages up to 6 dedicated Tx-IN FIFOs (one for each IN-configured EP) to reduce the application load
- It features soft disconnect capability

## 35.3 OTG\_HS functional description

*Figure 410* shows the OTG\_HS interface block diagram.

Figure 410. USB OTG interface block diagram



1. The USB DMA cannot directly address the internal Flash memory.

### 35.3.1 OTG pins

Table 206. OTG\_HS input/output pins

| Signal name         | Signal type          | Description                                |
|---------------------|----------------------|--------------------------------------------|
| OTG_HS_DP           | Digital input/output | USB OTG D+ line                            |
| OTG_HS_DM           | Digital input/output | USB OTG D- line                            |
| OTG_HS_ID           | Digital input        | USB OTG ID                                 |
| OTG_HS_VBUS         | Analog input         | USB OTG VBUS                               |
| OTG_HS_SOF          | Digital output       | USB OTG Start Of Frame (visibility)        |
| OTG_HS_ULPI_CK      | Digital input        | USB OTG ULPI clock                         |
| OTG_HS_ULPI_DIR     | Digital input        | USB OTG ULPI data bus direction control    |
| OTG_HS_ULPI_STP     | Digital output       | USB OTG ULPI data stream stop              |
| OTG_HS_ULPI_NXT     | Digital input        | USB OTG ULPI next data stream request      |
| OTG_HS_ULPI_D[0..7] | Digital input/output | USB OTG ULPI 8-bit bi-directional data bus |

### 35.3.2 High-speed OTG PHY

The USB OTG HS core embeds an ULPI interface to connect an external HS phy.

### 35.3.3 Embedded Full-speed OTG PHY

The full-speed OTG PHY includes the following components:

- FS/LS transceiver module used by both host and Device. It directly drives transmission and reception on the single-ended USB lines.
- Integrated ID pull-up resistor used to sample the ID line for A/B Device identification.
- DP/DM integrated pull-up and pull-down resistors controlled by the OTG\_HS core depending on the current role of the device. As a peripheral, it enables the DP pull-up resistor to signal full-speed peripheral connections as soon as  $V_{BUS}$  is sensed to be at a valid level (B-session valid). In host mode, pull-down resistors are enabled on both DP/DM. Pull-up and pull-down resistors are dynamically switched when the peripheral role is changed via the host negotiation protocol (HNP).
- Pull-up/pull-down resistor ECN circuit  
The DP pull-up consists of 2 resistors controlled separately from the OTG\_HS as per the resistor Engineering Change Notice applied to USB Rev2.0. The dynamic trimming of the DP pull-up strength allows to achieve a better noise rejection and Tx/Rx signal quality.
- $V_{BUS}$  sensing comparators with hysteresis used to detect  $V_{BUS\_VALID}$ , A-B Session Valid and session-end voltage thresholds. They are used to drive the session request protocol (SRP), detect valid startup and end-of-session conditions, and constantly monitor the  $V_{BUS}$  supply during USB operations.
- $V_{BUS}$  pulsing method circuit used to charge/discharge  $V_{BUS}$  through resistors during the SRP (weak drive).

**Caution:** To guarantee a correct operation for the USB OTG HS peripheral, the AHB frequency should be higher than 30 MHz.

## 35.4 OTG dual-role device

### 35.4.1 ID line detection

The host or peripheral (the default) role depends on the level of the ID input line. It is determined when the USB cable is plugged in and depends on which side of the USB cable is connected to the micro-AB receptacle:

- If the B-side of the USB cable is connected with a floating ID wire, the integrated pull-up resistor detects a high ID level and the default peripheral role is confirmed. In this configuration the OTG\_HS conforms to the FSM standard described in section 6.8.2. On-The-Go B-device of the USB On-The-Go Supplement, Revision 1.3.
- If the A-side of the USB cable is connected with a grounded ID, the OTG\_HS issues an ID line status change interrupt (CIDSCHG bit in the OTG\_HS\_GINTSTS register) for host software initialization, and automatically switches to host role. In this configuration the OTG\_HS conforms to the FSM standard described by section 6.8.1: On-The-Go A-Device of the USB On-The-Go Supplement, Revision 1.3.

### 35.4.2 HNP dual role device

The HNP capable bit in the Global USB configuration register (HNPCAP bit in the OTG\_HS\_GUSBCFG register) configures the OTG\_HS core to dynamically change from A-host to A-device role and vice-versa, or from B-device to B-host role and vice-versa, according to the host negotiation protocol (HNP). The current device status is defined by the

combination of the Connector ID Status bit in the Global OTG control and status register (CIDSTS bit in OTG\_HS\_GOTGCTL) and the current mode of operation bit in the global interrupt and status register (CMOD bit in OTG\_HS\_GINTSTS).

The HNP programming model is described in detail in [Section 35.13: OTG\\_HS programming model](#).

### 35.4.3 SRP dual-role device

The SRP capable bit in the global USB configuration register (SRPCAP bit in OTG\_HS\_GUSBCFG) configures the OTG\_HS core to switch  $V_{BUS}$  off for the A-device in order to save power. The A-device is always in charge of driving  $V_{BUS}$  regardless of the OTG\_HS role (host or peripheral). The SRP A/B-device program model is described in detail in [Section 35.13: OTG\\_HS programming model](#).

## 35.5 USB functional description in peripheral mode

The OTG\_HS operates as an USB peripheral in the following circumstances:

- OTG B-device  
OTG B-device default state if the B-side of USB cable is plugged in
- OTG A-device  
OTG A-device state after the HNP switches the OTG\_HS to peripheral role
- B-Device  
If the ID line is present, functional and connected to the B-side of the USB cable, and the HNP-capable bit in the Global USB Configuration register (HNPCAP bit in OTG\_HS\_GUSBCFG) is cleared (see On-The-Go specification Revision 1.3 section 6.8.3).
- Peripheral only (see [Figure 388: USB peripheral-only connection](#))  
The force peripheral mode bit in the Global USB configuration register (FDMOD in OTG\_HS\_GUSBCFG) is set to 1, forcing the OTG\_HS core to operate in USB peripheral-only mode (see On-The-Go specification Revision 1.3 section 6.8.3). In this case, the ID line is ignored even if it is available on the USB connector.

*Note:* To build a bus-powered device architecture in the B-Device or peripheral-only configuration, an external regulator must be added to generate the  $V_{DD}$  supply voltage from  $V_{BUS}$ .

### 35.5.1 SRP-capable peripheral

The SRP capable bit in the Global USB configuration register (SRPCAP bit in OTG\_HS\_GUSBCFG) configures the OTG\_HS to support the session request protocol (SRP). As a result, it allows the remote A-device to save power by switching  $V_{BUS}$  off when the USB session is suspended.

The SRP peripheral mode program model is described in detail in [Section : B-device session request protocol](#).

## 35.5.2 Peripheral states

### Powered state

The  $V_{BUS}$  input detects the B-session valid voltage used to put the USB peripheral in the Powered state (see USB2.0 specification section 9.1). The OTG\_HS then automatically connects the DP pull-up resistor to signal full-speed device connection to the host, and generates the session request interrupt (SRQINT bit in OTG\_HS\_GINTSTS) to notify the Powered state. The  $V_{BUS}$  input also ensures that valid  $V_{BUS}$  levels are supplied by the host during USB operations. If  $V_{BUS}$  drops below the B-session valid voltage (for example because power disturbances occurred or the host port has been switched off), the OTG\_HS automatically disconnects and the session end detected (SEDET bit in OTG\_HS\_GOTGINT) interrupt is generated to notify that the OTG\_HS has exited the Powered state.

In Powered state, the OTG\_HS expects a reset from the host. No other USB operations are possible. When a reset is received, the reset detected interrupt (USBRST in OTG\_HS\_GINTSTS) is generated. When the reset is complete, the enumeration done interrupt (ENUMDNE bit in OTG\_HS\_GINTSTS) is generated and the OTG\_HS enters the Default state.

### Soft disconnect

The Powered state can be exited by software by using the soft disconnect feature. The DP pull-up resistor is removed by setting the Soft disconnect bit in the device control register (SDIS bit in OTG\_HS\_DCTL), thus generating a device disconnect detection interrupt on the host side even though the USB cable was not really unplugged from the host port.

### Default state

In Default state the OTG\_HS expects to receive a SET\_ADDRESS command from the host. No other USB operations are possible. When a valid SET\_ADDRESS command is decoded on the USB, the application writes the corresponding number into the device address field in the device configuration register (DAD bit in OTG\_HS\_DCFG). The OTG\_HS then enters the address state and is ready to answer host transactions at the configured USB address.

### Suspended state

The OTG\_HS peripheral constantly monitors the USB activity. When the USB remains idle for 3 ms, the early suspend interrupt (ESUSP bit in OTG\_HS\_GINTSTS) is issued. It is confirmed 3 ms later, if appropriate, by generating a suspend interrupt (USBSUSP bit in OTG\_HS\_GINTSTS). The device suspend bit is then automatically set in the device status register (SUSPSTS bit in OTG\_HS\_DSTS) and the OTG\_HS enters the Suspended state.

The device can also exit from the Suspended state by itself. In this case the application sets the remote wake-up signaling bit in the device control register (RWUSIG bit in OTG\_HS\_DCTL) and clears it after 1 to 15 ms.

When a resume signaling is detected from the host, the resume interrupt (WKUPINT bit in OTG\_HS\_GINTSTS) is generated and the device suspend bit is automatically cleared.

### 35.5.3 Peripheral endpoints

The OTG\_HS core instantiates the following USB endpoints:

- Control endpoint 0  
This endpoint is bidirectional and handles control messages only.  
It has a separate set of registers to handle IN and OUT transactions, as well as dedicated control (OTG\_HS\_DIEPCTL0/OTG\_HS\_DOEPCTL0), transfer configuration (OTG\_HS\_DIEPTSIZ0/OTG\_HS\_DOEPSIZ0), and status-interrupt (OTG\_HS\_DIEPINTx/OTG\_HS\_DOEPINT0) registers. The bits available inside the control and transfer size registers slightly differ from other endpoints.
- 5 IN endpoints
  - They can be configured to support the isochronous, bulk or interrupt transfer type.
  - They feature dedicated control (OTG\_HS\_DIEPCTLx), transfer configuration (OTG\_HS\_DIEPTSIZx), and status-interrupt (OTG\_HS\_DIEPINTx) registers.
  - The Device IN endpoints common interrupt mask register (OTG\_HS\_DIEPMSK) allows to enable/disable a single endpoint interrupt source on all of the IN endpoints (EP0 included).
  - They support incomplete isochronous IN transfer interrupt (IISOIXFR bit in OTG\_HS\_GINTSTS). This interrupt is asserted when there is at least one isochronous IN endpoint for which the transfer is not completed in the current frame. This interrupt is asserted along with the end of periodic frame interrupt (OTG\_HS\_GINTSTS/EOPF).
- 5 OUT endpoints
  - They can be configured to support the isochronous, bulk or interrupt transfer type.
  - They feature dedicated control (OTG\_HS\_DOEPCTLx), transfer configuration (OTG\_HS\_DOEPSIZx) and status-interrupt (OTG\_HS\_DOEPINTx) registers.
  - The Device Out endpoints common interrupt mask register (OTG\_HS\_DOEPMSK) allows to enable/disable a single endpoint interrupt source on all OUT endpoints (EP0 included).
  - They support incomplete isochronous OUT transfer interrupt (INCOMPISOOUT bit in OTG\_HS\_GINTSTS). This interrupt is asserted when there is at least one isochronous OUT endpoint on which the transfer is not completed in the current frame. This interrupt is asserted along with the end of periodic frame interrupt (OTG\_HS\_GINTSTS/EOPF).

## Endpoint controls

The following endpoint controls are available through the device endpoint-x IN/OUT control register (DIEPCTLx/DOEPCTLx):

- Endpoint enable/disable
- Endpoint activation in current configuration
- Program the USB transfer type (isochronous, bulk, interrupt)
- Program the supported packet size
- Program the Tx-FIFO number associated with the IN endpoint
- Program the expected or transmitted data0/data1 PID (bulk/interrupt only)
- Program the even/odd frame during which the transaction is received or transmitted (isochronous only)
- Optionally program the NAK bit to always send a negative acknowledge to the host regardless of the FIFO status
- Optionally program the STALL bit to always stall host tokens to that endpoint
- Optionally program the Snoop mode for OUT endpoint where the received data CRC is not checked

## Endpoint transfer

The device endpoint-x transfer size registers (DIEPTSIZx/DOEPTSIZx) allow the application to program the transfer size parameters and read the transfer status.

The programming operation must be performed before setting the endpoint enable bit in the endpoint control register.

Once the endpoint is enabled, these fields are read-only as the OTG FS core updates them with the current transfer status.

The following transfer parameters can be programmed:

- Transfer size in bytes
- Number of packets constituting the overall transfer size.

## Endpoint status/interrupt

The device endpoint-x interrupt registers (DIEPINTx/DOPEPINTx) indicate the status of an endpoint with respect to USB- and AHB-related events. The application must read these registers when the OUT endpoint interrupt bit or the IN endpoint interrupt bit in the core interrupt register (OEPINT bit in OTG\_HS\_GINTSTS or IEPINT bit in OTG\_HS\_GINTSTS, respectively) is set. Before the application can read these registers, it must first read the device all endpoints interrupt register (OTG\_HS\_DAINR) to get the exact endpoint number for the device endpoint-x interrupt register. The application must clear the appropriate bit in this register to clear the corresponding bits in the DAINR and GINTSTS registers.

The peripheral core provides the following status checks and interrupt generation:

- Transfer completed interrupt, indicating that data transfer has completed on both the application (AHB) and USB sides
- Setup stage done (control-out only)
- Associated transmit FIFO is half or completely empty (in endpoints)
- NAK acknowledge transmitted to the host (isochronous-in only)
- IN token received when Tx-FIFO was empty (bulk-in/interrupt-in only)
- OUT token received when endpoint was not yet enabled
- Babble error condition detected
- Endpoint disable by application is effective
- Endpoint NAK by application is effective (isochronous-in only)
- More than 3 back-to-back setup packets received (control-out only)
- Timeout condition detected (control-in only)
- Isochronous out packet dropped without generating an interrupt

## 35.6 USB functional description on host mode

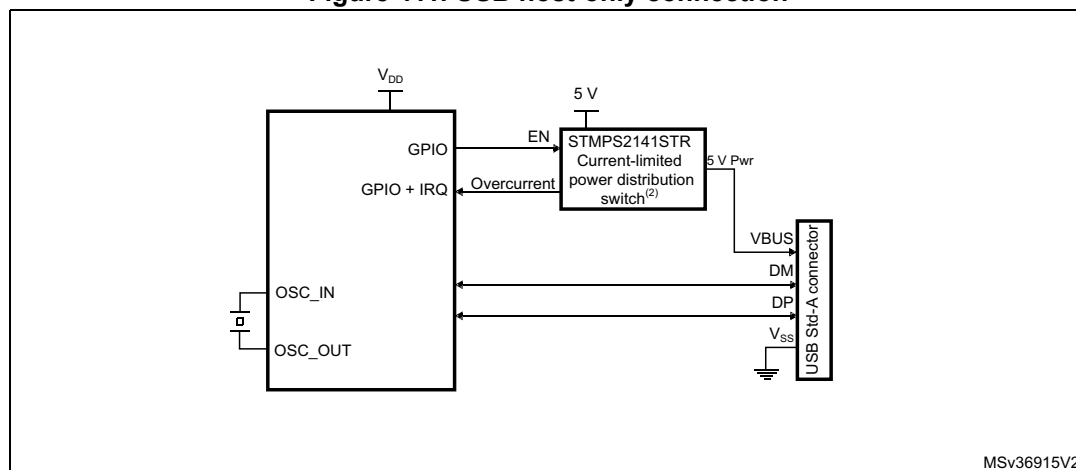
This section gives the functional description of the OTG\_HS in the USB host mode. The OTG\_HS works as a USB host in the following circumstances:

- OTG A-host  
OTG A-device default state when the A-side of the USB cable is plugged in
- OTG B-host  
OTG B-device after HNP switching to the host role
- A-device  
If the ID line is present, functional and connected to the A-side of the USB cable, and the HNP-capable bit is cleared in the Global USB Configuration register (HNPCAP bit in OTG\_HS\_GUSBCFG). Integrated pull-down resistors are automatically set on the DP/DM lines.
- Host only ([Figure 389: USB host-only connection](#)).  
The force host mode bit in the global USB configuration register (FHMOD bit in OTG\_HS\_GUSBCFG) forces the OTG\_HS core to operate in USB host-only mode. In this case, the ID line is ignored even if it is available on the USB connector. Integrated pull-down resistors are automatically set on the OTG\_HS\_FS\_DP/OTG\_HS\_FS\_DM lines.

**Note:** On-chip 5 V  $V_{BUS}$  generation is not supported. As a result, a charge pump or a basic power switch (if a 5 V supply is available on the application board) must be added externally to drive the 5 V  $V_{BUS}$  line. The external charge pump can be driven by any GPIO output. This is required for the OTG A-host, A-device and host-only configurations.

The  $V_{BUS}$  input ensures that valid  $V_{BUS}$  levels are supplied by the charge pump during USB operations while the charge pump overcurrent output can be input to any GPIO pin configured to generate port interrupts. The overcurrent ISR must promptly disable the  $V_{BUS}$  generation.

Figure 411. USB host-only connection



### 35.6.1 SRP-capable host

SRP support is available through the SRP capable bit in the global USB configuration register (SRPCAP bit in OTG\_HS\_GUSBCFG). When the SRP feature is enabled, the host can save power by switching off the  $V_{BUS}$  power while the USB session is suspended. The

SRP host mode program model is described in detail in [Section : A-device session request protocol](#).

## 35.6.2 USB host states

### Host port power

On-chip 5 V  $V_{BUS}$  generation is not supported. As a result, a charge pump or a basic power switch (if a 5 V supply voltage is available on the application board) must be added externally to drive the 5 V  $V_{BUS}$  line. The external charge pump can be driven by any GPIO output. When the application powers on  $V_{BUS}$  through the selected GPIO, it must also set the port power bit in the host port control and status register (PPWR bit in OTG\_HS\_HPRT).

### $V_{BUS}$ valid

When SRP or HNP is enabled the  $V_{BUS}$  sensing pin (PB13) pin should be connected to  $V_{BUS}$ . The  $V_{BUS}$  input ensures that valid  $V_{BUS}$  levels are supplied by the charge pump during USB operations. Any unforeseen  $V_{BUS}$  voltage drop below the  $V_{BUS}$  valid threshold (4.25 V) generates an OTG interrupt triggered by the session end detected bit (SEDET bit in OTG\_HS\_GOTGINT). The application must then switch the  $V_{BUS}$  power off and clear the port power bit.

When HNP and SRP are both disabled, the  $V_{BUS}$  sensing pin (PB13) should not be connected to  $V_{BUS}$ . This pin can be used as GPIO.

The charge pump overcurrent flag can also be used to prevent electrical damage. Connect the overcurrent flag output from the charge pump to any GPIO input, and configure it to generate a port interrupt on the active level. The overcurrent ISR must promptly disable the  $V_{BUS}$  generation and clear the port power bit.

### Detection of peripheral connection by the host

If SRP or HNP are enabled, even if USB peripherals or B-devices can be attached at any time, the OTG\_HS does not detect a bus connection until the end of the  $V_{BUS}$  sensing ( $V_{BUS}$  over 4.75 V).

When  $V_{BUS}$  is at a valid level and a remote B-device is attached, the OTG\_HS core issues a host port interrupt triggered by the device connected bit in the host port control and status register (PCDET bit in OTG\_HS\_HPRT).

When HNP and SRP are both disabled, USB peripherals or B-device are detected as soon as they are connected. The OTG\_HS core issues a host port interrupt triggered by the device connected bit in the host port control and status (PCDET bit in OTG\_HS\_HPRT).

### Detection of peripheral disconnection by the host

The peripheral disconnection event triggers the disconnect detected interrupt (DISCINT bit in OTG\_HS\_GINTSTS).

### Host enumeration

After detecting a peripheral connection, the host must start the enumeration process by issuing an USB reset and configuration commands to the new peripheral.

Before sending an USB reset, the application waits for the OTG interrupt triggered by the debounce done bit (DBCDNE bit in OTG\_HS\_GOTGINT), which indicates that the bus is

stable again after the electrical debounce caused by the attachment of a pull-up resistor on OTG\_HS\_FS\_DP (full speed) or OTG\_HS\_FS\_DM (low speed).

The application issues an USB reset (single-ended zero) via the USB by keeping the port reset bit set in the Host port control and status register (PRST bit in OTG\_HS\_HPRT) for a minimum of 10 ms and a maximum of 20 ms. The application monitors the time and then clears the port reset bit.

Once the USB reset sequence has completed, the host port interrupt is triggered by the port enable/disable change bit (PENCHNG bit in OTG\_HS\_HPRT) to inform the application that the speed of the enumerated peripheral can be read from the port speed field in the host port control and status register (PSPD bit in OTG\_HS\_HPRT), and that the host is starting to drive SOFs (full speed) or keep-alive tokens (low speed). The host is then ready to complete the peripheral enumeration by sending peripheral configuration commands.

### Host suspend

The application can decide to suspend the USB activity by setting the port suspend bit in the host port control and status register (PSUSP bit in OTG\_HS\_HPRT). The OTG\_HS core stops sending SOFs and enters the Suspended state.

The Suspended state can be exited on the remote device initiative (remote wake-up). In this case the remote wake-up interrupt (WKUPINT bit in OTG\_HS\_GINTSTS) is generated upon detection of a remote wake-up event, the port resume bit in the host port control and status register (PRES bit in OTG\_HS\_HPRT) is set, and a resume signaling is automatically issued on the USB. The application must monitor the resume window duration, and then clear the port resume bit to exit the Suspended state and restart the SOF.

If the Suspended state is exited on the host initiative, the application must set the port resume bit to start resume signaling on the host port, monitor the resume window duration and then clear the port resume bit.

### 35.6.3 Host channels

The OTG\_HS core instantiates 12 host channels. Each host channel supports an USB host transfer (USB pipe). The host is not able to support more than 8 transfer requests simultaneously. If more than 8 transfer requests are pending from the application, the host controller driver (HCD) must re-allocate channels when they become available, that is, after receiving the transfer completed and channel halted interrupts.

Each host channel can be configured to support IN/OUT and any type of periodic/nonperiodic transaction. Each host channel has dedicated control (HCCHARx), transfer configuration (HCTSIZx) and status/interrupt (HCINTx) registers with associated mask (HCINTMSKx) registers.

### Host channel controls

The following host channel controls are available through the host channel-x characteristics register (HCCHARx):

- Channel enable/disable
- Program the HS/FS/LS speed of target USB peripheral
- Program the address of target USB peripheral
- Program the endpoint number of target USB peripheral
- Program the transfer IN/OUT direction
- Program the USB transfer type (control, bulk, interrupt, isochronous)
- Program the maximum packet size (MPS)
- Program the periodic transfer to be executed during odd/even frames

### Host channel transfer

The host channel transfer size registers (HCTSIZx) allow the application to program the transfer size parameters, and read the transfer status.

The programming operation must be performed before setting the channel enable bit in the host channel characteristics register. Once the endpoint is enabled, the packet count field is read-only as the OTG HS core updates it according to the current transfer status.

The following transfer parameters can be programmed:

- Transfer size in bytes
- Number of packets constituting the overall transfer size
- Initial data PID

### Host channel status/interrupt

The host channel-x interrupt register (HCINTx) indicates the status of an endpoint with respect to USB- and AHB-related events. The application must read these register when the host channels interrupt bit in the core interrupt register (HCINT bit in OTG\_HS\_GINTSTS) is set. Before the application can read these registers, it must first read the host all channels interrupt (HCAINT) register to get the exact channel number for the host channel-x interrupt register. The application must clear the appropriate bit in this register to clear the corresponding bits in the HCAINT and GINTSTS registers. The mask bits for each interrupt source of each channel are also available in the OTG\_HS\_HCINTMSK-x register.

The host core provides the following status checks and interrupt generation:

- Transfer completed interrupt, indicating that the data transfer is complete on both the application (AHB) and USB sides
- Channel stopped due to transfer completed, USB transaction error or disable command from the application
- Associated transmit FIFO half or completely empty (IN endpoints)
- ACK response received
- NAK response received
- STALL response received
- USB transaction error due to CRC failure, timeout, bit stuff error, false EOP
- Babble error
- Frame overrun
- Data toggle error

#### 35.6.4 Host scheduler

The host core features a built-in hardware scheduler which is able to autonomously re-order and manage the USB the transaction requests posted by the application. At the beginning of each frame the host executes the periodic (isochronous and interrupt) transactions first, followed by the nonperiodic (control and bulk) transactions to achieve the higher level of priority granted to the isochronous and interrupt transfer types by the USB specification.

The host processes the USB transactions through request queues (one for periodic and one for nonperiodic). Each request queue can hold up to 8 entries. Each entry represents a pending transaction request from the application, and holds the IN or OUT channel number along with other information to perform a transaction on the USB. The order in which the requests are written to the queue determines the sequence of the transactions on the USB interface.

At the beginning of each frame, the host processes the periodic request queue first, followed by the nonperiodic request queue. The host issues an incomplete periodic transfer interrupt (IPXFR bit in OTG\_HS\_GINTSTS) if an isochronous or interrupt transaction scheduled for the current frame is still pending at the end of the current frame. The OTG HS core is fully responsible for the management of the periodic and nonperiodic request queues. The periodic transmit FIFO and queue status register (HPTXSTS) and nonperiodic transmit FIFO and queue status register (HNPTXSTS) are read-only registers which can be used by the application to read the status of each request queue. They contain:

- The number of free entries currently available in the periodic (nonperiodic) request queue (8 max)
- Free space currently available in the periodic (nonperiodic) Tx-FIFO (out-transactions)
- IN/OUT token, host channel number and other status information.

As request queues can hold a maximum of 8 entries each, the application can push to schedule host transactions in advance with respect to the moment they physically reach the USB for a maximum of 8 pending periodic transactions plus 8 pending nonperiodic transactions.

To post a transaction request to the host scheduler (queue) the application must check that there is at least 1 entry available in the periodic (nonperiodic) request queue by reading the PTXQSAV bits in the OTG\_HS\_HNPTXSTS register or NPTQSAV bits in the OTG\_HS\_HNPTXSTS register.

## 35.7 SOF trigger

The OTG FS core allows to monitor, track and configure SOF framing in the host and peripheral. It also features an SOF pulse output connectivity.

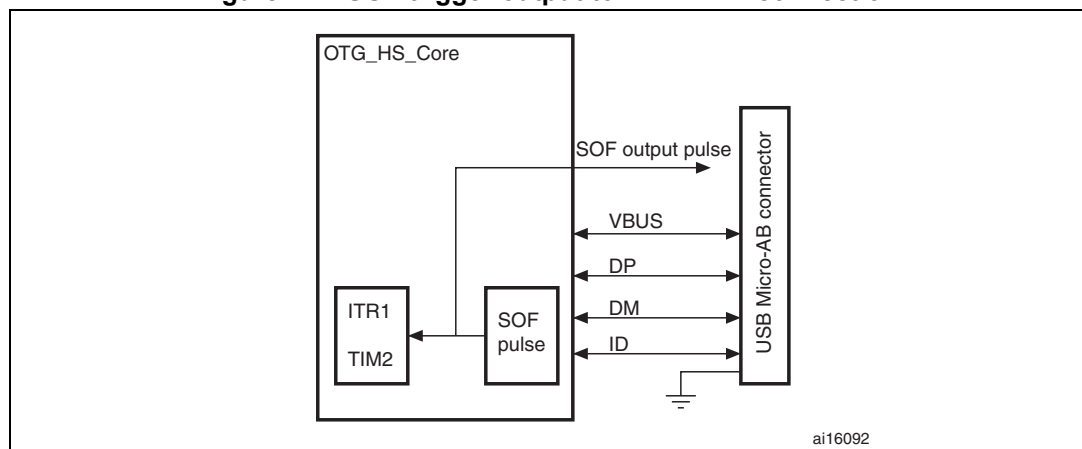
These capabilities are particularly useful to implement adaptive audio clock generation techniques, where the audio peripheral needs to synchronize to the isochronous stream provided by the PC, or the host needs trimming its framing rate according to the requirements of the audio peripheral.

### 35.7.1 Host SOFs

In host mode the number of PHY clocks occurring between the generation of two consecutive SOF (FS) or keep-alive (LS) tokens is programmable in the host frame interval register (OTG\_HS\_HFIR), thus providing application control over the SOF framing period. An interrupt is generated at any start of frame (SOF bit in OTG\_HS\_GINTSTS). The current frame number and the time remaining until the next SOF are tracked in the host frame number register (OTG\_HS\_HFNUM).

An SOF pulse signal is generated at any SOF starting token and with a width of 20 HCLK cycles. It can be made available externally on the SOF pin using the SOFOUTEN bit in the global control and configuration register. The SOF pulse is also internally connected to the input trigger of timer 2 (TIM2), so that the input capture feature, the output compare feature and the timer can be triggered by the SOF pulse. The TIM2 connection is enabled through ITR1\_RMP bits of TIM2\_OR register.

**Figure 412. SOF trigger output to TIM2 ITR1 connection**



### 35.7.2 Peripheral SOFs

In peripheral mode, the start of frame interrupt is generated each time an SOF token is received on the USB (SOF bit in OTG\_HS\_GINTSTS). The corresponding frame number can be read from the device status register (FNSOF bit in OTG\_HS\_DSTS). An SOF pulse signal with a width of 20 HCLK cycles is also generated and can be made available externally on the SOF pin by using the SOF output enable bit in the global control and configuration register (SOFOUTEN bit in OTG\_HS\_GCCFG). The SOF pulse signal is also internally connected to the TIM2 input trigger, so that the input capture feature, the output compare feature and the timer can be triggered by the SOF pulse (see [Figure 412](#)). The TIM2 connection is enabled through ITR1\_RMP bits of TIM2\_OR register.

The end of periodic frame interrupt (GINTSTS/EOPF) is used to notify the application when 80%, 85%, 90% or 95% of the time frame interval elapsed depending on the periodic frame interval field in the device configuration register (PFIVL bit in OTG\_HS\_DCFG).

This feature can be used to determine if all of the isochronous traffic for that frame is complete.

## 35.8 OTG\_HS low-power modes

[Table 207](#) below defines the STM32 low power modes and their compatibility with the OTG.

**Table 207. Compatibility of STM32 low power modes with the OTG**

| Mode    | Description                                                                                                 | USB compatibility                        |
|---------|-------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Run     | MCU fully active                                                                                            | Required when USB not in suspend state.  |
| Sleep   | USB suspend exit causes the device to exit Sleep mode. Peripheral registers content is kept.                | Available while USB is in suspend state. |
| Stop    | USB suspend exit causes the device to exit Stop mode. Peripheral registers content is kept <sup>(1)</sup> . | Available while USB is in suspend state. |
| Standby | Powered-down. The peripheral must be reinitialized after exiting Standby mode.                              | Not compatible with USB applications.    |

1. Within Stop mode there are different possible settings. Some restrictions may also exist, refer to [Section 5: Power controller \(PWR\)](#) to understand which (if any) restrictions apply when using OTG.

The following bits and procedures reduce power consumption.

The power consumption of the OTG PHY is controlled by three bits in the general core configuration register:

- PHY power down (GCCFG/PWRDWN)  
This bit switches on/off the PHY full-speed transceiver module. It must be preliminarily set to allow any USB operation.
- A-VBUS sensing enable (GCCFG/VBUSASEN)  
This bit switches on/off the  $V_{BUS}$  comparators associated with A-device operations. It must be set when in A-device (USB host) mode and during HNP.
- B-VBUS sensing enable (GCCFG/VBUSASEN)  
This bit switches on/off the  $V_{BUS}$  comparators associated with B-device operations. It must be set when in B-device (USB peripheral) mode and during HNP.  
Power reduction techniques are available in the USB suspended state, when the USB session is not yet valid or the device is disconnected.
- Stop PHY clock (STPPCLK bit in OTG\_HS\_PCGCCTL)
  - When setting the stop PHY clock bit in the clock gating control register, most of the clock domain internal to the OTG high-speed core is switched off by clock gating.

The dynamic power consumption due to the USB clock switching activity is cut even if the clock input is kept running by the application

- Most of the transceiver is also disabled, and only the part in charge of detecting the asynchronous resume or remote wake-up event is kept alive.
- Gate HCLK (GATEHCLK bit in OTG\_HS\_PCGCCTL)
 

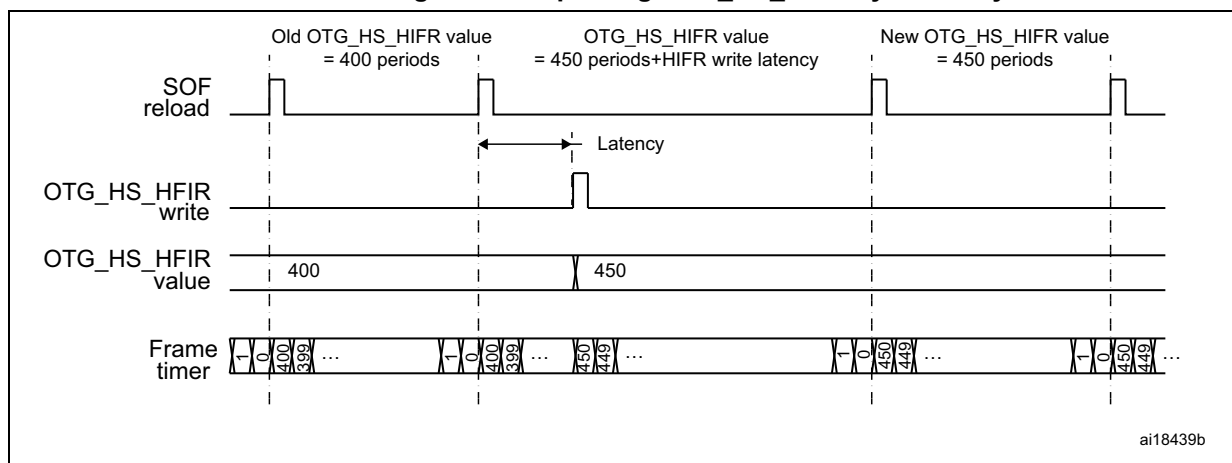
When setting the Gate HCLK bit in the clock gating control register, most of the system clock domain internal to the OTG\_HS core is switched off by clock gating. Only the register read and write interface is kept alive. The dynamic power consumption due to the USB clock switching activity is cut even if the system clock is kept running by the application for other purposes.
- USB system stop
  - When the OTG\_HS is in USB suspended state, the application can decide to drastically reduce the overall power consumption by shutting down all the clock sources in the system. USB System Stop is activated by first setting the Stop PHY clock bit and then configuring the system deep sleep mode in the powercontrol system module (PWR).
  - The OTG\_HS core automatically reactivates both system and USB clocks by asynchronous detection of remote wake-up (as an host) or resume (as a Device) signaling on the USB.

### 35.9 Dynamic update of the OTG\_HS\_HFIR register

The USB core embeds a dynamic trimming capability of micro-SOF framing period in host mode allowing to synchronize an external device with the micro-SOF frames.

When the OTG\_HS\_HFIR register is changed within a current micro-SOF frame, the SOF period correction is applied in the next frame as described in [Figure 413](#).

**Figure 413. Updating OTG\_HS\_HFIR dynamically**



## 35.10 FIFO RAM allocation

### 35.10.1 Peripheral mode

#### Receive FIFO RAM

For Receive FIFO RAM, the application should allocate RAM for SETUP packets: 10 locations must be reserved in the receive FIFO to receive SETUP packets on control endpoints. These locations are reserved for SETUP packets and are not used by the core to write any other data.

One location must be allocated for Global OUT NAK. Status information are also written to the FIFO along with each received packet. Therefore, a minimum space of  $(\text{Largest Packet Size} / 4) + 1$  must be allocated to receive packets. If a high-bandwidth endpoint or multiple isochronous endpoints are enabled, at least two spaces of  $(\text{Largest Packet Size} / 4) + 1$  must be allotted to receive back-to-back packets. Typically, two  $(\text{Largest Packet Size} / 4) + 1$  spaces are recommended so that when the previous packet is being transferred to AHB, the USB can receive the subsequent packet.

Along with each endpoints last packet, transfer complete status information are also pushed to the FIFO. Typically, one location for each OUT endpoint is recommended.

#### Transmit FIFO RAM

For Transmit FIFO RAM, the minimum RAM space required for each IN Endpoint Transmit FIFO is the maximum packet size for this IN endpoint.

*Note: More space allocated in the transmit IN Endpoint FIFO results in a better performance on the USB.*

### 35.10.2 Host mode

#### Receive FIFO RAM

For Receive FIFO RAM allocation, Status information are written to the FIFO along with each received packet. Therefore, a minimum space of  $(\text{Largest Packet Size} / 4) + 1$  must be allocated to receive packets. If a high-bandwidth channel or multiple isochronous channels are enabled, at least two spaces of  $(\text{Largest Packet Size} / 4) + 1$  must be allocated to receive back-to-back packets. Typically, two  $(\text{Largest Packet Size} / 4) + 1$  spaces are recommended so that when the previous packet is being transferred to AHB, the USB can receive the subsequent packet.

Along with each host channels last packet, transfer complete status information are also pushed to the FIFO. As a consequence, one location must be allocated to store this data.

#### Transmit FIFO RAM

For Transmit FIFO RAM allocation, the minimum amount of RAM required for the host nonperiodic Transmit FIFO is the largest maximum packet size for all supported nonperiodic OUT channels. Typically, a space corresponding to two Largest Packet Size is recommended, so that when the current packet is being transferred to the USB, the AHB can transmit the subsequent packet.

The minimum amount of RAM required for Host periodic Transmit FIFO is the largest maximum packet size for all supported periodic OUT channels. If there is at least one High

Bandwidth Isochronous OUT endpoint, then the space must be at least two times the maximum packet size for that channel.

*Note: More space allocated in the Transmit nonperiodic FIFO results in better performance on the USB.*

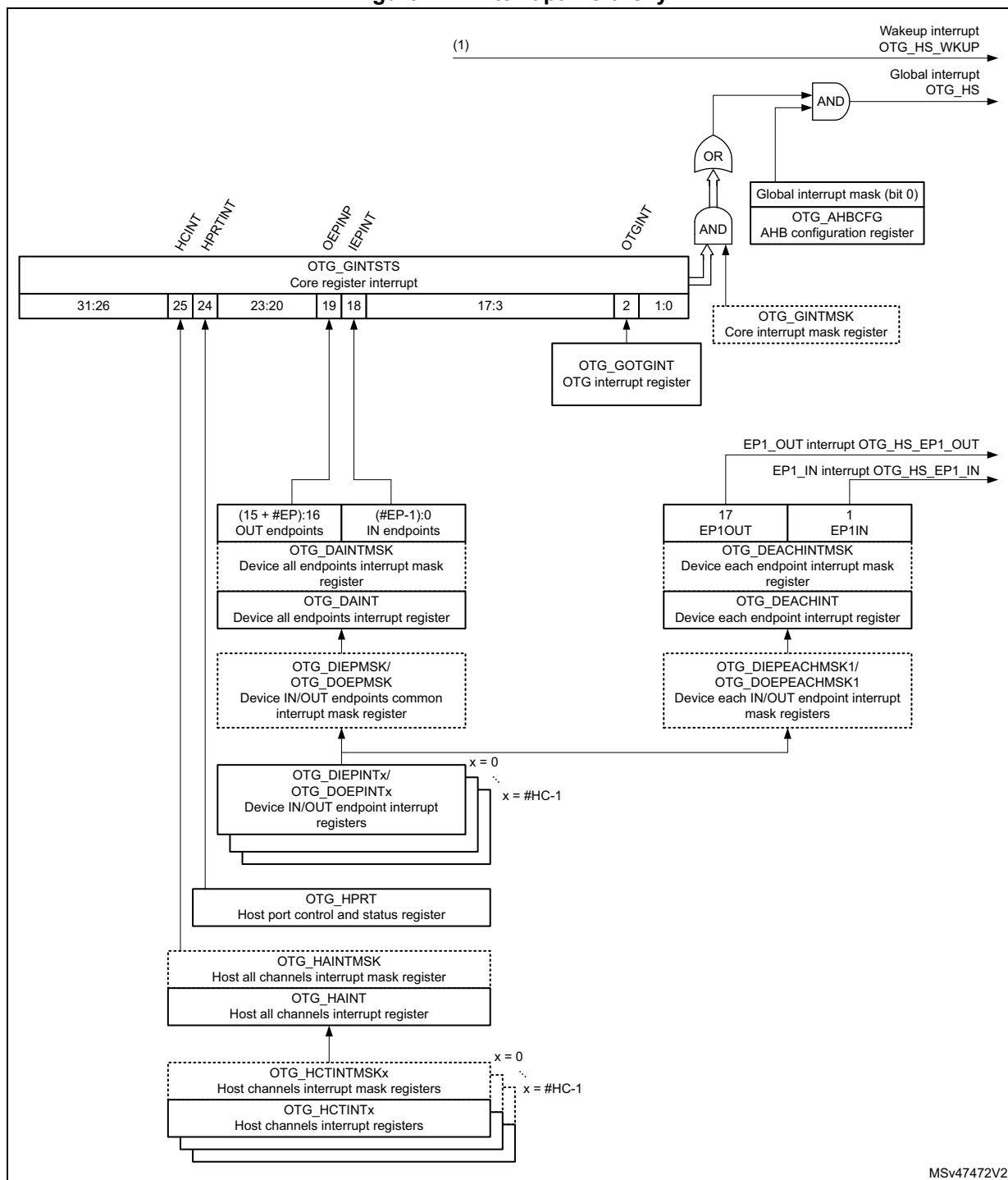
*When operating in DMA mode, the DMA address register for each host channel (HCDMA<sub>n</sub>) is stored in the SPRAM (FIFO). One location for each channel must be reserved for this.*

## 35.11 OTG\_HS interrupts

When the OTG\_HS controller is operating in one mode, either peripheral or host, the application must not access registers from the other mode. If an illegal access occurs, a mode mismatch interrupt is generated and reflected in the Core interrupt register (MMIS bit in the OTG\_HS\_GINTSTS register). When the core switches from one mode to the other, the registers in the new mode of operation must be reprogrammed as they would be after a power-on reset.

*Figure 414* shows the interrupt hierarchy.

### Figure 414. Interrupt hierarchy



1. OTG\_HS\_WKUP becomes active (high state) when resume condition occurs during L1 SLEEP or L2 SUSPEND states.

## 35.12 OTG\_HS control and status registers

By reading from and writing to the control and status registers (CSRs) through the AHB slave interface, the application controls the OTG\_HS controller. These registers are 32 bits wide, and the addresses are 32-bit block aligned. The OTG\_HS registers must be accessed by words (32 bits). CSRs are classified as follows:

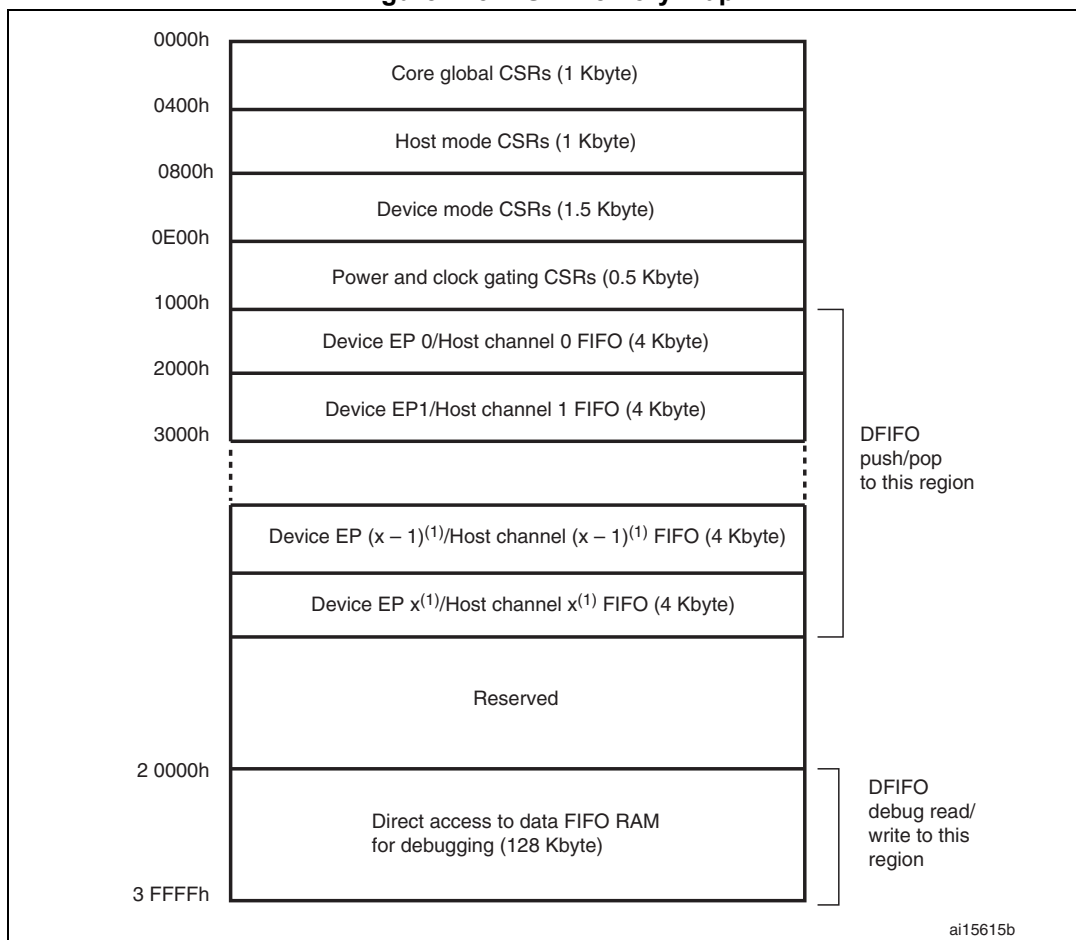
- Core global registers
- Host-mode registers
- Host global registers
- Host port CSRs
- Host channel-specific registers
- Device-mode registers
- Device global registers
- Device endpoint-specific registers
- Power and clock-gating registers
- Data FIFO (DFIFO) access registers

Only the Core global, Power and clock-gating, Data FIFO access, and host port control and status registers can be accessed in both host and peripheral modes. When the OTG\_HS controller is operating in one mode, either peripheral or host, the application must not access registers from the other mode. If an illegal access occurs, a mode mismatch interrupt is generated and reflected in the Core interrupt register (MMIS bit in the OTG\_HS\_GINTSTS register). When the core switches from one mode to the other, the registers in the new mode of operation must be reprogrammed as they would be after a power-on reset.

### 35.12.1 CSR memory map

The host and peripheral mode registers occupy different addresses. All registers are implemented in the AHB clock domain.

Figure 415. CSR memory map



1.  $x = 5$  in peripheral mode and  $x = 11$  in host mode.

### Global CSR map

These registers are available in both host and peripheral modes.

Table 208. Core global control and status registers (CSRs)

| Acronym        | Address offset | Register name                                                                    |
|----------------|----------------|----------------------------------------------------------------------------------|
| OTG_HS_GOTGCTL | 0x000          | <a href="#">OTG_HS control and status register (OTG_HS_GOTGCTL) on page 1396</a> |
| OTG_HS_GOTGINT | 0x004          | <a href="#">OTG_HS interrupt register (OTG_HS_GOTGINT) on page 1397</a>          |
| OTG_HS_GAHBCFG | 0x008          | <a href="#">OTG_HS AHB configuration register (OTG_HS_GAHBCFG) on page 1399</a>  |
| OTG_HS_GUSBCFG | 0x00C          | <a href="#">OTG_HS USB configuration register (OTG_HS_GUSBCFG) on page 1400</a>  |
| OTG_HS_GRSTCTL | 0x010          | <a href="#">OTG_HS reset register (OTG_HS_GRSTCTL) on page 1403</a>              |
| OTG_HS_GINTSTS | 0x014          | <a href="#">OTG_HS core interrupt register (OTG_HS_GINTSTS) on page 1406</a>     |
| OTG_HS_GINTMSK | 0x018          | <a href="#">OTG_HS interrupt mask register (OTG_HS_GINTMSK) on page 1410</a>     |

Table 208. Core global control and status registers (CSRs) (continued)

| Acronym                             | Address offset                 | Register name                                                                                                                                 |
|-------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| OTG_HS_GRXSTSR                      | 0x01C                          | <i>OTG_HS Receive status debug read/OTG status read and pop registers (OTG_HS_GRXSTSR/OTG_HS_GRXSTSP) on page 1413</i>                        |
| OTG_HS_GRXSTSP                      | 0x020                          |                                                                                                                                               |
| OTG_HS_GRXFSIZ                      | 0x024                          | <i>OTG_HS Receive FIFO size register (OTG_HS_GRXFSIZ) on page 1414</i>                                                                        |
| OTG_HS_GNPTXFSIZ/<br>OTG_HS_TX0FSIZ | 0x028                          | <i>OTG_HS nonperiodic transmit FIFO size/Endpoint 0 transmit FIFO size register (OTG_HS_GNPTXFSIZ/OTG_HS_TX0FSIZ) on page 1415</i>            |
| OTG_HS_GNPTXSTS                     | 0x02C                          | <i>OTG_HS nonperiodic transmit FIFO/queue status register (OTG_HS_GNPTXSTS) on page 1415</i>                                                  |
| OTG_HS_GCCFG                        | 0x038                          | <i>OTG_HS general core configuration register (OTG_HS_GCCFG) on page 1416</i>                                                                 |
| OTG_HS_CID                          | 0x03C                          | <i>OTG_HS core ID register (OTG_HS_CID) on page 1417</i>                                                                                      |
| OTG_HS_HPTXFSIZ                     | 0x100                          | <i>OTG_HS Host periodic transmit FIFO size register (OTG_HS_HPTXFSIZ) on page 1417</i>                                                        |
| OTG_HS_DIEPTXF <sub>x</sub>         | 0x104<br>0x108<br>...<br>0x114 | <i>OTG_HS device IN endpoint transmit FIFO size register (OTG_HS_DIEPTXF<sub>x</sub>) (x = 1..5, where x is the FIFO_number) on page 1418</i> |

### Host-mode CSR map

These registers must be programmed every time the core changes to host mode.

Table 209. Host-mode control and status registers (CSRs)

| Acronym         | Offset address | Register name                                                                                 |
|-----------------|----------------|-----------------------------------------------------------------------------------------------|
| OTG_HS_HCFG     | 0x400          | <i>OTG_HS host configuration register (OTG_HS_HCFG) on page 1418</i>                          |
| OTG_HS_HFIR     | 0x404          | <i>OTG_HS Host frame interval register (OTG_HS_HFIR) on page 1420</i>                         |
| OTG_HS_HFNUM    | 0x408          | <i>OTG_HS host frame number/frame time remaining register (OTG_HS_HFNUM) on page 1420</i>     |
| OTG_HS_HPTXSTS  | 0x410          | <i>OTG_HS Host periodic transmit FIFO/queue status register (OTG_HS_HPTXSTS) on page 1421</i> |
| OTG_HS_HAINT    | 0x414          | <i>OTG_HS Host all channels interrupt register (OTG_HS_HAINT) on page 1422</i>                |
| OTG_HS_HAINTMSK | 0x418          | <i>OTG_HS host all channels interrupt mask register (OTG_HS_HAINTMSK) on page 1422</i>        |
| OTG_HS_HPRT     | 0x440          | <i>OTG_HS host port control and status register (OTG_HS_HPRT) on page 1423</i>                |

Table 209. Host-mode control and status registers (CSRs) (continued)

| Acronym          | Offset address                 | Register name                                                                                                              |
|------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| OTG_HS_HCCHARx   | 0x500<br>0x520<br>...<br>0x660 | <i>OTG_HS host channel-x characteristics register (OTG_HS_HCCHARx) (x = 0..11, where x = Channel_number) on page 1425</i>  |
| OTG_HS_HCSPLTx   | 0x504                          | <i>OTG_HS host channel-x split control register (OTG_HS_HCSPLTx) (x = 0..11, where x = Channel_number) on page 1427</i>    |
| OTG_HS_HCINTx    | 0x508                          | <i>OTG_HS host channel-x interrupt register (OTG_HS_HCINTx) (x = 0..11, where x = Channel_number) on page 1428</i>         |
| OTG_HS_HCINTMSKx | 0x50C                          | <i>OTG_HS host channel-x interrupt mask register (OTG_HS_HCINTMSKx) (x = 0..11, where x = Channel_number) on page 1429</i> |
| OTG_HS_HCTSIZx   | 0x510                          | <i>OTG_HS host channel-x transfer size register (OTG_HS_HCTSIZx) (x = 0..11, where x = Channel_number) on page 1430</i>    |
| OTG_HS_HCDMAx    | 0x514                          | <i>OTG_HS host channel-x DMA address register (OTG_HS_HCDMAx) (x = 0..11, where x = Channel_number) on page 1431</i>       |

### Device-mode CSR map

These registers must be programmed every time the core changes to peripheral mode.

Table 210. Device-mode control and status registers

| Acronym           | Offset address | Register name                                                                                    |
|-------------------|----------------|--------------------------------------------------------------------------------------------------|
| OTG_HS_DCFG       | 0x800          | <i>OTG_HS device configuration register (OTG_HS_DCFG) on page 1431</i>                           |
| OTG_HS_DCTL       | 0x804          | <i>OTG_HS device control register (OTG_HS_DCTL) on page 1433</i>                                 |
| OTG_HS_DSTS       | 0x808          | <i>OTG_HS device status register (OTG_HS_DSTS) on page 1435</i>                                  |
| OTG_HS_DIEPMSK    | 0x810          | <i>OTG_HS device IN endpoint common interrupt mask register (OTG_HS_DIEPMSK) on page 1436</i>    |
| OTG_HS_DOEPMSK    | 0x814          | <i>OTG_HS device OUT endpoint common interrupt mask register (OTG_HS_DOEPMSK) on page 1437</i>   |
| OTG_HS_DAINTh     | 0x818          | <i>OTG_HS device all endpoints interrupt register (OTG_HS_DAINTh) on page 1438</i>               |
| OTG_HS_DAINThMSK  | 0x81C          | <i>OTG_HS all endpoints interrupt mask register (OTG_HS_DAINThMSK) on page 1439</i>              |
| OTG_HS_DVBUSDIS   | 0x828          | <i>OTG_HS device <math>V_{BUS}</math> discharge time register (OTG_HS_DVBUSDIS) on page 1439</i> |
| OTG_HS_DVBUSPULSE | 0x82C          | <i>OTG_HS device <math>V_{BUS}</math> pulsing time register (OTG_HS_DVBUSPULSE) on page 1440</i> |
| OTG_HS_DTHRCTL    | 0x830          | <i>OTG_HS Device threshold control register (OTG_HS_DTHRCTL) on page 1441</i>                    |

Table 210. Device-mode control and status registers (continued)

| Acronym                             | Offset address                 | Register name                                                                                                                               |
|-------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| OTG_HS_DIEPEMPMSK                   | 0x834                          | <i>OTG_HS device IN endpoint FIFO empty interrupt mask register: (OTG_HS_DIEPEMPMSK) on page 1442</i>                                       |
| OTG_HS_DEACHINT                     | 0x838                          | <i>OTG_HS device each endpoint interrupt register (OTG_HS_DEACHINT) on page 1442</i>                                                        |
| OTG_HS_DEACHINTMSK                  | 0x83C                          | <i>OTG_HS device each endpoint interrupt register mask (OTG_HS_DEACHINTMSK) on page 1443</i>                                                |
| OTG_HS_DIEPEACHMSK1                 | 0x844                          | <i>OTG_HS device each in endpoint-1 interrupt register (OTG_HS_DIEPEACHMSK1) on page 1443</i>                                               |
| OTG_HS_DOEPEACHMSK1                 | 0x884                          | <i>OTG_HS device each OUT endpoint-1 interrupt register (OTG_HS_DOEPEACHMSK1) on page 1444</i>                                              |
| OTG_HS_DIEPCTLx                     | 0x900<br>0x920<br>...<br>0x9A0 | <i>OTG device endpoint-x control register (OTG_HS_DIEPCTLx) (x = 0..5, where x = Endpoint_number) on page 1445</i>                          |
| OTG_HS_DIEPINTx                     | 0x908                          | <i>OTG_HS device endpoint-x interrupt register (OTG_HS_DIEPINTx) (x = 0..5, where x = Endpoint_number) on page 1452</i>                     |
| OTG_HS_DIEPTSIZE0                   | 0x910                          | <i>OTG_HS device IN endpoint 0 transfer size register (OTG_HS_DIEPTSIZE0) on page 1455</i>                                                  |
| OTG_HS_DIEPDMAx/<br>OTG_HS_DOEPDMAx | 0x914/0xB14                    | <i>OTG_HS device endpoint-x DMA address register (OTG_HS_DIEPDMAx / OTG_HS_DOEPDMAx) (x = 0..5, where x = Endpoint_number) on page 1459</i> |
| OTG_HS_DTXFSTSx                     | 0x918                          | <i>OTG_HS device IN endpoint transmit FIFO status register (OTG_HS_DTXFSTSx) (x = 0..5, where x = Endpoint_number) on page 1458</i>         |
| OTG_HS_DIEPTSIZEx                   | 0x930<br>0x950<br>...<br>0x9B0 | <i>OTG_HS device endpoint-x transfer size register (OTG_HS_DIEPTSIZEx) (x = 1..5, where x = Endpoint_number) on page 1457</i>               |
| OTG_HS_DOEPCTL0                     | 0xB00                          | <i>OTG_HS device control OUT endpoint 0 control register (OTG_HS_DOEPCTL0) on page 1448</i>                                                 |
| OTG_HS_DOEPSIZE0                    | 0xB10                          | <i>OTG_HS device OUT endpoint 0 transfer size register (OTG_HS_DOEPSIZE0) on page 1456</i>                                                  |
| OTG_HS_DOEPCTLx                     | 0xB20<br>0xB40<br>...<br>0xBA0 | <i>OTG_HS device endpoint-x control register (OTG_HS_DOEPCTLx) (x = 1..5, where x = Endpoint_number) on page 1449</i>                       |

**Table 210. Device-mode control and status registers (continued)**

| Acronym         | Offset address | Register name                                                                                                                   |
|-----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| OTG_HS_DOEPINTx | 0xB08          | <i>OTG_HS device endpoint-x interrupt register<br/>(OTG_HS_DOEPINTx) (x = 0..5, where x = Endpoint_number) on page 1454</i>     |
|                 | 0xB28          |                                                                                                                                 |
|                 | ...            |                                                                                                                                 |
|                 | 0xBA8          |                                                                                                                                 |
| OTG_HS_DOEPSIZx | 0xB30          | <i>OTG_HS device endpoint-x transfer size register<br/>(OTG_HS_DOEPSIZx) (x = 1..5, where x = Endpoint_number) on page 1458</i> |
|                 | 0xB50          |                                                                                                                                 |
|                 | ...            |                                                                                                                                 |
|                 | 0xBB0          |                                                                                                                                 |

### Data FIFO (DFIFO) access register map

These registers, available in both host and peripheral modes, are used to read or write the FIFO space for a specific endpoint or a channel, in a given direction. If a host channel is of type IN, the FIFO can only be read on the channel. Similarly, if a host channel is of type OUT, the FIFO can only be written on the channel.

**Table 211. Data FIFO (DFIFO) access register map**

| FIFO access register section                                                                                                                                                              | Address range | Access |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|
| Device IN Endpoint 0/Host OUT Channel 0: DFIFO Write Access<br>Device OUT Endpoint 0/Host IN Channel 0: DFIFO Read Access                                                                 | 0x1000–0x1FFC | w<br>r |
| Device IN Endpoint 1/Host OUT Channel 1: DFIFO Write Access<br>Device OUT Endpoint 1/Host IN Channel 1: DFIFO Read Access                                                                 | 0x2000–0x2FFC | w<br>r |
| ...                                                                                                                                                                                       | ...           | ...    |
| Device IN Endpoint x <sup>(1)</sup> /Host OUT Channel x <sup>(1)</sup> : DFIFO Write Access<br>Device OUT Endpoint x <sup>(1)</sup> /Host IN Channel x <sup>(1)</sup> : DFIFO Read Access | 0xX000–0xXFFC | w<br>r |

1. Where x is 5 in peripheral mode and 11 in host mode.

### Power and clock gating CSR map

There is a single register for power and clock gating. It is available in both host and peripheral modes.

**Table 212. Power and clock gating control and status registers**

| Register name                           | Acronym        | Offset address: 0xE00–0xFFF |
|-----------------------------------------|----------------|-----------------------------|
| Power and clock gating control register | OTG_HS_PCGCCTL | 0xE00–0xE04                 |
| Reserved                                | -              | 0xE05–0xFFF                 |

## 35.12.2 OTG\_HS global registers

These registers are available in both host and peripheral modes, and do not need to be reprogrammed when switching between these modes.

Bit values in the register descriptions are expressed in binary unless otherwise specified.

### OTG\_HS control and status register (OTG\_HS\_GOTGCTL)

Address offset: 0x000

Reset value: 0x0001 0000

The OTG control and status register controls the behavior and reflects the status of the OTG function of the core.

|          |    |    |    |    |    |    |    |    |    |    |    |       |       |      |        |          |    |    |    |        |         |       |        |          |   |   |   |     |        |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|-------|-------|------|--------|----------|----|----|----|--------|---------|-------|--------|----------|---|---|---|-----|--------|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19    | 18    | 17   | 16     | 15       | 14 | 13 | 12 | 11     | 10      | 9     | 8      | 7        | 6 | 5 | 4 | 3   | 2      | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    | BSVLD | ASVLD | DBCT | CIDSTS | Reserved |    |    |    | DHNPEN | HSHNPEN | HNPBQ | HNGSCS | Reserved |   |   |   | SRQ | SRQSCS |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    | r     | r     | r    | r      |          |    |    |    | r      | r       | r     | r      |          |   |   |   | r   | r      |   |   |

Bits 31:20 Reserved, must be kept at reset value.

**Bit 19 BSVLD:** B-session valid

Indicates the peripheral mode transceiver status.

0: B-session is not valid.

1: B-session is valid.

In OTG mode, you can use this bit to determine if the device is connected or disconnected.

*Note: Only accessible in peripheral mode.*

**Bit 18 ASVLD:** A-session valid

Indicates the host mode transceiver status.

0: A-session is not valid

1: A-session is valid

*Note: Only accessible in host mode.*

**Bit 17 DBCT:** Long/short debounce time

Indicates the debounce time of a detected connection.

0: Long debounce time, used for physical connections (100 ms + 2.5  $\mu$ s)

1: Short debounce time, used for soft connections (2.5  $\mu$ s)

*Note: Only accessible in host mode.*

**Bit 16 CIDSTS:** Connector ID status

Indicates the connector ID status on a connect event.

0: The OTG\_HS controller is in A-device mode

1: The OTG\_HS controller is in B-device mode

*Note: Accessible in both peripheral and host modes.*

Bits 15:12 Reserved, must be kept at reset value.

**Bit 11 DHNPEN:** Device HNP enabled

The application sets this bit when it successfully receives a SetFeature.SetHNPEnable command from the connected USB host.

0: HNP is not enabled in the application

1: HNP is enabled in the application

*Note: Only accessible in peripheral mode.*

**Bit 10 HSHNPEN:** Host set HNP enable

The application sets this bit when it has successfully enabled HNP (using the SetFeature.SetHNPEnable command) on the connected device.

0: Host Set HNP is not enabled

1: Host Set HNP is enabled

*Note: Only accessible in host mode.*

**Bit 9 HNPRQ:** HNP request

The application sets this bit to initiate an HNP request to the connected USB host. The application can clear this bit by writing a 0 when the host negotiation success status change bit in the OTG interrupt register (HNSSCHG bit in OTG\_HS\_GOTGINT) is set. The core clears this bit when the HNSSCHG bit is cleared.

0: No HNP request

1: HNP request

*Note: Only accessible in peripheral mode.*

**Bit 8 HNGSCS:** Host negotiation success

The core sets this bit when host negotiation is successful. The core clears this bit when the HNP Request (HNPRQ) bit in this register is set.

0: Host negotiation failure

1: Host negotiation success

*Note: Only accessible in peripheral mode.*

Bits 7:2 Reserved, must be kept at reset value.

**Bit 1 SRQ:** Session request

The application sets this bit to initiate a session request on the USB. The application can clear this bit by writing a 0 when the host negotiation success status change bit in the OTG Interrupt register (HNSSCHG bit in OTG\_HS\_GOTGINT) is set. The core clears this bit when the HNSSCHG bit is cleared.

If you use the USB 1.1 full-speed serial transceiver interface to initiate the session request, the application must wait until  $V_{BUS}$  discharges to 0.2 V, after the B-Session Valid bit in this register (BSVLD bit in OTG\_HS\_GOTGCTL) is cleared.

0: No session request

1: Session request

*Note: Only accessible in peripheral mode.*

**Bit 0 SRQSCS:** Session request success

The core sets this bit when a session request initiation is successful.

0: Session request failure

1: Session request success

*Note: Only accessible in peripheral mode.*

**OTG\_HS interrupt register (OTG\_HS\_GOTGINT)**

Address offset: 0x04

Reset value: 0x0000 0000

The application reads this register whenever there is an OTG interrupt and clears the bits in this register to clear the OTG interrupt.

|          |    |    |    |    |    |    |    |    |    |    |    |        |         |        |          |    |    |    |    |    |    |   |         |         |          |   |   |   |       |      |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|--------|---------|--------|----------|----|----|----|----|----|----|---|---------|---------|----------|---|---|---|-------|------|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19     | 18      | 17     | 16       | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8       | 7       | 6        | 5 | 4 | 3 | 2     | 1    | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    | DBCONE | ADTOCHG | HNGDET | Reserved |    |    |    |    |    |    |   | HNSSCHG | SRSSCHG | Reserved |   |   |   | SEDET | Res. |   |
|          |    |    |    |    |    |    |    |    |    |    |    | rc_w1  | rc_w1   | rc_w1  |          |    |    |    |    |    |    |   | rc_w1   | rc_w1   |          |   |   |   | rc_w1 |      |   |

Bits 31:20 Reserved, must be kept at reset value.

Bit 19 **DBCONE**: Debounce done

The core sets this bit when the debounce is completed after the device connect. The application can start driving USB reset after seeing this interrupt. This bit is only valid when the HNP Capable or SRP Capable bit is set in the Core USB Configuration register (HNPCAP bit or SRPCAP bit in OTG\_HS\_GUSBCFG, respectively).

*Note: Only accessible in host mode.*

Bit 18 **ADTOCHG**: A-device timeout change

The core sets this bit to indicate that the A-device has timed out while waiting for the B-device to connect.

*Note: Accessible in both peripheral and host modes.*

Bit 17 **HNGDET**: Host negotiation detected

The core sets this bit when it detects a host negotiation request on the USB.

*Note: Accessible in both peripheral and host modes.*

Bits 16:10 Reserved, must be kept at reset value.

Bit 9 **HNSSCHG**: Host negotiation success status change

The core sets this bit on the success or failure of a USB host negotiation request. The application must read the host negotiation success bit of the OTG Control and Status register (HNGSCS in OTG\_HS\_GOTGCTL) to check for success or failure.

*Note: Accessible in both peripheral and host modes.*

Bits 7:3 Reserved, must be kept at reset value.

Bit 8 **SRSSCHG**: Session request success status change

The core sets this bit on the success or failure of a session request. The application must read the session request success bit in the OTG Control and status register (SRQSCS bit in OTG\_HS\_GOTGCTL) to check for success or failure.

*Note: Accessible in both peripheral and host modes.*

Bit 2 **SEDET**: Session end detected

The core sets this bit to indicate that the level of the voltage on  $V_{BUS}$  is no longer valid for a B-device session when  $V_{BUS} < 0.8$  V.

Bits 1:0 Reserved, must be kept at reset value.

**OTG\_HS AHB configuration register (OTG\_HS\_GAHBCFG)**

Address offset: 0x008

Reset value: 0x0000 0000

This register can be used to configure the core after power-on or a change in mode. This register mainly contains AHB system-related configuration parameters. Do not change this register after the initial programming. The application must program this register before starting any transactions on either the AHB or the USB.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |          |         |          |       |         |    |    |    |   |      |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|----------|---------|----------|-------|---------|----|----|----|---|------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8        | 7       | 6        | 5     | 4       | 3  | 2  | 1  | 0 |      |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   | PTXFELVL | TXFELVL | Reserved | DMAEN | HBSTLEN |    |    |    |   | GINT |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   | rw       | rw      |          | rw    | rw      | rw | rw | rw |   |      |

Bits 31:20 Reserved, must be kept at reset value.

Bit 8 **PTXFELVL**: Periodic Tx FIFO empty level

Indicates when the periodic Tx FIFO empty interrupt bit in the Core interrupt register (PTXFE bit in OTG\_HS\_GINTSTS) is triggered.

0: PTXFE (in OTG\_HS\_GINTSTS) interrupt indicates that the Periodic Tx FIFO is half empty  
1: PTXFE (in OTG\_HS\_GINTSTS) interrupt indicates that the Periodic Tx FIFO is completely empty

*Note: Only accessible in host mode.*

Bit 7 **TXFELVL**: Tx FIFO empty level

In peripheral mode, this bit indicates when the IN endpoint Transmit FIFO empty interrupt (TXFE in OTG\_HS\_DIEPINTx) is triggered.

0: TXFE (in OTG\_HS\_DIEPINTx) interrupt indicates that the IN Endpoint Tx FIFO is half empty  
1: TXFE (in OTG\_HS\_DIEPINTx) interrupt indicates that the IN Endpoint Tx FIFO is completely empty

*Note: Only accessible in peripheral mode.*

Bit 6 Reserved, must be kept at reset value.

Bits 5 **DMAEN**: DMA enable

0: The core operates in slave mode  
1: The core operates in DMA mode

Bits 4:1 **HBSTLEN**: Burst length/type

0000 Single  
0001 INCR  
0011 INCR4  
0101 INCR8  
0111 INCR16  
Others: Reserved

Bit 0 **GINT**: Global interrupt mask

This bit is used to mask or unmask the interrupt line assertion to the application. Irrespective of this bit setting, the interrupt status registers are updated by the core.

0: Mask the interrupt assertion to the application.  
1: Unmask the interrupt assertion to the application

*Note: Accessible in both peripheral and host modes.*

**OTG\_HS USB configuration register (OTG\_HS\_GUSBCFG)**

Address offset: 0x00C

Reset value: 0x0000 1440

This register can be used to configure the core after power-on or a changing to host mode or peripheral mode. It contains USB and USB-PHY related configuration parameters. The application must program this register before starting any transactions on either the AHB or the USB. Do not make changes to this register after the initial programming.

| 31     | 30    | 29    | 28       | 27 | 26 | 25 | 24     | 23   | 22   | 21    | 20         | 19         | 18      | 17     | 16      | 15       | 14      | 13       | 12   | 11 | 10 | 9      | 8  | 7      | 6        | 5      | 4        | 3 | 2 | 1     | 0 |  |
|--------|-------|-------|----------|----|----|----|--------|------|------|-------|------------|------------|---------|--------|---------|----------|---------|----------|------|----|----|--------|----|--------|----------|--------|----------|---|---|-------|---|--|
| CTXPKT | FDMOD | FHMOD | Reserved |    |    |    | ULPIPD | PTCI | PCCI | TSDFS | ULPIEVBUSI | ULPIEVBUSD | ULPICSM | ULPIAR | ULPIFSL | Reserved | PHYLPCS | Reserved | TRDT |    |    | HNPCAP |    | SRPCAP | Reserved | PHYSEL | Reserved |   |   | TOTAL |   |  |
| rw     | rw    | rw    |          |    |    |    | rw     | rw   | rw   | rw    | rw         | rw         | rw      | rw     | rw      | rw       | rw      | rw       | rw   | rw | rw | rw     | rw | rw     | rw       | rw     |          |   |   |       |   |  |

Bit 31 **CTXPKT**: Corrupt Tx packet

This bit is for debug purposes only. Never set this bit to 1.

*Note: Accessible in both peripheral and host modes.*

Bit 30 **FDMOD**: Forced peripheral mode

Writing a 1 to this bit forces the core to peripheral mode irrespective of the OTG\_HS\_ID input pin.

0: Normal mode

1: Forced peripheral mode

After setting the force bit, the application must wait at least 25 ms before the change takes effect.

*Note: Accessible in both peripheral and host modes.*

Bit 29 **FHMOD**: Forced host mode

Writing a 1 to this bit forces the core to host mode irrespective of the OTG\_HS\_ID input pin.

0: Normal mode

1: Forced host mode

After setting the force bit, the application must wait at least 25 ms before the change takes effect.

*Note: Accessible in both peripheral and host modes.*

Bits 28:26 Reserved, must be kept at reset value.

Bit 25 **ULPIPD**: ULPI interface protect disable

This bit controls the circuitry built in the PHY to protect the ULPI interface when the link tri-states stp and data. Any pull-up or pull-down resistors employed by this feature can be disabled. Refer to the ULPI specification for more details.

0: Enables the interface protection circuit

1: Disables the interface protection circuit

Bit 24 **PTCI**: Indicator pass through

This bit controls whether the complement output is qualified with the internal  $V_{BUS}$  valid comparator before being used in the  $V_{BUS}$  state in the RX CMD. Refer to the ULPI specification for more details.

0: Complement Output signal is qualified with the Internal  $V_{BUS}$  valid comparator

1: Complement Output signal is not qualified with the Internal  $V_{BUS}$  valid comparator

- Bit 23 **PCCI**: Indicator complement  
 This bit controls the PHY to invert the ExternalVbusIndicator input signal, and generate the complement output. Refer to the ULPI specification for more details.  
 0: PHY does not invert the ExternalVbusIndicator signal  
 1: PHY inverts ExternalVbusIndicator signal
- Bit 22 **TSDPS**: TermSel DLine pulsing selection  
 This bit selects utmi\_termselect to drive the data line pulse during SRP (session request protocol).  
 0: Data line pulsing using utmi\_txvalid (default)  
 1: Data line pulsing using utmi\_termsel
- Bit 21 **ULPIEBUSI**: ULPI external V<sub>BUS</sub> indicator  
 This bit indicates to the ULPI PHY to use an external V<sub>BUS</sub> overcurrent indicator.  
 0: PHY uses an internal V<sub>BUS</sub> valid comparator  
 1: PHY uses an external V<sub>BUS</sub> valid comparator
- Bit 20 **ULPIEBUSD**: ULPI External V<sub>BUS</sub> Drive  
 This bit selects between internal or external supply to drive 5 V on V<sub>BUS</sub>, in the ULPI PHY.  
 0: PHY drives V<sub>BUS</sub> using internal charge pump (default)  
 1: PHY drives V<sub>BUS</sub> using external supply.
- Bit 19 **ULPICSM**: ULPI Clock SuspendM  
 This bit sets the ClockSuspendM bit in the interface control register on the ULPI PHY. This bit applies only in the serial and carkit modes.  
 0: PHY powers down the internal clock during suspend  
 1: PHY does not power down the internal clock
- Bit 18 **ULPIAR**: ULPI Auto-resume  
 This bit sets the AutoResume bit in the interface control register on the ULPI PHY.  
 0: PHY does not use AutoResume feature  
 1: PHY uses AutoResume feature
- Bit 17 **ULPIFSL**: ULPI FS/LS select  
 The application uses this bit to select the FS/LS serial interface for the ULPI PHY. This bit is valid only when the FS serial transceiver is selected on the ULPI PHY.  
 0: ULPI interface  
 1: ULPI FS/LS serial interface
- Bit 16 Reserved, must be kept at reset value.
- Bit 15 **PHYLPCS**: PHY Low-power clock select  
 This bit selects either 480 MHz or 48 MHz (low-power) PHY mode. In FS and LS modes, the PHY can usually operate on a 48 MHz clock to save power.  
 0: 480 MHz internal PLL clock  
 1: 48 MHz external clock  
 In 480 MHz mode, the UTMI interface operates at either 60 or 30 MHz, depending on whether the 8- or 16-bit data width is selected. In 48 MHz mode, the UTMI interface operates at 48 MHz in FS and LS modes.
- Bit 14 Reserved, must be kept at reset value.
- Bits 13:10 **TRDT**: USB turnaround time  
 These bits allow to set the turnaround time in PHY clocks. They must be configured according to [Table 213: TRDT values](#), depending on the application AHB frequency. Higher TRDT values allow stretching the USB response time to IN tokens in order to compensate for longer AHB read access latency to the Data FIFO.

Bit 9 **HNPCAP**: HNP-capable

The application uses this bit to control the OTG\_HS controller's HNP capabilities.

0: HNP capability is not enabled

1: HNP capability is enabled

*Note: Accessible in both peripheral and host modes.*

Bit 8 **SRPCAP**: SRP-capable

The application uses this bit to control the OTG\_HS controller's SRP capabilities. If the core operates as a nonSRP-capable B-device, it cannot request the connected A-device (host) to activate  $V_{BUS}$  and start a session.

0: SRP capability is not enabled

1: SRP capability is enabled

*Note: Accessible in both peripheral and host modes.*

## Bit 7 Reserved, must be kept at reset value.

Bit 6 **PHYSEL**: USB 2.0 high-speed ULPI PHY or USB 1.1 full-speed serial transceiver select

0: USB 2.0 high-speed ULPI PHY

1: USB 1.1 full-speed serial transceiver

## Bits 5:3 Reserved, must be kept at reset value.

Bits 2:0 **TOTAL**: FS timeout calibration

The number of PHY clocks that the application programs in this field is added to the full-speed interpacket timeout duration in the core to account for any additional delays introduced by the PHY. This can be required, because the delay introduced by the PHY in generating the line state condition can vary from one PHY to another.

The USB standard timeout value for full-speed operation is 16 to 18 (inclusive) bit times. The application must program this field based on the speed of enumeration. The number of bit times added per PHY clock is 0.25 bit times.

Table 213. TRDT values

| AHB frequency range (MHz) |     | TRDT minimum value |
|---------------------------|-----|--------------------|
| Min.                      | Max |                    |
| 30                        | -   | 0x9                |

**OTG\_HS reset register (OTG\_HS\_GRSTCTL)**

Address offset: 0x010

Reset value: 0x8000 0000

The application uses this register to reset various hardware features inside the core.

| 31     | 30     | 29       | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9      | 8  | 7       | 6       | 5        | 4     | 3     | 2     | 1 | 0 |
|--------|--------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|---------|---------|----------|-------|-------|-------|---|---|
| AHBIDL | DMAREQ | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | TXFNUM |    | TXFFLSH | RXFFLSH | Reserved | FCRST | HSRST | CSRST |   |   |
| r      | r      |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rw     | rs | rs      | rs      |          |       |       |       |   |   |

**Bit 31 AHBIDL:** AHB master idle

Indicates that the AHB master state machine is in the Idle condition.

*Note:* Accessible in both peripheral and host modes.**Bit 30 DMAREQ:** DMA request signal

This bit indicates that the DMA request is in progress. Used for debug.

**Bits 29:11** Reserved, must be kept at reset value.**Bits 10:6 TXFNUM:** TxFIFO number

This is the FIFO number that must be flushed using the TxFIFO Flush bit. This field must not be changed until the core clears the TxFIFO Flush bit.

00000:

- Nonperiodic TxFIFO flush in host mode
- Tx FIFO 0 flush in peripheral mode

00001:

- Periodic TxFIFO flush in host mode
- TXFIFO 1 flush in peripheral mode

00010: TXFIFO 2 flush in peripheral mode

...

00101: TXFIFO 15 flush in peripheral mode

10000: Flush all the transmit FIFOs in peripheral or host mode.

*Note:* Accessible in both peripheral and host modes.**Bit 5 TXFFLSH:** TxFIFO flush

This bit selectively flushes a single or all transmit FIFOs, but cannot do so if the core is in the midst of a transaction.

The application must write this bit only after checking that the core is neither writing to the TxFIFO nor reading from the TxFIFO. Verify using these registers:

- Read: the NAK effective interrupt ensures the core is not reading from the FIFO
- Write: the AHBIDL bit in OTG\_HS\_GRSTCTL ensures that the core is not writing anything to the FIFO

*Note:* Accessible in both peripheral and host modes.

Bit 4 **RXFFLSH**: RxFIFO flush

The application can flush the entire RxFIFO using this bit, but must first ensure that the core is not in the middle of a transaction.

The application must only write to this bit after checking that the core is neither reading from the RxFIFO nor writing to the RxFIFO.

The application must wait until the bit is cleared before performing any other operation. This bit requires 8 clocks (slowest of PHY or AHB clock) to be cleared.

*Note: Accessible in both peripheral and host modes.*

## Bit 3    Reserved, must be kept at reset value.

**Bit 2 FCRST:** Host frame counter reset

The application writes this bit to reset the (micro) frame number counter inside the core. When the (micro) frame counter is reset, the subsequent SOF sent out by the core has a frame number of 0.

*Note:* Only accessible in host mode.

**Bit 1 HSRST:** HCLK soft reset

The application uses this bit to flush the control logic in the AHB Clock domain. Only AHB Clock Domain pipelines are reset.

FIFOs are not flushed with this bit.

All state machines in the AHB clock domain are reset to the Idle state after terminating the transactions on the AHB, following the protocol.

CSR control bits used by the AHB clock domain state machines are cleared.

To clear this interrupt, status mask bits that control the interrupt status and are generated by the AHB clock domain state machine are cleared.

Because interrupt status bits are not cleared, the application can get the status of any core events that occurred after it set this bit.

This is a self-clearing bit that the core clears after all necessary logic is reset in the core. This can take several clocks, depending on the core's current state.

*Note:* Accessible in both peripheral and host modes.

**Bit 0 CSRST:** Core soft reset

Resets the HCLK and PCLK domains as follows:

Clears the interrupts and all the CSR register bits except for the following bits:

- RSTPDMODL bit in OTG\_HS\_PCGCCTL
- GAYEHCLK bit in OTG\_HS\_PCGCCTL
- PWRCLMP bit in OTG\_HS\_PCGCCTL
- STPPCLK bit in OTG\_HS\_PCGCCTL
- FSLSPCS bit in OTG\_HS\_HCFG
- DSPD bit in OTG\_HS\_DCFG

All module state machines (except for the AHB slave unit) are reset to the Idle state, and all the transmit FIFOs and the receive FIFO are flushed.

Any transactions on the AHB Master are terminated as soon as possible, after completing the last data phase of an AHB transfer. Any transactions on the USB are terminated immediately.

The application can write to this bit any time it wants to reset the core. This is a self-clearing bit and the core clears this bit after all the necessary logic is reset in the core, which can take several clocks, depending on the current state of the core. Once this bit has been cleared, the software must wait at least 3 PHY clocks before accessing the PHY domain (synchronization delay). The software must also check that bit 31 in this register is set to 1 (AHB Master is Idle) before starting any operation.

Typically, the software reset is used during software development and also when you dynamically change the PHY selection bits in the above listed USB configuration registers. When you change the PHY, the corresponding clock for the PHY is selected and used in the PHY domain. Once a new clock is selected, the PHY domain has to be reset for proper operation.

*Note:* Accessible in both peripheral and host modes.

**OTG\_HS core interrupt register (OTG\_HS\_GINTSTS)**

Address offset: 0x014

Reset value: 0x0400 0020

This register interrupts the application for system-level events in the current mode (peripheral mode or host mode).

Some of the bits in this register are valid only in host mode, while others are valid in peripheral mode only. This register also indicates the current mode. To clear the interrupt status bits of the rc\_w1 type, the application must write 1 into the bit.

The FIFO status interrupts are read-only; once software reads from or writes to the FIFO while servicing these interrupts, FIFO interrupt conditions are cleared automatically.

The application must clear the OTG\_HS\_GINTSTS register at initialization before unmasking the interrupt bit to avoid any interrupts generated prior to initialization.

| 31       | 30 | 29 | 28 | 27       | 26    | 25 | 24 | 23 | 22        | 21                  | 20 | 19 | 18 | 17       | 16       | 15   | 14 | 13 | 12    | 11      | 10      | 9      | 8       | 7     | 6        | 5        | 4     | 3     | 2     | 1        | 0      |        |       |        |      |      |
|----------|----|----|----|----------|-------|----|----|----|-----------|---------------------|----|----|----|----------|----------|------|----|----|-------|---------|---------|--------|---------|-------|----------|----------|-------|-------|-------|----------|--------|--------|-------|--------|------|------|
| WKUINT   |    |    |    | Reserved | PTXFE |    |    |    | DATAFSUSP | IPXFR/INCOMP/ISOOUT |    |    |    | Reserved | Reserved | EOFP |    |    |       | ISOODRP | ENUMDNE | USBRST | USBSUSP | ESUSP | Reserved | GONAKEFF |       |       |       | GINAKEFF | NPTXFE | RXFLVL | SOF   | OTGINT | MMIS | CMOD |
| SRQINT   |    |    |    |          |       |    |    |    |           |                     |    |    |    |          |          |      |    |    |       |         |         |        |         |       |          |          |       |       |       |          |        |        |       |        |      |      |
| DISCINT  |    |    |    |          |       |    |    |    |           |                     |    |    |    |          |          |      |    |    |       |         |         |        |         |       |          |          |       |       |       |          |        |        |       |        |      |      |
| CIDSCHG  |    |    |    |          |       |    |    |    |           |                     |    |    |    |          |          |      |    |    |       |         |         |        |         |       |          |          |       |       |       |          |        |        |       |        |      |      |
| Reserved |    |    |    | r        | r     |    |    |    | rc_w1     | r                   |    |    |    | rc_w1    | rc_w1    |      |    |    | rc_w1 | rc_w1   | rc_w1   | rc_w1  | rc_w1   | rc_w1 | rc_w1    | rc_w1    | rc_w1 | rc_w1 | rc_w1 | rc_w1    | rc_w1  | rc_w1  | rc_w1 |        |      |      |

Bit 31 **WKUPINT**: Resume/remote wake-up detected interrupt

In peripheral mode, this interrupt is asserted when a resume is detected on the USB. In host mode, this interrupt is asserted when a remote wake-up is detected on the USB.

*Note: Accessible in both peripheral and host modes.*

Bit 30 **SRQINT**: Session request/new session detected interrupt

In host mode, this interrupt is asserted when a session request is detected from the device. In peripheral mode, this interrupt is asserted when  $V_{BUS}$  is in the valid range for a B-device device. Accessible in both peripheral and host modes.

Bit 29 **DISCINT**: Disconnect detected interrupt

Asserted when a device disconnect is detected.

*Note: Only accessible in host mode.*

Bit 28 **CIDSCHG**: Connector ID status change

The core sets this bit when there is a change in connector ID status.

*Note: Accessible in both peripheral and host modes.*

Bit 27 Reserved, must be kept at reset value.

Bit 26 **PTXFE**: Periodic Tx FIFO empty

Asserted when the periodic transmit FIFO is either half or completely empty and there is space for at least one entry to be written in the periodic request queue. The half or completely empty status is determined by the periodic Tx FIFO empty level bit in the Core AHB configuration register (PTXFELVL bit in OTG\_HS\_GAHBCFG).

*Note: Only accessible in host mode.*

**Bit 25 HCINT:** Host channels interrupt

The core sets this bit to indicate that an interrupt is pending on one of the channels of the core (in host mode). The application must read the host all channels interrupt (OTG\_HS\_HAINT) register to determine the exact number of the channel on which the interrupt occurred, and then read the corresponding host channel-x interrupt (OTG\_HS\_HCINTx) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the OTG\_HS\_HCINTx register to clear this bit.

*Note: Only accessible in host mode.*

**Bit 24 HPRTINT:** Host port interrupt

The core sets this bit to indicate a change in port status of one of the OTG\_HS controller ports in host mode. The application must read the host port control and status (OTG\_HS\_HPRT) register to determine the exact event that caused this interrupt. The application must clear the appropriate status bit in the host port control and status register to clear this bit.

*Note: Only accessible in host mode.*

Bits 23 Reserved, must be kept at reset value.

**Bit 22 DATAFSUSP:** Data fetch suspended

This interrupt is valid only in DMA mode. This interrupt indicates that the core has stopped fetching data for IN endpoints due to the unavailability of TxFIFO space or request queue space. This interrupt is used by the application for an endpoint mismatch algorithm. For example, after detecting an endpoint mismatch, the application:

- Sets a global nonperiodic IN NAK handshake
- Disables IN endpoints
- Flushes the FIFO
- Determines the token sequence from the IN token sequence learning queue
- Re-enables the endpoints
- Clears the global nonperiodic IN NAK handshake If the global nonperiodic IN NAK is cleared, the core has not yet fetched data for the IN endpoint, and the IN token is received: the core generates an “IN token received when FIFO empty” interrupt. The OTG then sends a NAK response to the host. To avoid this scenario, the application can check the FetSusp interrupt in OTG\_HS\_GINTSTS, which ensures that the FIFO is full before clearing a global NAK handshake. Alternatively, the application can mask the “IN token received when FIFO empty” interrupt when clearing a global IN NAK handshake.

**Bit 21 IPXFR:** Incomplete periodic transfer

In host mode, the core sets this interrupt bit when there are incomplete periodic transactions still pending, which are scheduled for the current frame.

*Note: Only accessible in host mode.*

**INCOMPISOOUT:** Incomplete isochronous OUT transfer

In peripheral mode, the core sets this interrupt to indicate that there is at least one isochronous OUT endpoint on which the transfer is not completed in the current frame. This interrupt is asserted along with the End of periodic frame interrupt (EOPF) bit in this register.

*Note: Only accessible in peripheral mode.*

**Bit 20 IISOXFR:** Incomplete isochronous IN transfer

The core sets this interrupt to indicate that there is at least one isochronous IN endpoint on which the transfer is not completed in the current frame. This interrupt is asserted along with the End of periodic frame interrupt (EOPF) bit in this register.

*Note: Only accessible in peripheral mode.*

**Bit 19 OEPINT:** OUT endpoint interrupt

The core sets this bit to indicate that an interrupt is pending on one of the OUT endpoints of the core (in peripheral mode). The application must read the device all endpoints interrupt (OTG\_HS\_DAINTh) register to determine the exact number of the OUT endpoint on which the interrupt occurred, and then read the corresponding device OUT Endpoint-x Interrupt (OTG\_HS\_DOEPINTx) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the corresponding OTG\_HS\_DOEPINTx register to clear this bit.

*Note: Only accessible in peripheral mode.*

**Bit 18 IEPINT:** IN endpoint interrupt

The core sets this bit to indicate that an interrupt is pending on one of the IN endpoints of the core (in peripheral mode). The application must read the device All Endpoints Interrupt (OTG\_HS\_DAINTh) register to determine the exact number of the IN endpoint on which the interrupt occurred, and then read the corresponding device IN Endpoint-x interrupt (OTG\_HS\_DIEPINTx) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the corresponding OTG\_HS\_DIEPINTx register to clear this bit.

*Note: Only accessible in peripheral mode.*

Bits 17:16 Reserved, must be kept at reset value.

**Bit 15 EOPF:** End of periodic frame interrupt

Indicates that the period specified in the periodic frame interval field of the device configuration register (PFIVL bit in OTG\_HS\_DCFG) has been reached in the current frame.

*Note: Only accessible in peripheral mode.*

**Bit 14 ISOODRP:** Isochronous OUT packet dropped interrupt

The core sets this bit when it fails to write an isochronous OUT packet into the Rx FIFO because the Rx FIFO does not have enough space to accommodate a maximum size packet for the isochronous OUT endpoint.

*Note: Only accessible in peripheral mode.*

**Bit 13 ENUMDNE:** Enumeration done

The core sets this bit to indicate that speed enumeration is complete. The application must read the device Status (OTG\_HS\_DSTS) register to obtain the enumerated speed.

*Note: Only accessible in peripheral mode.*

**Bit 12 USBRST:** USB reset

The core sets this bit to indicate that a reset is detected on the USB.

*Note: Only accessible in peripheral mode.*

**Bit 11 USBSUSP:** USB suspend

The core sets this bit to indicate that a suspend was detected on the USB. The core enters the Suspended state when there is no activity on the data lines for a period of 3 ms.

*Note: Only accessible in peripheral mode.*

**Bit 10 ESUSP:** Early suspend

The core sets this bit to indicate that an Idle state has been detected on the USB for 3 ms.

*Note: Only accessible in peripheral mode.*

Bits 9:8 Reserved, must be kept at reset value.

**Bit 7 GONAKEFF:** Global OUT NAK effective

Indicates that the Set global OUT NAK bit in the Device control register (SGONAK bit in OTG\_HS\_DCTL), set by the application, has taken effect in the core. This bit can be cleared by writing the Clear global OUT NAK bit in the Device control register (CGONAK bit in OTG\_HS\_DCTL).

*Note: Only accessible in peripheral mode.*

**Bit 6 GINAKEFF:** Global IN nonperiodic NAK effective

Indicates that the Set global nonperiodic IN NAK bit in the Device control register (SGINAK bit in OTG\_HS\_DCTL), set by the application, has taken effect in the core. That is, the core has sampled the Global IN NAK bit set by the application. This bit can be cleared by clearing the Clear global nonperiodic IN NAK bit in the Device control register (CGINAK bit in OTG\_HS\_DCTL).

This interrupt does not necessarily mean that a NAK handshake is sent out on the USB. The STALL bit takes precedence over the NAK bit.

*Note: Only accessible in peripheral mode.*

**Bit 5 NPTXFE:** Nonperiodic TxFIFO empty

This interrupt is asserted when the nonperiodic TxFIFO is either half or completely empty, and there is space in at least one entry to be written to the nonperiodic transmit request queue. The half or completely empty status is determined by the nonperiodic TxFIFO empty level bit in the OTG\_HS\_GAHBCFG register (TXFELVL bit in OTG\_HS\_GAHBCFG).

*Note: Only accessible in host mode.*

**Bit 4 RXFLVL:** RxFIFO nonempty

Indicates that there is at least one packet pending to be read from the RxFIFO.

*Note: Accessible in both host and peripheral modes.*

**Bit 3 SOF:** Start of frame

In host mode, the core sets this bit to indicate that an SOF (FS), or Keep-Alive (LS) is transmitted on the USB. The application must write a 1 to this bit to clear the interrupt.

In peripheral mode, the core sets this bit to indicate that an SOF token has been received on the USB. The application can read the Device Status register to get the current frame number. This interrupt is seen only when the core is operating in FS.

*Note: Accessible in both host and peripheral modes.*

**Bit 2 OTGINT:** OTG interrupt

The core sets this bit to indicate an OTG protocol event. The application must read the OTG Interrupt Status (OTG\_HS\_GOTGINT) register to determine the exact event that caused this interrupt. The application must clear the appropriate status bit in the OTG\_HS\_GOTGINT register to clear this bit.

*Note: Accessible in both host and peripheral modes.*

**Bit 1 MMIS:** Mode mismatch interrupt

The core sets this bit when the application is trying to access:

A host mode register, when the core is operating in peripheral mode

A peripheral mode register, when the core is operating in host mode

The register access is completed on the AHB with an OKAY response, but is ignored by the core internally and does not affect the operation of the core.

*Note: Accessible in both host and peripheral modes.*

**Bit 0 CMOD:** Current mode of operation

Indicates the current mode.

0: Peripheral mode

1: Host mode

*Note: Accessible in both host and peripheral modes.*

**OTG\_HS interrupt mask register (OTG\_HS\_GINTMSK)**

Address offset: 0x018

Reset value: 0x0000 0000

This register works with the Core interrupt register to interrupt the application. When an interrupt bit is masked, the interrupt associated with that bit is not generated. However, the Core Interrupt (OTG\_HS\_GINTSTS) register bit corresponding to that interrupt is still set.

| 31 | 30   | 29 | 28    | 27 | 26      | 25 | 24       | 23       | 22     | 21 | 20   | 19 | 18    | 17       | 16     | 15 | 14              | 13 | 12       | 11 | 10      | 9  | 8       | 7        | 6     | 5  | 4        | 3  | 2        | 1  | 0      |    |          |    |        |          |           |    |           |    |         |    |         |    |      |    |        |    |       |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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Bit 31 **WUIM**: Resume/remote wake-up detected interrupt mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Accessible in both host and peripheral modes.*

Bit 30 **SRQIM**: Session request/new session detected interrupt mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Accessible in both host and peripheral modes.*

Bit 29 **DISCINT**: Disconnect detected interrupt mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Accessible in both host and peripheral modes.*

Bit 28 **CIDSCHGM**: Connector ID status change mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Accessible in both host and peripheral modes.*

Bit 27 Reserved, must be kept at reset value.

Bit 26 **PTXFEM**: Periodic Tx FIFO empty mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Only accessible in host mode.*

Bit 25 **HCIM**: Host channels interrupt mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Only accessible in host mode.*

Bit 24 **PRTIM**: Host port interrupt mask

0: Masked interrupt

1: Unmasked interrupt

*Note: Only accessible in host mode.*

Bit 23 Reserved, must be kept at reset value.

- Bit 22 **FSUSPM**: Data fetch suspended mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 21 **IPXFRM**: Incomplete periodic transfer mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in host mode.*  
**IISOXFRM**: Incomplete isochronous OUT transfer mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 20 **IISOIXFRM**: Incomplete isochronous IN transfer mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 19 **OEPINT**: OUT endpoints interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 18 **IEPINT**: IN endpoints interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bits 17:16 Reserved, must be kept at reset value.
- Bit 15 **EOPFM**: End of periodic frame interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 14 **ISOODRPM**: Isochronous OUT packet dropped interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 13 **ENUMDNEM**: Enumeration done mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 12 **USBRST**: USB reset mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 11 **USBSUSPM**: USB suspend mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*

- Bit 10 **ESUSPM**: Early suspend mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bits 9:8 Reserved, must be kept at reset value.
- Bit 7 **GONAKEFFM**: Global OUT NAK effective mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 6 **GINAKEFFM**: Global nonperiodic IN NAK effective mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Only accessible in peripheral mode.*
- Bit 5 **NPTXFEM**: Nonperiodic Tx FIFO empty mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Accessible in both peripheral and host modes.*
- Bit 4 **RXFLVLM**: Receive FIFO nonempty mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Accessible in both peripheral and host modes.*
- Bit 3 **SOFM**: Start of frame mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Accessible in both peripheral and host modes.*
- Bit 2 **OTGINT**: OTG interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Accessible in both peripheral and host modes.*
- Bit 1 **MMISM**: Mode mismatch interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt  
*Note: Accessible in both peripheral and host modes.*
- Bit 0 Reserved, must be kept at reset value.

## OTG\_HS Receive status debug read/OTG status read and pop registers (OTG\_HS\_GRXSTSR/OTG\_HS\_GRXSTSP)

Address offset for Read: 0x01C

Address offset for Pop: 0x020

Reset value: 0x0000 0000

A read to the Receive status debug read register returns the contents of the top of the Receive FIFO. A read to the Receive status read and pop register additionally pops the top data entry out of the RxFIFO.

The receive status contents must be interpreted differently in host and peripheral modes. The core ignores the receive status pop/read when the receive FIFO is empty and returns a value of 0x0000 0000. The application must only pop the Receive Status FIFO when the Receive FIFO nonempty bit of the Core interrupt register (RXFLVL bit in OTG\_HS\_GINTSTS) is asserted.

### Host mode:

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20     | 19 | 18   | 17   | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9     | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------|----|----|----|----|----|----|----|----|----|----|--------|----|------|------|----|----|----|----|----|----|----|-------|---|---|---|---|---|---|---|---|---|
| Reserved |    |    |    |    |    |    |    |    |    |    | PKTSTS |    | DPID | BCNT |    |    |    |    |    |    |    | CHNUM |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    | r      |    | r    | r    |    |    |    |    |    |    |    | r     |   |   |   |   |   |   |   |   |   |

Bits 31:21 Reserved, must be kept at reset value.

Bits 20:17 **PKTSTS**: Packet status

Indicates the status of the received packet

0010: IN data packet received

0011: IN transfer completed (triggers an interrupt)

0101: Data toggle error (triggers an interrupt)

0111: Channel halted (triggers an interrupt)

Others: Reserved

Bits 16:15 **DPID**: Data PID

Indicates the Data PID of the received packet

00: DATA0

10: DATA1

01: DATA2

11: MDATA

Bits 14:4 **BCNT**: Byte count

Indicates the byte count of the received IN data packet.

Bits 3:0 **CHNUM**: Channel number

Indicates the channel number to which the current received packet belongs.

**Peripheral mode:**

|          |    |    |    |    |    |    |        |    |        |    |      |      |    |    |    |    |    |    |    |       |    |   |   |   |   |   |   |   |   |   |   |
|----------|----|----|----|----|----|----|--------|----|--------|----|------|------|----|----|----|----|----|----|----|-------|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24     | 23 | 22     | 21 | 20   | 19   | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11    | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    | FRMNUM |    | PKTSTS |    | DPID | BCNT |    |    |    |    |    |    |    | EPNUM |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    | r      |    | r      |    | r    | r    |    |    |    |    |    |    |    | r     |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:25 Reserved, must be kept at reset value.

Bits 24:21 **FRMNUM**: Frame number

This is the least significant 4 bits of the frame number in which the packet is received on the USB. This field is supported only when isochronous OUT endpoints are supported.

Bits 20:17 **PKTSTS**: Packet status

Indicates the status of the received packet

0001: Global OUT NAK (triggers an interrupt)

0010: OUT data packet received

0011: OUT transfer completed (triggers an interrupt)

0100: SETUP transaction completed (triggers an interrupt)

0110: SETUP data packet received

Others: Reserved

Bits 16:15 **DPID**: Data PID

Indicates the Data PID of the received OUT data packet

00: DATA0

10: DATA1

01: DATA2

11: MDATA

Bits 14:4 **BCNT**: Byte count

Indicates the byte count of the received data packet.

Bits 3:0 **EPNUM**: Endpoint number

Indicates the endpoint number to which the current received packet belongs.

**OTG\_HS Receive FIFO size register (OTG\_HS\_GRXFSIZ)**

Address offset: 0x024

Reset value: 0x0000 0400

The application can program the RAM size that must be allocated to the RxFIFO.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15   | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RXFD |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rw   |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **RXFD**: RxFIFO depth

This value is in terms of 32-bit words.

Minimum value is 16

Maximum value is 1024

The power-on reset value of this register is specified as the largest Rx data FIFO depth.

### OTG\_HS nonperiodic transmit FIFO size/Endpoint 0 transmit FIFO size register (OTG\_HS\_GNPTXFSIZ/OTG\_HS\_TX0FSIZ)

Address offset: 0x028

Reset value: 0x0000 0200

#### Host mode:

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |         |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15      | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| NPTXFD |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NPTXFSA |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| rw     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rw      |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:16 **NPTXFD**: Nonperiodic TxFIFO depth

This value is in terms of 32-bit words.

Minimum value is 16

Maximum value is 1024

Bits 15:0 **NPTXFSA**: Nonperiodic transmit RAM start address

This field contains the memory start address for nonperiodic transmit FIFO RAM.

#### Peripheral mode:

|       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31    | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| TX0FD |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | TX0FSA |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r/rw  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r/rw   |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:16 **TX0FD**: Endpoint 0 TxFIFO depth

This value is in terms of 32-bit words.

Minimum value is 16

Maximum value is 256

Bits 15:0 **TX0FSA**: Endpoint 0 transmit RAM start address

This field contains the memory start address for Endpoint 0 transmit FIFO RAM.

### OTG\_HS nonperiodic transmit FIFO/queue status register (OTG\_HS\_GNPTXSTS)

Address offset: 0x02C

Reset value: 0x0008 0400

**Note:** *In peripheral mode, this register is not valid.*

This read-only register contains the free space information for the nonperiodic TxFIFO and the nonperiodic transmit request queue.

|          |          |    |    |    |    |    |    |    |          |    |    |    |    |    |    |    |          |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|----------|----------|----|----|----|----|----|----|----|----------|----|----|----|----|----|----|----|----------|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30       | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22       | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14       | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reserved | NPTXQTOP |    |    |    |    |    |    |    | NPTQXSAV |    |    |    |    |    |    |    | NPTXFSAV |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|          | r        |    |    |    |    |    |    |    | r        |    |    |    |    |    |    |    | r        |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

- Bit 31 Reserved, must be kept at reset value.
- Bits 30:24 **NPTXQTOP**: Top of the nonperiodic transmit request queue  
Entry in the nonperiodic Tx request queue that is currently being processed by the MAC.  
Bits [30:27]: Channel/endpoint number  
Bits [26:25]:  
– 00: IN/OUT token  
– 01: Zero-length transmit packet (device IN/host OUT)  
– 10: PING/CSPLIT token  
– 11: Channel halt command  
Bit [24]: Terminate (last entry for selected channel/endpoint)
- Bits 23:16 **NPTQXSAV**: Nonperiodic transmit request queue space available  
Indicates the amount of free space available in the nonperiodic transmit request queue.  
This queue holds both IN and OUT requests in host mode. Peripheral mode has only IN requests.  
00: Nonperiodic transmit request queue is full  
01: dx1 location available  
10: dx2 locations available  
b $xn$ : dx $n$  locations available ( $0 \leq n \leq dx8$ )  
Others: Reserved
- Bits 15:0 **NPTXFSAV**: Nonperiodic TxFIFO space available  
Indicates the amount of free space available in the nonperiodic TxFIFO.  
Values are in terms of 32-bit words.  
00: Nonperiodic TxFIFO is full  
01: dx1 word available  
10: dx2 words available  
0 $xn$ : dx $n$  words available (where  $0 \leq n \leq dx1024$ )  
Others: Reserved

OTG\_HS general core configuration register (OTG\_HS\_GCCFG)

Address offset: 0x038  
Reset value: 0x0000 XXXX

|          |    |    |    |    |    |    |    |    |    |            |          |           |          |          |         |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|------------|----------|-----------|----------|----------|---------|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21         | 20       | 19        | 18       | 17       | 16      | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    | NOVBUSSENS | SOFOUTEN | VBUSBSSEN | VBUSASEN | I2CPADEN | .PWRDWN | Reserved |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    | rW         | rW       | rW        | rW       | rW       | rW      |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    | rW         | rW       | rW        | rW       | rW       | rW      |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |



Bits 31:22 Reserved, must be kept at reset value.

Bit 21 **NOVBUSSENS**: V<sub>BUS</sub> sensing disable option

When this bit is set, V<sub>BUS</sub> is considered internally to be always at V<sub>BUS</sub> valid level (5 V). This option removes the need for a dedicated V<sub>BUS</sub> pad, and leave this pad free to be used for other purposes such as a shared functionality. V<sub>BUS</sub> connection can be remapped on another general purpose input pad and monitored by software.

This option is only suitable for host-only or device-only applications.

0: V<sub>BUS</sub> sensing available by hardware

1: V<sub>BUS</sub> sensing not available by hardware.

Bit 20 **SOFOUTEN**: SOF output enable

0: SOF pulse not available on PAD

1: SOF pulse available on PAD

Bit 19 **VBUSSEN**: Enable the V<sub>BUS</sub> sensing “B” device

0: V<sub>BUS</sub> sensing “B” disabled

1: V<sub>BUS</sub> sensing “B” enabled

Bit 18 **VBUSASEN**: Enable the V<sub>BUS</sub> sensing “A” device

0: V<sub>BUS</sub> sensing “A” disabled

1: V<sub>BUS</sub> sensing “A” enabled

Bit 17 **I2CPADEN**: Enable I<sup>2</sup>C bus connection for the external I<sup>2</sup>C PHY interface.

0: I<sup>2</sup>C bus disabled

1: I<sup>2</sup>C bus enabled

Bit 16 **PWRDWN**: Power down

Used to activate the transceiver in transmission/reception

0: Power down active

1: Power down deactivated (“Transceiver active”)

Bits 15:0 Reserved, must be kept at reset value.

### OTG\_HS core ID register (OTG\_HS\_CID)

Address offset: 0x03C

Reset value: 0x0000 1100

This is a register containing the Product ID as reset value.

|            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| PRODUCT_ID |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| rw         | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw |

Bits 31:0 **PRODUCT\_ID**: Product ID field

Application-programmable ID field.

### OTG\_HS Host periodic transmit FIFO size register (OTG\_HS\_HPTXFSIZ)

Address offset: 0x100

Reset value: 0x0200 0800

|       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31    | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15    | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| PTXFD |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | PTXSA |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r     | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r     | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:16 **PTXFD**: Host periodic Tx FIFO depth

This value is in terms of 32-bit words.

Minimum value is 16

Maximum value is 512

Bits 15:0 **PTXSA**: Host periodic Tx FIFO start address

The power-on reset value of this register is the sum of the largest Rx data FIFO depth and largest nonperiodic Tx data FIFO depth.

### OTG\_HS device IN endpoint transmit FIFO size register (OTG\_HS\_DIEPTXFx) (x = 1..5, where x is the FIFO\_number)

Address offset:  $0x104 + 0x04 * (x - 1)$

Reset value: 0x02000400

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| INEPTXFD |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | INEPTXSA |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r        | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r        | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:16 **INEPTXFD**: IN endpoint Tx FIFO depth

This value is in terms of 32-bit words.

Minimum value is 16

Maximum value is 512

The power-on reset value of this register is specified as the largest IN endpoint FIFO number depth.

Bits 15:0 **INEPTXSA**: IN endpoint FIFOx transmit RAM start address

This field contains the memory start address for IN endpoint transmit FIFOx. The address must be aligned with a 32-bit memory location.

## 35.12.3 Host-mode registers

Bit values in the register descriptions are expressed in binary unless otherwise specified.

Host-mode registers affect the operation of the core in the host mode. Host mode registers must not be accessed in peripheral mode, as the results are undefined. Host mode registers can be categorized as follows:

### OTG\_HS host configuration register (OTG\_HS\_HCFG)

Address offset: 0x400

Reset value: 0x0000 0000

This register configures the core after power-on. Do not change to this register after initializing the host.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |       |    |         |   |    |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|-------|----|---------|---|----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4     | 3  | 2       | 1 | 0  |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   | FSLSS |    | FSLSPCS |   |    |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   | r     | rw |         |   | rw |

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **FSLSS**: FS- and LS-only support

The application uses this bit to control the core's enumeration speed. Using this bit, the application can make the core enumerate as an FS host, even if the connected device supports HS traffic. Do not make changes to this field after initial programming.

0: HS/FS/LS, based on the maximum speed supported by the connected device

1: FS/LS-only, even if the connected device can support HS (read-only)

Bits 1:0 **FSLSPCS**: FS/LS PHY clock select

When the core is in FS host mode:

01: PHY clock is running at 48 MHz

Others: Reserved

When the core is in LS host mode:

00: Reserved

01: PHY clock is running at 48 MHz.

10: Select 6 MHz PHY clock frequency

11: Reserved

*Note: The FSLSPCS bit must be set on a connection event according to the speed of the connected device. A software reset must be performed after changing this bit.*

**OTG\_HS Host frame interval register (OTG\_HS\_HFIR)**

Address offset: 0x404

Reset value: 0x0000 EA60

This register stores the frame interval information for the current speed to which the OTG\_HS controller has enumerated.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15    | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | FRIVL |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r/w   | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w |

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **FRIVL**: Frame interval

The value that the application programs to this field specifies the interval between two consecutive SOFs (FS), micro-SOFs (HS) or Keep-Alive tokens (LS). This field contains the number of PHY clocks that constitute the required frame interval. The application can write a value to this register only after the Port enable bit of the host port control and status register (PENA bit in OTG\_HS\_HPRT) has been set. If no value is programmed, the core calculates the value based on the PHY clock specified in the FS/LS PHY Clock Select field of the Host configuration register (FSLSPCS in OTG\_HS\_HCFG):

frame duration × PHY clock frequency

– Frame interval = 1 ms × (FRIVL - 1)

*Note: The FRIVL bit can be modified whenever the application needs to change the Frame interval time.*

**OTG\_HS host frame number/frame time remaining register (OTG\_HS\_HFNUM)**

Address offset: 0x408

Reset value: 0x0000 3FFF

This register indicates the current frame number. It also indicates the time remaining (in terms of the number of PHY clocks) in the current frame.

|       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31    | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15    | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| FTREM |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | FRNUM |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r     | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r     | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:16 **FTREM**: Frame time remaining

Indicates the amount of time remaining in the current frame, in terms of PHY clocks. This field decrements on each PHY clock. When it reaches zero, this field is reloaded with the value in the Frame interval register and a new SOF is transmitted on the USB.

Bits 15:0 **FRNUM**: Frame number

This field increments when a new SOF is transmitted on the USB, and is cleared to 0 when it reaches 0x3FFF.

## OTG\_HS\_Host periodic transmit FIFO/queue status register (OTG\_HS\_HPTXSTS)

Address offset: 0x410

Reset value: 0x0008 0100

This read-only register contains the free space information for the periodic TxFIFO and the periodic transmit request queue.

|         |    |    |    |    |    |    |    |         |    |    |    |    |    |    |    |          |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|---------|----|----|----|----|----|----|----|---------|----|----|----|----|----|----|----|----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31      | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23      | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| PTXQTOP |    |    |    |    |    |    |    | PTXQSAV |    |    |    |    |    |    |    | PTXFSAVL |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r       | r  | r  | r  | r  | r  | r  | r  | r       | r  | r  | r  | r  | r  | r  | r  | r        | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:24 **PTXQTOP**: Top of the periodic transmit request queue

This indicates the entry in the periodic Tx request queue that is currently being processed by the MAC.

This register is used for debugging.

Bit [31]: Odd/Even frame

– 0: send in even (micro) frame

– 1: send in odd (micro) frame

Bits [30:27]: Channel/endpoint number

Bits [26:25]: Type

– 00: IN/OUT

– 01: Zero-length packet

– 11: Disable channel command

Bit [24]: Terminate (last entry for the selected channel/endpoint)

Bits 23:16 **PTXQSAV**: Periodic transmit request queue space available

Indicates the number of free locations available to be written in the periodic transmit request queue. This queue holds both IN and OUT requests.

00: Periodic transmit request queue is full

01: dx1 location available

10: dx2 locations available

bxn: dxn locations available ( $0 \leq dxn \leq PTXFD$ )

Others: Reserved

Bits 15:0 **PTXFSAVL**: Periodic transmit data FIFO space available

Indicates the number of free locations available to be written to in the periodic TxFIFO.

Values are in terms of 32-bit words

0000: Periodic TxFIFO is full

0001: dx1 word available

0010: dx2 words available

bxn: dxn words available (where  $0 \leq dxn \leq dx512$ )

Others: Reserved

**OTG\_HS Host all channels interrupt register (OTG\_HS\_HAINT)**

Address offset: 0x414

Reset value: 0x0000 000

When a significant event occurs on a channel, the host all channels interrupt register interrupts the application using the host channels interrupt bit of the Core interrupt register (HCINT bit in OTG\_HS\_GINTSTS). This is shown in [Figure 414](#). There is one interrupt bit per channel, up to a maximum of 16 bits. Bits in this register are set and cleared when the application sets and clears bits in the corresponding host channel-x interrupt register.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15    | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | HAINT |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r     | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **HAINT**: Channel interrupts

One bit per channel: Bit 0 for Channel 0, bit 15 for Channel 15

**OTG\_HS host all channels interrupt mask register (OTG\_HS\_HAINTMSK)**

Address offset: 0x418

Reset value: 0x0000 0000

The host all channel interrupt mask register works with the host all channel interrupt register to interrupt the application when an event occurs on a channel. There is one interrupt mask bit per channel, up to a maximum of 16 bits.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15     | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | HAINTM |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r/w    | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w |

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **HAINTM**: Channel interrupt mask

0: Masked interrupt

1: Unmasked interrupt

One bit per channel: Bit 0 for channel 0, bit 15 for channel 15

**OTG\_HS host port control and status register (OTG\_HS\_HPRT)**

Address offset: 0x440

Reset value: 0x0000 0000

This register is available only in host mode. Currently, the OTG host supports only one port.

A single register holds USB port-related information such as USB reset, enable, suspend, resume, connect status, and test mode for each port. It is shown in [Figure 414](#). The rc\_w1 bits in this register can trigger an interrupt to the application through the host port interrupt bit of the core interrupt register (HPRTINT bit in OTG\_HS\_GINTSTS). On a Port Interrupt, the application must read this register and clear the bit that caused the interrupt. For the rc\_w1 bits, the application must write a 1 to the bit to clear the interrupt.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18   | 17 | 16    | 15 | 14 | 13 | 12   | 11    | 10 | 9        | 8    | 7     | 6    | 5      | 4    | 3       | 2     | 1     | 0     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|------|----|-------|----|----|----|------|-------|----|----------|------|-------|------|--------|------|---------|-------|-------|-------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    | PSPD |    | PTCTL |    |    |    | PPWR | PLSTS |    | Reserved | PRST | PSUSP | PRES | POCHNG | POCA | PENCHNG | PENA  | PCDET | PCSTS |
|          |    |    |    |    |    |    |    |    |    |    |    |    | r    | r  | rw    | rw | rw | rw | rw   | r     | r  |          | rw   | rs    | rw   | rc_w1  | r    | rc_w1   | rc_w0 | rc_w1 | r     |

Bits 31:19 Reserved, must be kept at reset value.

Bits 18:17 **PSPD**: Port speed

Indicates the speed of the device attached to this port.

00: High speed

01: Full speed

10: Low speed

11: Reserved

Bits 16:13 **PTCTL**: Port test control

The application writes a nonzero value to this field to put the port into a Test mode, and the corresponding pattern is signaled on the port.

0000: Test mode disabled

0001: Test\_J mode

0010: Test\_K mode

0011: Test\_SE0\_NAK mode

0100: Test\_Packet mode

0101: Test\_Force\_Enable

Others: Reserved

Bit 12 **PPWR**: Port power

The application uses this field to control power to this port, and the core clears this bit on an overcurrent condition.

0: Power off

1: Power on

Bits 11:10 **PLSTS**: Port line status

Indicates the current logic level USB data lines

Bit [10]: Logic level of OTG\_HS\_FS\_DP

Bit [11]: Logic level of OTG\_HS\_FS\_DM

Bit 9 Reserved, must be kept at reset value.

**Bit 8 PRST:** Port reset

When the application sets this bit, a reset sequence is started on this port. The application must time the reset period and clear this bit after the reset sequence is complete.

0: Port not in reset

1: Port in reset

The application must leave this bit set for a minimum duration of at least 10 ms to start a reset on the port. The application can leave it set for another 10 ms in addition to the required minimum duration, before clearing the bit, even though there is no maximum limit set by the USB standard.

High speed: 50 ms

Full speed/Low speed: 10 ms

**Bit 7 PSUSP:** Port suspend

The application sets this bit to put this port in Suspend mode. The core only stops sending SOFs when this is set. To stop the PHY clock, the application must set the Port clock stop bit, which asserts the suspend input pin of the PHY.

The read value of this bit reflects the current suspend status of the port. This bit is cleared by the core after a remote wake-up signal is detected or the application sets the Port reset bit or Port resume bit in this register or the Resume/remote wake-up detected interrupt bit or Disconnect detected interrupt bit in the Core interrupt register (WKUINT or DISCINT in OTG\_HS\_GINTSTS, respectively).

0: Port not in Suspend mode

1: Port in Suspend mode

**Bit 6 PRES:** Port resume

The application sets this bit to drive resume signaling on the port. The core continues to drive the resume signal until the application clears this bit.

If the core detects a USB remote wake-up sequence, as indicated by the Port resume/remote wake-up detected interrupt bit of the Core interrupt register (WKUINT bit in OTG\_HS\_GINTSTS), the core starts driving resume signaling without application intervention and clears this bit when it detects a disconnect condition. The read value of this bit indicates whether the core is currently driving resume signaling.

0: No resume driven

1: Resume driven

**Bit 5 POCCHNG:** Port overcurrent change

The core sets this bit when the status of the Port overcurrent active bit (bit 4) in this register changes.

**Bit 4 POCA:** Port overcurrent active

Indicates the overcurrent condition of the port.

0: No overcurrent condition

1: Overcurrent condition

**Bit 3 PENCHNG:** Port enable/disable change

The core sets this bit when the status of the Port enable bit [2] in this register changes.

Bit 2 **PENA**: Port enable

A port is enabled only by the core after a reset sequence, and is disabled by an overcurrent condition, a disconnect condition, or by the application clearing this bit. The application cannot set this bit by a register write. It can only clear it to disable the port. This bit does not trigger any interrupt to the application.

0: Port disabled

1: Port enabled

Bit 1 **PCDET**: Port connect detected

The core sets this bit when a device connection is detected to trigger an interrupt to the application using the host port interrupt bit in the Core interrupt register (HPRTINT bit in OTG\_HS\_GINTSTS). The application must write a 1 to this bit to clear the interrupt.

Bit 0 **PCSTS**: Port connect status

0: No device is attached to the port

1: A device is attached to the port

### OTG\_HS host channel-x characteristics register (OTG\_HS\_HCCHARx) (x = 0..11, where x = Channel\_number)

Address offset:  $0x500 + 0x20 * x$

Reset value: 0x0000 0000

| 31    | 30    | 29     | 28  | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18    | 17 | 16    | 15       | 14    | 13    | 12 | 11 | 10 | 9     | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |    |  |
|-------|-------|--------|-----|----|----|----|----|----|----|----|----|----|-------|----|-------|----------|-------|-------|----|----|----|-------|----|----|----|----|----|----|----|----|----|----|--|
| CHENA | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP |    | LSDEV | Reserved | EPDIR | EPNUM |    |    |    | MPSIZ |    |    |    |    |    |    |    |    |    |    |  |
| rs    | rs    | rw     | rw  | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw    | rw | rw    |          | rw    | rw    | rw | rw | rw | rw    | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw |  |

Bit 31 **CHENA**: Channel enable

This field is set by the application and cleared by the OTG host.

0: Channel disabled

1: Channel enabled

Bit 30 **CHDIS**: Channel disable

The application sets this bit to stop transmitting/receiving data on a channel, even before the transfer for that channel is complete. The application must wait for the Channel disabled interrupt before treating the channel as disabled.

Bit 29 **ODDFRM**: Odd frame

This field is set (reset) by the application to indicate that the OTG host must perform a transfer in an odd frame. This field is applicable for only periodic (isochronous and interrupt) transactions.

0: Even (micro) frame

1: Odd (micro) frame

Bits 28:22 **DAD**: Device address

This field selects the specific device serving as the data source or sink.

Bits 21:20 **MC**: Multi Count (MC) / Error Count (EC)

- When the split enable bit (SPLITEN) in the host channel-x split control register (OTG\_HS\_HCSPLTx) is reset (0), this field indicates to the host the number of transactions that must be executed per micro-frame for this periodic endpoint. For nonperiodic transfers, this field specifies the number of packets to be fetched for this channel before the internal DMA engine changes arbitration.  
00: Reserved This field yields undefined results  
01: 1 transaction  
b10: 2 transactions to be issued for this endpoint per micro-frame  
11: 3 transactions to be issued for this endpoint per micro-frame.
- When the SPLITEN bit is set (1) in OTG\_HS\_HCSPLTx, this field indicates the number of immediate retries to be performed for a periodic split transaction on transaction errors. This field must be set to at least 01.

Bits 19:18 **EPTYP**: Endpoint type

Indicates the transfer type selected.

- 00: Control
- 01: Isochronous
- 10: Bulk
- 11: Interrupt

Bit 17 **LSDEV**: Low-speed device

This field is set by the application to indicate that this channel is communicating to a low-speed device.

Bit 16 Reserved, must be kept at reset value.

Bit 15 **EPDIR**: Endpoint direction

Indicates whether the transaction is IN or OUT.

- 0: OUT
- 1: IN

Bits 14:11 **EPNUM**: Endpoint number

Indicates the endpoint number on the device serving as the data source or sink.

Bits 10:0 **MPSIZ**: Maximum packet size

Indicates the maximum packet size of the associated endpoint.

### OTG\_HS host channel-x split control register (OTG\_HS\_HCSPLTx) (x = 0..11, where x = Channel\_number)

Address offset:  $0x504 + 0x20 * x$

Reset value: 0x0000 0000

| 31      | 30       | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16        | 15      | 14 | 13      | 12 | 11 | 10 | 9  | 8  | 7       | 6  | 5  | 4  | 3  | 2  | 1  | 0 |
|---------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|-----------|---------|----|---------|----|----|----|----|----|---------|----|----|----|----|----|----|---|
| SPLITEN | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | COMPLSPLT | XACTPOS |    | HUBADDR |    |    |    |    |    | PRTADDR |    |    |    |    |    |    |   |
|         |          |    |    |    |    |    |    |    |    |    |    |    |    |    | rW        | rW      | rW | rW      | rW | rW | rW | rW | rW | rW      | rW | rW | rW | rW | rW | rW |   |

Bit 31 **SPLITEN**: Split enable

The application sets this bit to indicate that this channel is enabled to perform split transactions.

Bits 30:17 **Reserved**, must be kept at reset value.

Bit 16 **COMPLSPLT**: Do complete split

The application sets this bit to request the OTG host to perform a complete split transaction.

Bits 15:14 **XACTPOS**: Transaction position

This field is used to determine whether to send all, first, middle, or last payloads with each OUT transaction.

11: All. This is the entire data payload of this transaction (which is less than or equal to 188 bytes)

10: Begin. This is the first data payload of this transaction (which is larger than 188 bytes)

00: Mid. This is the middle payload of this transaction (which is larger than 188 bytes)

01: End. This is the last payload of this transaction (which is larger than 188 bytes)

Bits 13:7 **HUBADDR**: Hub address

This field holds the device address of the transaction translator's hub.

Bits 6:0 **PRTADDR**: Port address

This field is the port number of the recipient transaction translator.

**OTG\_HS host channel-x interrupt register (OTG\_HS\_HCINTx) (x = 0..11, where x = Channel\_number)**Address offset:  $0x508 + 0x20 * x$ 

Reset value: 0x0000 0000

This register indicates the status of a channel with respect to USB- and AHB-related events. It is shown in [Figure 414](#). The application must read this register when the host channels interrupt bit in the Core interrupt register (HCINT bit in OTG\_HS\_GINTSTS) is set. Before the application can read this register, it must first read the host all channels interrupt (OTG\_HS\_HAINT) register to get the exact channel number for the host channel-x interrupt register. The application must clear the appropriate bit in this register to clear the corresponding bits in the OTG\_HS\_HAINT and OTG\_HS\_GINTSTS registers.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10    | 9     | 8     | 7     | 6     | 5     | 4     | 3     | 2      | 1     | 0     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | DTERR | FRMOR | BBERR | TXERR | NYET  | ACK   | NAK   | STALL | AHBERR | CHH   | XFRC  |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rc_w1 | rc_w1 | rc_w1 | rc_w1 | rc_w1 | rc_w1 | rc_w1 | rc_w1 | rc_w1  | rc_w1 | rc_w1 |

Bits 31:11 Reserved, must be kept at reset value.

Bit 10 **DTERR**: Data toggle error

Bit 9 **FRMOR**: Frame overrun

Bit 8 **BBERR**: Babble error

Bit 7 **TXERR**: Transaction error

Indicates one of the following errors occurred on the USB.

CRC check failure

Timeout

Bit stuff error

False EOP

Bit 6 **NYET**: Response received interrupt

Bit 5 **ACK**: ACK response received/transmitted interrupt

Bit 4 **NAK**: NAK response received interrupt

Bit 3 **STALL**: STALL response received interrupt

Bit 2 **AHBERR**: AHB error

This error is generated only in Internal DMA mode when an AHB error occurs during an AHB read/write operation. The application can read the corresponding DMA channel address register to get the error address.

Bit 1 **CHH**: Channel halted

Indicates the transfer completed abnormally either because of any USB transaction error or in response to disable request by the application.

Bit 0 **XFRC**: Transfer completed

Transfer completed normally without any errors.

### OTG\_HS host channel-x interrupt mask register (OTG\_HS\_HCINTMSKx) (x = 0..11, where x = Channel\_number)

Address offset:  $0x50C + 0x20 * x$

Reset value: 0x0000 0000

This register reflects the mask for each channel status described in the previous section.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10     | 9      | 8      | 7      | 6    | 5    | 4    | 3      | 2       | 1    | 0    |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|--------|--------|--------|------|------|------|--------|---------|------|------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFRM |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rW     | rW     | rW     | rW     | rW   | rW   | rW   | rW     | rW      | rW   | rW   |

Bits 31:11 Reserved, must be kept at reset value.

Bit 10 **DTERRM**: Data toggle error mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 9 **FRMORM**: Frame overrun mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 8 **BBERRM**: Babble error mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 7 **TXERRM**: Transaction error mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 6 **NYET**: response received interrupt mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 5 **ACKM**: ACK response received/transmitted interrupt mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 4 **NAKM**: NAK response received interrupt mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 3 **STALLM**: STALL response received interrupt mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 2 **AHBERRM**: AHB error mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 1 **CHHM**: Channel halted mask

- 0: Masked interrupt
- 1: Unmasked interrupt

Bit 0 **XFRM**: Transfer completed mask

- 0: Masked interrupt
- 1: Unmasked interrupt

### OTG\_HS host channel-x transfer size register (OTG\_HS\_HCTSIZx) (x = 0..11, where x = Channel\_number)

Address offset: 0x510 + 0x20 \* x

Reset value: 0x0000 0000

| 31     | 30   | 29 | 28     | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18     | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0 |
|--------|------|----|--------|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|
| DOPING | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |
|        |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |
| rW     | rW   | rW | rW     | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW     | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW |   |

Bit 31 **DOPING**: Do ping

This bit is used only for OUT transfers. Setting this field to 1 directs the host to do PING protocol.

*Note: Do not set this bit for IN transfers. If this bit is set for IN transfers it disables the channel.*

Bits 30:29 **DPID**: Data PID

The application programs this field with the type of PID to use for the initial transaction. The host maintains this field for the rest of the transfer.

- 00: DATA0
- 01: DATA2
- 10: DATA1
- 11: MDATA (noncontrol)/SETUP (control)

Bits 28:19 **PKTCNT**: Packet count

This field is programmed by the application with the expected number of packets to be transmitted (OUT) or received (IN).

The host decrements this count on every successful transmission or reception of an OUT/IN packet. Once this count reaches zero, the application is interrupted to indicate normal completion.

Bits 18:0 **XFRSIZ**: Transfer size

For an OUT, this field is the number of data bytes the host sends during the transfer.

For an IN, this field is the buffer size that the application has reserved for the transfer. The application is expected to program this field as an integer multiple of the maximum packet size for IN transactions (periodic and nonperiodic).

### OTG\_HS host channel-x DMA address register (OTG\_HS\_HCDMAx) (x = 0..11, where x = Channel\_number)

Address offset:  $0x514 + 0x20 * x$

Reset value: 0x0000 0000

|         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31      | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| DMAADDR |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| rW      | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW |

Bits 31:0 **DMAADDR**: DMA address

This field holds the start address in the external memory from which the data for the endpoint must be fetched or to which it must be stored. This register is incremented on every AHB transaction.

## 35.12.4 Device-mode registers

### OTG\_HS device configuration register (OTG\_HS\_DCFG)

Address offset: 0x800

Reset value: 0x0220 0000

This register configures the core in peripheral mode after power-on or after certain control commands or enumeration. Do not make changes to this register after initial programming.

|          |    |    |    |    |    |           |    |          |          |    |    |    |    |    |    |    |       |    |     |    |    |    |    |    |          |          |    |      |   |   |   |
|----------|----|----|----|----|----|-----------|----|----------|----------|----|----|----|----|----|----|----|-------|----|-----|----|----|----|----|----|----------|----------|----|------|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25        | 24 | 23       | 22       | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14    | 13 | 12  | 11 | 10 | 9  | 8  | 7  | 6        | 5        | 4  | 3    | 2 | 1 | 0 |
| Reserved |    |    |    |    |    | PERSCHIVL |    | Reserved | Reserved |    |    |    |    |    |    |    | PFIVL |    | DAD |    |    |    |    |    | Reserved | NZLSOHSK |    | DSPD |   |   |   |
|          |    |    |    |    |    |           |    |          |          |    |    |    |    |    |    |    |       |    |     |    |    |    |    |    |          |          |    |      |   |   |   |
|          |    |    |    |    |    | rW        | rW |          |          |    |    |    |    |    |    |    | rW    |    | rW  | rW | rW | rW | rW | rW |          | rW       | rW | rW   |   |   |   |

Bits 31:26 Reserved, must be kept at reset value.

Bits 25:24 **PERSCHIVL**: Periodic scheduling interval

This field specifies the amount of time the Internal DMA engine must allocate for fetching periodic IN endpoint data. Based on the number of periodic endpoints, this value must be specified as 25, 50 or 75% of the (micro)frame.

- When any periodic endpoints are active, the internal DMA engine allocates the specified amount of time in fetching periodic IN endpoint data
- When no periodic endpoint is active, then the internal DMA engine services nonperiodic endpoints, ignoring this field
- After the specified time within a (micro)frame, the DMA switches to fetching nonperiodic endpoints

00: 25% of (micro)frame

01: 50% of (micro)frame

10: 75% of (micro)frame

11: Reserved

Bits 23:13 Reserved, must be kept at reset value.

Bits 12:11 **PFIVL**: Periodic (micro)frame interval

Indicates the time within a (micro) frame at which the application must be notified using the end of periodic (micro) frame interrupt. This can be used to determine if all the isochronous traffic for that frame is complete.

00: 80% of the frame interval

01: 85% of the frame interval

10: 90% of the frame interval

11: 95% of the frame interval

Bits 10:4 **DAD**: Device address

The application must program this field after every SetAddress control command.

Bit 3 Reserved, must be kept at reset value.

Bit 2 **NZLSOHSK**: Nonzero-length status OUT handshake

The application can use this field to select the handshake the core sends on receiving a nonzero-length data packet during the OUT transaction of a control transfer's Status stage.

1: Send a STALL handshake on a nonzero-length status OUT transaction and do not send the received OUT packet to the application.

0: Send the received OUT packet to the application (zero-length or nonzero-length) and send a handshake based on the NAK and STALL bits for the endpoint in the device endpoint control register.

Bits 1:0 **DSPD**: Device speed

Indicates the speed at which the application requires the core to enumerate, or the maximum speed the application can support. However, the actual bus speed is determined only after the chirp sequence is completed, and is based on the speed of the USB host to which the core is connected.

00: High speed

01: Full speed using external ULPI PHY

10: Reserved

11: Full speed using internal embedded PHY

**OTG\_HS device control register (OTG\_HS\_DCTL)**

Address offset: 0x804

Reset value: 0x0000 0000

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11       | 10     | 9      | 8      | 7      | 6    | 5  | 4  | 3      | 2      | 1    | 0      |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----------|--------|--------|--------|--------|------|----|----|--------|--------|------|--------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | POPRGDNE | CGONAK | SGONAK | CGINAK | SGINAK | TCTL |    |    | GONSTS | GINSTS | SDIS | RWUSIG |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RW       | W      | W      | W      | W      | RW   | RW | RW | R      | R      | RW   | RW     |

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 **POPRGDNE**: Power-on programming done

The application uses this bit to indicate that register programming is completed after a wake-up from power down mode.

Bit 10 **CGONAK**: Clear global OUT NAK

Writing 1 to this field clears the Global OUT NAK.

Bit 9 **SGONAK**: Set global OUT NAK

Writing 1 to this field sets the Global OUT NAK.

The application uses this bit to send a NAK handshake on all OUT endpoints.

The application must set this bit only after making sure that the Global OUT NAK effective bit in the Core interrupt register (GONAKEFF bit in OTG\_HS\_GINTSTS) is cleared.

Bit 8 **CGINAK**: Clear global IN NAK

Writing 1 to this field clears the Global IN NAK.

Bit 7 **SGINAK**: Set global IN NAK

Writing 1 to this field sets the Global nonperiodic IN NAK. The application uses this bit to send a NAK handshake on all nonperiodic IN endpoints.

The application must set this bit only after making sure that the Global IN NAK effective bit in the Core interrupt register (GINAKEFF bit in OTG\_HS\_GINTSTS) is cleared.

Bits 6:4 **TCTL**: Test control

000: Test mode disabled

001: Test\_J mode

010: Test\_K mode

011: Test\_SE0\_NAK mode

100: Test\_Packet mode

101: Test\_Force\_Enable

Others: Reserved

Bit 3 **GONSTS**: Global OUT NAK status

0: A handshake is sent based on the FIFO Status and the NAK and STALL bit settings.

1: No data is written to the Rx FIFO, irrespective of space availability. Sends a NAK handshake on all packets, except on SETUP transactions. All isochronous OUT packets are dropped.

Bit 2 **GINSTS**: Global IN NAK status

0: A handshake is sent out based on the data availability in the transmit FIFO.

1: A NAK handshake is sent out on all nonperiodic IN endpoints, irrespective of the data availability in the transmit FIFO.

Bit 1 **SDIS**: Soft disconnect

The application uses this bit to signal the USB OTG core to perform a soft disconnect. As long as this bit is set, the host does not see that the device is connected, and the device does not receive signals on the USB. The core stays in the disconnected state until the application clears this bit.

0: Normal operation. When this bit is cleared after a soft disconnect, the core generates a device connect event to the USB host. When the device is reconnected, the USB host restarts device enumeration.

1: The core generates a device disconnect event to the USB host.

Bit 0 **RWUSIG**: Remote wake-up signaling

When the application sets this bit, the core initiates remote signaling to wake up the USB host. The application must set this bit to instruct the core to exit the Suspend state. As specified in the USB 2.0 specification, the application must clear this bit 1 ms to 15 ms after setting it.

[Table 214](#) contains the minimum duration (according to device state) for which the Soft disconnect (SDIS) bit must be set for the USB host to detect a device disconnect. To accommodate clock jitter, it is recommended that the application add some extra delay to the specified minimum duration.

**Table 214. Minimum duration for soft disconnect**

| Operating speed | Device state                                    | Minimum duration   |
|-----------------|-------------------------------------------------|--------------------|
| High speed      | Not Idle or Suspended (Performing transactions) | 125 $\mu$ s        |
| Full speed      | Suspended                                       | 1 ms + 2.5 $\mu$ s |
| Full speed      | Idle                                            | 2.5 $\mu$ s        |
| Full speed      | Not Idle or Suspended (Performing transactions) | 2.5 $\mu$ s        |

**OTG\_HS device status register (OTG\_HS\_DSTS)**

Address offset: 0x808

Reset value: 0x0000 0010

This register indicates the status of the core with respect to USB-related events. It must be read on interrupts from the device all interrupts (OTG\_HS\_DAINR) register.

|          |    |    |    |    |    |    |    |    |    |       |    |    |    |    |    |    |    |    |    |          |    |   |   |      |         |   |         |   |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|-------|----|----|----|----|----|----|----|----|----|----------|----|---|---|------|---------|---|---------|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21    | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11       | 10 | 9 | 8 | 7    | 6       | 5 | 4       | 3 | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    | FNSOF |    |    |    |    |    |    |    |    |    | Reserved |    |   |   | EERR | ENUMSPD |   | SUSPSTS |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    |       |    |    |    |    |    |    |    |    |    |          |    |   |   | r    | r       | r | r       | r | r | r | r |

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:8 **FNSOF**: Frame number of the received SOF

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **EERR**: Erratic error

The core sets this bit to report any erratic errors.

Due to erratic errors, the OTG\_HS controller goes into Suspended state and an interrupt is generated to the application with Early suspend bit of the Core interrupt register (ESUSP bit in OTG\_HS\_GINTSTS). If the early suspend is asserted due to an erratic error, the application can only perform a soft disconnect recover.

Bits 2:1 **ENUMSPD**: Enumerated speed

Indicates the speed at which the OTG\_HS controller has come up after speed detection through a chirp sequence.

00: High speed

01: Reserved

10: Reserved

11: Full speed (PHY clock is running at 48 MHz)

Others: reserved

Bit 0 **SUSPSTS**: Suspend status

In peripheral mode, this bit is set as long as a Suspend condition is detected on the USB.

The core enters the Suspended state when there is no activity on the USB data lines for a period of 3 ms. The core comes out of the suspend:

- When there is an activity on the USB data lines
- When the application writes to the Remote wake-up signaling bit in the Device control register (RWUSIG bit in OTG\_HS\_DCTL).

## OTG\_HS device IN endpoint common interrupt mask register (OTG\_HS\_DIEPMSK)

Address offset: 0x810

Reset value: 0x0000 0000

This register works with each of the Device IN endpoint interrupt (OTG\_HS\_DIEPINTx) registers for all endpoints to generate an interrupt per IN endpoint. The IN endpoint interrupt for a specific status in the OTG\_HS\_DIEPINTx register can be masked by writing to the corresponding bit in this register. Status bits are masked by default.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13   | 12       | 11 | 10 | 9 | 8 | 7 | 6      | 5        | 4       | 3       | 2         | 1   | 0       |      |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|----------|----|----|---|---|---|--------|----------|---------|---------|-----------|-----|---------|------|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NAKM | Reserved |    |    |   |   |   | TXFURM | Reserved | INEPNEM | INEPNMM | ITTXFEMSK | TOM | AHBERRM | EPDM | XFCRM |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |          |    |    |   |   |   |        |          |         |         |           |     |         |      |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

Bits 31:10 Reserved, must be kept at reset value.

Bit 13 **NAKM**: NAK interrupt mask

0: Masked interrupt

1: Unmasked interrupt

Bits 12:9 Reserved, must be kept at reset value.

Bit 8 **TXFURM**: FIFO underrun mask

0: Masked interrupt

1: Unmasked interrupt

Bit 7 Reserved, must be kept at reset value.

Bit 6 **INENEM**: IN endpoint NAK effective mask

0: Masked interrupt

1: Unmasked interrupt

Bit 5 **INENMM**: IN token received with EP mismatch mask

0: Masked interrupt

1: Unmasked interrupt

Bit 4 **ITTXFEMSK**: IN token received when Tx FIFO empty mask

0: Masked interrupt

1: Unmasked interrupt

Bit 3 **TOM**: Timeout condition mask (nonisochronous endpoints)

0: Masked interrupt

1: Unmasked interrupt

- Bit 2 **AHBERRM**: AHB error mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 1 **EPDM**: Endpoint disabled interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 0 **XFRM**: Transfer completed interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

### OTG\_HS device OUT endpoint common interrupt mask register (OTG\_HS\_DOEPMASK)

Address offset: 0x814

Reset value: 0x0000 0000

This register works with each of the Device OUT endpoint interrupt (OTG\_HS\_DOEPINTx) registers for all endpoints to generate an interrupt per OUT endpoint. The OUT endpoint interrupt for a specific status in the OTG\_HS\_DOEPINTx register can be masked by writing into the corresponding bit in this register. Status bits are masked by default.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14      | 13     | 12    | 11       | 10 | 9 | 8    | 7        | 6       | 5         | 4      | 3     | 2       | 1    | 0    |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---------|--------|-------|----------|----|---|------|----------|---------|-----------|--------|-------|---------|------|------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYETMSK | NAKMSK | BERRM | Reserved |    |   | OPEM | Reserved | B2BSTUP | STSPHSRXM | OTEPDM | STUPM | AHBERRM | EPDM | XFRM |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r/w     | r/w    | r/w   |          |    |   | r/w  |          | r/w     | r/w       | r/w    | r/w   | r/w     | r/w  | r/w  |

Bits 31:15 Reserved, must be kept at reset value.

- Bit 14 **NYETMSK**: NYET interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

- Bit 13 **NAKMSK**: NAK interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

- Bit 12 **BERRM**: Babble error interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bits 11:9 Reserved, must be kept at reset value.

- Bit 8 **OPEM**: OUT packet error mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bit 7 Reserved, must be kept at reset value.

- Bit 6 **B2BSTUP**: Back-to-back SETUP packets received mask  
Applies to control OUT endpoints only.  
0: Masked interrupt  
1: Unmasked interrupt
- Bit 5 **STSPHSRXM**: Status phase received for control write mask  
0: Masked interrupt  
1: Unmasked interrupt
- Bit 4 **OTEPDM**: OUT token received when endpoint disabled mask  
Applies to control OUT endpoints only.  
0: Masked interrupt  
1: Unmasked interrupt
- Bit 3 **STUPM**: SETUP phase done mask  
Applies to control endpoints only.  
0: Masked interrupt  
1: Unmasked interrupt
- Bit 2 **AHBERRM**: AHB error mask  
0: Masked interrupt  
1: Unmasked interrupt
- Bit 1 **EPDM**: Endpoint disabled interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt
- Bit 0 **XFRM**: Transfer completed interrupt mask  
0: Masked interrupt  
1: Unmasked interrupt

### OTG\_HS device all endpoints interrupt register (OTG\_HS\_DAIN)

Address offset: 0x818

Reset value: 0x0000 0000

When a significant event occurs on an endpoint, a device all endpoints interrupt register interrupts the application using the Device OUT endpoints interrupt bit or Device IN endpoints interrupt bit of the Core interrupt register (OEPINT or IEPINT in OTG\_HS\_GINTSTS, respectively). There is one interrupt bit per endpoint, up to a maximum of 16 bits for OUT endpoints and 16 bits for IN endpoints. For a bidirectional endpoint, the corresponding IN and OUT interrupt bits are used. Bits in this register are set and cleared when the application sets and clears bits in the corresponding Device Endpoint-x interrupt register (OTG\_HS\_DIEPINTx/OTG\_HS\_DOEPINTx).

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| OEPINT |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | IEPINT |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r      | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r      | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:16 **OEPI**: OUT endpoint interrupt bits

One bit per OUT endpoint:

Bit 16 for OUT endpoint 0, bit 31 for OUT endpoint 15

Bits 15:0 **IEPI**: IN endpoint interrupt bits

One bit per IN endpoint:

Bit 0 for IN endpoint 0, bit 15 for endpoint 15

### OTG\_HS all endpoints interrupt mask register (OTG\_HS\_DAINMSK)

Address offset: 0x81C

Reset value: 0x0000 0000

The device endpoint interrupt mask register works with the device endpoint interrupt register to interrupt the application when an event occurs on a device endpoint. However, the device all endpoints interrupt (OTG\_HS\_DAIN) register bit corresponding to that interrupt is still set.

|      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31   | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15   | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| OEPM |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | IEPM |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| rW   | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW   | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW |

Bits 31:16 **OEPM**: OUT EP interrupt mask bits

One per OUT endpoint:

Bit 16 for OUT EP 0, bit 19 for OUT EP 3

0: Masked interrupt

1: Unmasked interrupt

Bits 15:0 **IEPM**: IN EP interrupt mask bits

One bit per IN endpoint:

Bit 0 for IN EP 0, bit 3 for IN EP 3

0: Masked interrupt

1: Unmasked interrupt

### OTG\_HS device $V_{BUS}$ discharge time register (OTG\_HS\_DVBUSDIS)

Address offset: 0x0828

Reset value: 0x0000 17D7

This register specifies the  $V_{BUS}$  discharge time after  $V_{BUS}$  pulsing during SRP.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15     | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | VBUSDT |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rW     | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW |

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **VBUSDT**: Device  $V_{BUS}$  discharge time

Specifies the  $V_{BUS}$  discharge time after  $V_{BUS}$  pulsing during SRP. This value equals:

$V_{BUS}$  discharge time in PHY clocks / 1 024

Depending on your  $V_{BUS}$  load, this value may need adjusting.

OTG\_HS device V<sub>BUS</sub> pulsing time register (OTG\_HS\_DVBUSPULSE)

Address offset: 0x082C

Reset value: 0x0000 05B8

This register specifies the V<sub>BUS</sub> pulsing time during SRP.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|----|----|----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11     | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | DVBUSP |    |    |    |    |    |    |    |    |    |    |    |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rW     | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW |

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **DVBUSP**: Device V<sub>BUS</sub> pulsing time  
Specifies the V<sub>BUS</sub> pulsing time during SRP. This value equals:  
V<sub>BUS</sub> pulsing time in PHY clocks / 1 024



**OTG\_HS Device threshold control register (OTG\_HS\_DTHRCTL)**

Address offset: 0x0830

Reset value: 0x0000 0000

| 31       | 30 | 29 | 28 | 27    | 26       | 25      | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15      | 14       | 13      | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1        | 0           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     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| Reserved |    |    |    | ARPEN | Reserved | RXTHRLN |    |    |    |    |    |    |    |    |    | RXTHREN | Reserved | TXTHRLN |    |    |    |   |   |   |   |   |   |   |   | ISOTHREN | NONISOTHREN |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     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 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|          |    |    |    |       |          |         |    |    |    |    |    |    |    |    |    |         |          |         |    |    |    |   |   |   |   |   |   |   |   |          |             | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w |

Bits 31:28 Reserved, must be kept at reset value.

Bit 27 **ARPEN**: Arbiter parking enable

This bit controls internal DMA arbiter parking for IN endpoints. When thresholding is enabled and this bit is set to one, then the arbiter parks on the IN endpoint for which there is a token received on the USB. This is done to avoid getting into underrun conditions. By default parking is enabled.

Bit 26 Reserved, must be kept at reset value.

Bits 25: 17 **RXTHRLLEN**: Receive threshold length

This field specifies the receive thresholding size in words. This field also specifies the amount of data received on the USB before the core can start transmitting on the AHB. The threshold length has to be at least eight words. The recommended value for RXTHRLLEN is to be the same as the programmed AHB burst length (HBSTLEN bit in OTG\_HS\_GAHBCFG).

Bit 16 **RXTHREN**: Receive threshold enable

When this bit is set, the core enables thresholding in the receive direction.

Bits 15: 11 Reserved, must be kept at reset value.

Bits 10:2 **TXTHRLLEN**: Transmit threshold length

This field specifies the transmit thresholding size in words. This field specifies the amount of data in bytes to be in the corresponding endpoint transmit FIFO, before the core can start transmitting on the USB. The threshold length has to be at least eight words. This field controls both isochronous and nonisochronous IN endpoint thresholds. The recommended value for TXTHRLLEN is to be the same as the programmed AHB burst length (HBSTLEN bit in OTG\_HS\_GAHBCFG).

Bit 1 **ISOTHREN**: ISO IN endpoint threshold enable

When this bit is set, the core enables thresholding for isochronous IN endpoints.

Bit 0 **NONISOTHREN**: Nonisochronous IN endpoints threshold enable

When this bit is set, the core enables thresholding for nonisochronous IN endpoints.

### OTG\_HS device IN endpoint FIFO empty interrupt mask register: (OTG\_HS\_DIEPEMPMSK)

Address offset: 0x834

Reset value: 0x0000 0000

This register is used to control the IN endpoint FIFO empty interrupt generation (TXFE\_OTG\_HS\_DIEPINTx).

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15        | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | INEPTXFEM |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r/w       | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w |

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **INEPTXFEM**: IN EP Tx FIFO empty interrupt mask bits

These bits act as mask bits for OTG\_HS\_DIEPINTx.

TXFE interrupt one bit per IN endpoint:

Bit 0 for IN endpoint 0, bit 15 for IN endpoint 15

0: Masked interrupt

1: Unmasked interrupt

### OTG\_HS device each endpoint interrupt register (OTG\_HS\_DEACHINT)

Address offset: 0x0838

Reset value: 0x0000 0000

There is one interrupt bit for endpoint 1 IN and one interrupt bit for endpoint 1 OUT.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |         |          |    |    |    |    |    |    |   |   |   |   |   |         |          |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|---------|----------|----|----|----|----|----|----|---|---|---|---|---|---------|----------|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17      | 16       | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4       | 3        | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | OEP1INT | Reserved |    |    |    |    |    |    |   |   |   |   |   | IEP1INT | Reserved |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    | r       |          |    |    |    |    |    |    |   |   |   |   |   | r       |          |   |   |   |

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **OEP1INT**: OUT endpoint 1 interrupt bit

Bits 16:2 Reserved, must be kept at reset value.

Bit 1 **IEP1INT**: IN endpoint 1 interrupt bit

Bit 0 Reserved, must be kept at reset value.

### OTG\_HS device each endpoint interrupt register mask (OTG\_HS\_DEACHINTMSK)

Address offset: 0x083C

Reset value: 0x0000 0000

There is one interrupt bit for endpoint 1 IN and one interrupt bit for endpoint 1 OUT.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |          |          |    |    |    |    |    |    |   |   |   |   |   |          |          |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----------|----------|----|----|----|----|----|----|---|---|---|---|---|----------|----------|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17       | 16       | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4        | 3        | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | OEP1INTM | Reserved |    |    |    |    |    |    |   |   |   |   |   | IEP1INTM | Reserved |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    | r/w      |          |    |    |    |    |    |    |   |   |   |   |   | r/w      |          |   |   |   |

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **OEP1INTM**: OUT Endpoint 1 interrupt mask bit

Bits 16:2 Reserved, must be kept at reset value.

Bit 1 **IEP1INTM**: IN Endpoint 1 interrupt mask bit

Bit 0 Reserved, must be kept at reset value.

### OTG\_HS device each in endpoint-1 interrupt register (OTG\_HS\_DIEPEACHMSK1)

Address offset: 0x844

Reset value: 0x0000 0000

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |          |    |    |   |     |     |        |          |        |        |           |     |         |      |       |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|----------|----|----|---|-----|-----|--------|----------|--------|--------|-----------|-----|---------|------|-------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13   | 12       | 11 | 10 | 9 | 8   | 7   | 6      | 5        | 4      | 3      | 2         | 1   | 0       |      |       |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NAKM | Reserved |    |    |   | BIM |     | TXFURM | Reserved | INENEM | INENMM | ITTXFEMSK | TOM | AHBERRM | EPDM | XFRDM |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r/w  |          |    |    |   | r/w | r/w | r/w    |          | r/w    | r/w    | r/w       | r/w | r/w     | r/w  |       |

Bits 31:14 Reserved, must be kept at reset value.

Bit 13 **NAKM**: NAK interrupt mask

0: Masked interrupt

1: unmasked interrupt

Bit 12:10 Reserved, must be kept at reset value.

Bit 9 **BIM**: BNA interrupt mask

0: Masked interrupt

1: Unmasked interrupt

Bit 8 **TXFURM**: FIFO underrun mask

0: Masked interrupt

1: Unmasked interrupt

Bit 7 Reserved, must be kept at reset value.

- Bit 6 **INEPNEM**: IN endpoint NAK effective mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 5 **INEPNMM**: IN token received with EP mismatch mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 4 **ITTXFEMSK**: IN token received when TxFIFO empty mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 3 **TOM**: Timeout condition mask (nonisochronous endpoints)  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 2 **AHBERRM**: AHB error mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 1 **EPDM**: Endpoint disabled interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt
- Bit 0 **XFRM**: Transfer completed interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

### OTG\_HS device each OUT endpoint-1 interrupt register (OTG\_HS\_DOEPEACHMSK1)

Address offset: 0x884

Reset value: 0x0000 0000

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14    | 13   | 12    | 11       | 10 | 9   | 8      | 7        | 6 | 5       | 4       | 3         | 2   | 1       | 0    |       |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|------|-------|----------|----|-----|--------|----------|---|---------|---------|-----------|-----|---------|------|-------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYETM | NAKM | BERRM | Reserved |    | BIM | TXFURM | Reserved |   | INEPNEM | INEPNMM | ITTXFEMSK | TOM | AHBERRM | EPDM | XFCRM |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rw    | rw   | rw    |          |    | rw  | rw     |          |   | rw      | rw      | rw        | rw  | rw      | rw   | rw    |

Bits 31:15 Reserved, must be kept at reset value.

- Bit 14 **NYETM**: NYET interrupt mask  
 0: Masked interrupt  
 1: unmasked interrupt

- Bit 13 **NAKM**: NAK interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

- Bit 12 **BERRM**: Bubble error interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bit 11:10 Reserved, must be kept at reset value.

Bit 9 **BIM**: BNA interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bit 8 **OPEM**: OUT packet error mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 **AHBERRM**: AHB error mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bit 1 **EPDM**: Endpoint disabled interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

Bit 0 **XFRM**: Transfer completed interrupt mask  
 0: Masked interrupt  
 1: Unmasked interrupt

**OTG device endpoint-x control register (OTG\_HS\_DIEPCTLx) (x = 0..5, where x = Endpoint\_number)**

Address offset: 0x900 + 0x20 \* x

Reset value: 0x0000 0000

The application uses this register to control the behavior of each logical endpoint other than endpoint 0.

| 31    | 30    | 29      | 28              | 27   | 26   | 25     | 24 | 23 | 22 | 21    | 20       | 19    | 18 | 17     | 16         | 15     | 14       | 13    | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
|-------|-------|---------|-----------------|------|------|--------|----|----|----|-------|----------|-------|----|--------|------------|--------|----------|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|
| EPENA | EPDIS | SODDFRM | SD0PID/SEVNFIRM | SNAK | CNAK | TXFNUM |    |    |    | Stall | Reserved | EPTYP |    | NAKSTS | EONUM/DPID | USBAEP | Reserved | MPSIZ |    |    |    |    |    |    |    |    |    |    |    |    |    |
| rs    | rs    | w       | w               | w    | w    | rw     | rw | rw | rw | rw/rs |          | rw    | rw | r      | r          | rw     |          | rw    | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw |

- Bit 31 **EPENA**: Endpoint enable  
 The application sets this bit to start transmitting data on an endpoint.  
 The core clears this bit before setting any of the following interrupts on this endpoint:
- SETUP phase done
  - Endpoint disabled
  - Transfer completed
- Bit 30 **EPDIS**: Endpoint disable  
 The application sets this bit to stop transmitting/receiving data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint disabled interrupt. The application must set this bit only if Endpoint enable is already set for this endpoint.
- Bit 29 **SODDFRM**: Set odd frame  
 Applies to isochronous IN and OUT endpoints only.  
 Writing to this field sets the Even/Odd frame (EONUM) field to odd frame.
- Bit 28 **SD0PID**: Set DATA0 PID  
 Applies to interrupt/bulk IN endpoints only.  
 Writing to this field sets the endpoint data PID (DPID) field in this register to DATA0.
- SEVNFRM**: Set even frame  
 Applies to isochronous IN endpoints only.  
 Writing to this field sets the Even/Odd frame (EONUM) field to even frame.
- Bit 27 **SNACK**: Set NAK  
 A write to this bit sets the NAK bit for the endpoint.  
 Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for OUT endpoints on a Transfer completed interrupt, or after a SETUP is received on the endpoint.
- Bit 26 **CNAK**: Clear NAK  
 A write to this bit clears the NAK bit for the endpoint.
- Bits 25:22 **TXFNUM**: TxFIFO number  
 These bits specify the FIFO number associated with this endpoint. Each active IN endpoint must be programmed to a separate FIFO number.  
 This field is valid only for IN endpoints.
- Bit 21 **STALL**: STALL handshake  
 Applies to noncontrol, nonisochronous IN endpoints only (access type is rw).  
 The application sets this bit to stall all tokens from the USB host to this endpoint. If a NAK bit, Global IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Only the application can clear this bit, never the core.
- Applies to control endpoints only (access type is rs).  
 The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
- Bit 20 Reserved, must be kept at reset value.

**Bits 19:18 EPTYP:** Endpoint type

This is the transfer type supported by this logical endpoint.

- 00: Control
- 01: Isochronous
- 10: Bulk
- 11: Interrupt

**Bit 17 NAKSTS:** NAK status

It indicates the following:

- 0: The core is transmitting nonNAK handshakes based on the FIFO status.
- 1: The core is transmitting NAK handshakes on this endpoint.

When either the application or the core sets this bit:

For nonisochronous IN endpoints: The core stops transmitting any data on an IN endpoint, even if there are data available in the TxFIFO.

For isochronous IN endpoints: The core sends out a zero-length data packet, even if there are data available in the TxFIFO.

Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.

**Bit 16 EONUM:** Even/odd frame

Applies to isochronous IN endpoints only.

Indicates the frame number in which the core transmits/receives isochronous data for this endpoint. The application must program the even/odd frame number in which it intends to transmit/receive isochronous data for this endpoint using the SEVNFRM and SODDFRM fields in this register.

- 0: Even frame
- 1: Odd frame

**DPID:** Endpoint data PID

Applies to interrupt/bulk IN endpoints only.

Contains the PID of the packet to be received or transmitted on this endpoint. The application must program the PID of the first packet to be received or transmitted on this endpoint, after the endpoint is activated. The application uses the SD0PID register field to program either DATA0 or DATA1 PID.

- 0: DATA0
- 1: DATA1

**Bit 15 USBAEP:** USB active endpoint

Indicates whether this endpoint is active in the current configuration and interface. The core clears this bit for all endpoints (other than EP 0) after detecting a USB reset. After receiving the SetConfiguration and SetInterface commands, the application must program endpoint registers accordingly and set this bit.

**Bits 14:11** Reserved, must be kept at reset value.

**Bits 10:0 MPSIZ:** Maximum packet size

The application must program this field with the maximum packet size for the current logical endpoint. This value is in bytes.

## OTG\_HS device control OUT endpoint 0 control register (OTG\_HS\_DOEPCTL0)

Address offset: 0xB00

Reset value: 0x0000 8000

This section describes the device control OUT endpoint 0 control register. Nonzero control endpoints use registers for endpoints 1–15.

| 31    | 30 | 29    | 28 | 27       | 26 | 25 | 24   | 23 | 22   | 21 | 20       | 19 | 18 | 17 | 16 | 15    | 14 | 13   | 12 | 11    | 10 | 9      | 8 | 7        | 6 | 5 | 4      | 3 | 2        | 1 | 0 |  |  |  |  |  |  |  |       |  |
|-------|----|-------|----|----------|----|----|------|----|------|----|----------|----|----|----|----|-------|----|------|----|-------|----|--------|---|----------|---|---|--------|---|----------|---|---|--|--|--|--|--|--|--|-------|--|
| EPENA |    | EPDIS |    | Reserved |    |    | SNAK |    | CNAK |    | Reserved |    |    |    |    | STALL |    | SNPM |    | EPTYP |    | NAKSTS |   | Reserved |   |   | USBAEP |   | Reserved |   |   |  |  |  |  |  |  |  | MPSIZ |  |
| w     | r  | w     | w  |          |    |    | rs   | rw | r    | r  |          |    |    |    |    | r     | r  |      |    |       |    |        |   |          |   |   |        |   |          |   |   |  |  |  |  |  |  |  |       |  |

Bit 31 **EPENA**: Endpoint enable

The application sets this bit to start transmitting data on endpoint 0.

The core clears this bit before setting any of the following interrupts on this endpoint:

- SETUP phase done
- Endpoint disabled
- Transfer completed

Bit 30 **EPDIS**: Endpoint disable

The application cannot disable control OUT endpoint 0.

Bits 29:28 Reserved, must be kept at reset value.

Bit 27 **SNAK**: Set NAK

A write to this bit sets the NAK bit for the endpoint.

Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit on a Transfer completed interrupt, or after a SETUP is received on the endpoint.

Bit 26 **CNAK**: Clear NAK

A write to this bit clears the NAK bit for the endpoint.

Bits 25:22 Reserved, must be kept at reset value.

Bit 21 **STALL**: STALL handshake

The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.

Bit 20 **SNPM**: Snoop mode

This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.

Bits 19:18 **EPTYP**: Endpoint type

Hardcoded to 2'b00 for control.

**Bit 17 NAKSTS:** NAK status

Indicates the following:

0: The core is transmitting nonNAK handshakes based on the FIFO status.

1: The core is transmitting NAK handshakes on this endpoint.

When either the application or the core sets this bit, the core stops receiving data, even if there is space in the RxFIFO to accommodate the incoming packet. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.

Bit 16 Reserved, must be kept at reset value.

**Bit 15 USBAEP:** USB active endpoint

This bit is always set to 1, indicating that a control endpoint 0 is always active in all configurations and interfaces.

Bits 14:2 Reserved, must be kept at reset value.

**Bits 1:0 MPSIZ:** Maximum packet size

The maximum packet size for control OUT endpoint 0 is the same as what is programmed in control IN endpoint 0.

00: 64 bytes

01: 32 bytes

10: 16 bytes

11: 8 bytes

**OTG\_HS device endpoint-x control register (OTG\_HS\_DOEPCTLx) (x = 1..5, where x = Endpoint\_number)**
Address offset for OUT endpoints:  $0xB00 + 0x20 * x$ 

Reset value: 0x0000 0000

The application uses this register to control the behavior of each logical endpoint other than endpoint 0.

| 31    | 30    | 29      | 28              | 27   | 26   | 25       | 24 | 23 | 22 | 21 | 20    | 19   | 18    | 17     | 16         | 15     | 14       | 13 | 12 | 11 | 10 | 9     | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
|-------|-------|---------|-----------------|------|------|----------|----|----|----|----|-------|------|-------|--------|------------|--------|----------|----|----|----|----|-------|----|----|----|----|----|----|----|----|----|
| EPENA | EPDIS | SODDFRM | SD0PID/SEVNF0RM | SNAK | CNAK | Reserved |    |    |    |    | Stall | SNPM | EPTYP | NAKSTS | EONUM/DPID | USBAEP | Reserved |    |    |    |    | MPSIZ |    |    |    |    |    |    |    |    |    |
| rs    | rs    | w       | w               | w    | w    |          |    |    |    |    | rw    | rw   | rw    | rw     | r          | r      | rw       |    |    |    |    |       | rw | rw | rw | rw | rw | rw | rw | rw | rw |

- Bit 31 **EPENA**: Endpoint enable  
 Applies to IN and OUT endpoints.  
 The application sets this bit to start transmitting data on an endpoint.  
 The core clears this bit before setting any of the following interrupts on this endpoint:
- SETUP phase done
  - Endpoint disabled
  - Transfer completed
- Bit 30 **EPDIS**: Endpoint disable  
 The application sets this bit to stop transmitting/receiving data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint disabled interrupt. The application must set this bit only if Endpoint enable is already set for this endpoint.
- Bit 29 **SODDFRM**: Set odd frame  
 Applies to isochronous OUT endpoints only.  
 Writing to this field sets the Even/Odd frame (EONUM) field to odd frame.
- Bit 28 **SD0PID**: Set DATA0 PID  
 Applies to interrupt/bulk OUT endpoints only.  
 Writing to this field sets the endpoint data PID (DPID) field in this register to DATA0.
- SEVNFRM**: Set even frame  
 Applies to isochronous OUT endpoints only.  
 Writing to this field sets the Even/Odd frame (EONUM) field to even frame.
- Bit 27 **SNACK**: Set NAK  
 A write to this bit sets the NAK bit for the endpoint.  
 Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for OUT endpoints on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
- Bit 26 **CNAK**: Clear NAK  
 A write to this bit clears the NAK bit for the endpoint.
- Bits 25:22 Reserved, must be kept at reset value.
- Bit 21 **STALL**: STALL handshake  
 Applies to noncontrol, nonisochronous OUT endpoints only (access type is rw).  
 The application sets this bit to stall all tokens from the USB host to this endpoint. If a NAK bit, Global IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Only the application can clear this bit, never the core.
- Bit 20 **SNPM**: Snoop mode  
 This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.
- Bits 19:18 **EPTYP**: Endpoint type  
 This is the transfer type supported by this logical endpoint.
- 00: Control
  - 01: Isochronous
  - 10: Bulk
  - 11: Interrupt

**Bit 17 NAKSTS:** NAK status

Indicates the following:

- 0: The core is transmitting nonNAK handshakes based on the FIFO status.
- 1: The core is transmitting NAK handshakes on this endpoint.

When either the application or the core sets this bit:

The core stops receiving any data on an OUT endpoint, even if there is space in the RxFIFO to accommodate the incoming packet.

Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.

**Bit 16 EONUM:** Even/odd frame

Applies to isochronous IN and OUT endpoints only.

Indicates the frame number in which the core transmits/receives isochronous data for this endpoint. The application must program the even/odd frame number in which it intends to transmit/receive isochronous data for this endpoint using the SEVNFRM and SODDFRM fields in this register.

- 0: Even frame
- 1: Odd frame

**DPID:** Endpoint data PID

Applies to interrupt/bulk OUT endpoints only.

Contains the PID of the packet to be received or transmitted on this endpoint. The application must program the PID of the first packet to be received or transmitted on this endpoint, after the endpoint is activated. The application uses the SD0PID register field to program either DATA0 or DATA1 PID.

- 0: DATA0
- 1: DATA1

**Bit 15 USBAEP:** USB active endpoint

Indicates whether this endpoint is active in the current configuration and interface. The core clears this bit for all endpoints (other than EP 0) after detecting a USB reset. After receiving the SetConfiguration and SetInterface commands, the application must program endpoint registers accordingly and set this bit.

Bits 14:11 Reserved, must be kept at reset value.

**Bits 10:0 MPSIZ:** Maximum packet size

The application must program this field with the maximum packet size for the current logical endpoint. This value is in bytes.

**OTG\_HS device endpoint-x interrupt register (OTG\_HS\_DIEPINTx) (x = 0..5, where x = Endpoint\_number)**Address offset:  $0x908 + 0x20 * x$ 

Reset value: 0x0000 0080

This register indicates the status of an endpoint with respect to USB- and AHB-related events. It is shown in [Figure 414](#). The application must read this register when the IN endpoints interrupt bit of the Core interrupt register (IEPINT in OTG\_HS\_GINTSTS) is set. Before the application can read this register, it must first read the device all endpoints interrupt (OTG\_HS\_DAINTE) register to get the exact endpoint number for the device endpoint-x interrupt register. The application must clear the appropriate bit in this register to clear the corresponding bits in the OTG\_HS\_DAINTE and OTG\_HS\_GINTSTS registers.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13  | 12       | 11 | 10        | 9        | 8 | 7     | 6 | 5          | 4     | 3      | 2      | 1      | 0     |        |        |      |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|----------|----|-----------|----------|---|-------|---|------------|-------|--------|--------|--------|-------|--------|--------|------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NAK | Reserved |    | PKTDRPSTS | Reserved |   |       |   | TXFIFOUDRN | TXFE  | INEPNE | INEPNM | ITTXFE | TOC   | AHBERR | EPDISD | XFRC |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     | rc_w1    |    |           | rc_w1    |   | rc_w1 | r | rc_w1      | rc_w1 | rc_w1  | rc_w1  | rc_w1  | rc_w1 | rc_w1  |        |      |

Bits 31:14 Reserved, must be kept at reset value.

Bit 13 **NAK**: NAK interrupt

The core generates this interrupt when a NAK is transmitted or received by the device. In case of isochronous IN endpoints the interrupt gets generated when a zero length packet is transmitted due to unavailability of data in the Tx FIFO.

Bit 12 Reserved, must be kept at reset value.

Bit 11 **PKTDRPSTS**: Packet dropped status

This bit indicates to the application that an ISOC OUT packet has been dropped. This bit does not have an associated mask bit and does not generate an interrupt.

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **TXFIFOUDRN**: Transmit Fifo Underrun (TxfifoUndrn) The core generates this interrupt when it detects a transmit FIFO underrun condition for this endpoint.

**Dependency**: This interrupt is valid only when Thresholding is enabled

Bit 7 **TXFE**: Transmit FIFO empty

This interrupt is asserted when the TxFIFO for this endpoint is either half or completely empty. The half or completely empty status is determined by the TxFIFO empty level bit in the Core AHB configuration register (TXFELVL bit in OTG\_HS\_GAHBCFG).

Bit 6 **INENPNE**: IN endpoint NAK effective

This bit can be cleared when the application clears the IN endpoint NAK by writing to the CNAK bit in OTG\_HS\_DIEPCTLx.

This interrupt indicates that the core has sampled the NAK bit set (either by the application or by the core). The interrupt indicates that the IN endpoint NAK bit set by the application has taken effect in the core.

This interrupt does not guarantee that a NAK handshake is sent on the USB. A STALL bit takes priority over a NAK bit.

- Bit 5 **INEPNM**: IN token received with EP mismatch  
Indicates that the data in the top of the non-periodic TxFIFO belongs to an endpoint other than the one for which the IN token was received. This interrupt is asserted on the endpoint for which the IN token was received.
- Bit 4 **ITTXFE**: IN token received when TxFIFO is empty  
Applies to nonperiodic IN endpoints only.  
Indicates that an IN token was received when the associated TxFIFO (periodic/nonperiodic) was empty. This interrupt is asserted on the endpoint for which the IN token was received.
- Bit 3 **TOC**: Timeout condition  
Applies only to Control IN endpoints.  
Indicates that the core has detected a timeout condition on the USB for the last IN token on this endpoint.
- Bit 2 **AHBERR**: AHB error  
This is generated only in internal DMA mode when there is an AHB error during an AHB read/write. The application can read the corresponding endpoint DMA address register to get the error address.
- Bit 1 **EPDISD**: Endpoint disabled interrupt  
This bit indicates that the endpoint is disabled per the application's request.
- Bit 0 **XFRC**: Transfer completed interrupt  
This field indicates that the programmed transfer is complete on the AHB as well as on the USB, for this endpoint.

**OTG\_HS device endpoint-x interrupt register (OTG\_HS\_DOEPINTx) (x = 0..5, where x = Endpoint\_number)**

Address offset: 0xB08 + 0x20 \* x

Reset value: 0x0000 0080

This register indicates the status of an endpoint with respect to USB- and AHB-related events. It is shown in [Figure 414](#). The application must read this register when the OUT Endpoints Interrupt bit of the Core interrupt register (OEPINT bit in OTG\_HS\_GINTSTS) is set. Before the application can read this register, it must first read the device all endpoints interrupt (OTG\_HS\_DAINTE) register to get the exact endpoint number for the device Endpoint-x interrupt register. The application must clear the appropriate bit in this register to clear the corresponding bits in the OTG\_HS\_DAINTE and OTG\_HS\_GINTSTS registers.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14    | 13    | 12    | 11       | 10 | 9 | 8         | 7        | 6       | 5        | 4       | 3     | 2      | 1      | 0     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|-------|-------|----------|----|---|-----------|----------|---------|----------|---------|-------|--------|--------|-------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET  | NAK   | BERR  | Reserved |    |   | OUTPKTERR | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP  | AHBERR | EPDISD | XFRC  |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rc_w1 | rc_w1 | rc_w1 |          |    |   | rc_w1     |          | rc_w1   |          | rc_w1   | rc_w1 | rc_w1  | rc_w1  | rc_w1 |

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **NYET**: NYET interrupt

The core generates this interrupt when a NYET response is transmitted for a nonisochronous OUT endpoint.

Bit 13 **NAK**: NAK input

The core generates this interrupt when a NAK is transmitted or received by the device. In case of isochronous IN endpoints the interrupt gets generated when a zero length packet is transmitted due to unavailability of data in the Tx FIFO.

Bit 12 **BERR**: Babble error interrupt

The core generates this interrupt when babble is received for the endpoint.

Bits 11:9 Reserved, must be kept at reset value.

Bit 8 **OUTPKTERR**: OUT packet error

This interrupt is asserted when the core detects an overflow or a CRC error for an OUT packet. This interrupt is valid only when thresholding is enabled.

Bit 7 Reserved, must be kept at reset value.

Bit 6 **B2BSTUP**: Back-to-back SETUP packets received

Applies to Control OUT endpoint only.  
This bit indicates that the core has received more than three back-to-back SETUP packets for this particular endpoint.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **OTEPDIS**: OUT token received when endpoint disabled

Applies only to control OUT endpoint.  
Indicates that an OUT token was received when the endpoint was not yet enabled. This interrupt is asserted on the endpoint for which the OUT token was received.

Bit 3 **STUP**: SETUP phase done

Applies to control OUT endpoints only.

Indicates that the SETUP phase for the control endpoint is complete and no more back-to-back SETUP packets were received for the current control transfer. On this interrupt, the application can decode the received SETUP data packet.

Bit 2 **AHBERR**: AHB error

This is generated only in internal DMA mode when there is an AHB error during an AHB read/write. The application can read the corresponding endpoint DMA address register to get the error address.

Bit 1 **EPDISD**: Endpoint disabled interrupt

This bit indicates that the endpoint is disabled per the application's request.

Bit 0 **XFRC**: Transfer completed interrupt

This field indicates that the programmed transfer is complete on the AHB as well as on the USB, for this endpoint.

**OTG\_HS device IN endpoint 0 transfer size register (OTG\_HS\_DIEPTSIZ0)**

Address offset: 0x910

Reset value: 0x0000 0000

The application must modify this register before enabling endpoint 0. Once endpoint 0 is enabled using the endpoint enable bit in the device control endpoint 0 control registers (EPENA in OTG\_HS\_DIEPCTL0), the core modifies this register. The application can only read this register once the core has cleared the Endpoint enable bit.

Nonzero endpoints use the registers for endpoints 1–15.

|          |    |    |    |    |    |    |    |    |    |    |        |    |          |    |    |    |    |    |    |    |    |   |   |        |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|----|----|----|--------|----|----------|----|----|----|----|----|----|----|----|---|---|--------|----|----|----|----|----|----|----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20     | 19 | 18       | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7      | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Reserved |    |    |    |    |    |    |    |    |    |    | PKTCNT |    | Reserved |    |    |    |    |    |    |    |    |   |   | XFRSIZ |    |    |    |    |    |    |    |
|          |    |    |    |    |    |    |    |    |    |    | rw     | rw |          |    |    |    |    |    |    |    |    |   |   | rw     | rw | rw | rw | rw | rw | rw | rw |

Bits 31:21 Reserved, must be kept at reset value.

Bits 20:19 **PKTCNT**: Packet count

Indicates the total number of USB packets that constitute the Transfer Size amount of data for endpoint 0.

This field is decremented every time a packet (maximum size or short packet) is read from the TxFIFO.

Bits 18:7 Reserved, must be kept at reset value.

Bits 6:0 **XFRSIZ**: Transfer size

Indicates the transfer size in bytes for endpoint 0. The core interrupts the application only after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet.

The core decrements this field every time a packet from the external memory is written to the TxFIFO.

**OTG\_HS device OUT endpoint 0 transfer size register (OTG\_HS\_DOEPTSIZE0)**

Address offset: 0xB10

Reset value: 0x0000 0000

The application must modify this register before enabling endpoint 0. Once endpoint 0 is enabled using the Endpoint enable bit in the device control endpoint 0 control registers (EPENA bit in OTG\_HS\_DOEPCTL0), the core modifies this register. The application can only read this register once the core has cleared the Endpoint enable bit.

Nonzero endpoints use the registers for endpoints 1–15.

| 31       | 30 | 29      | 28 | 27       | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19     | 18 | 17       | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5      | 4 | 3 | 2 | 1 | 0 |  |  |
|----------|----|---------|----|----------|----|----|----|----|----|----|----|--------|----|----------|----|----|----|----|----|----|----|---|---|---|---|--------|---|---|---|---|---|--|--|
| Reserved |    | STUPCNT |    | Reserved |    |    |    |    |    |    |    | PKTCNT |    | Reserved |    |    |    |    |    |    |    |   |   |   |   | XFRSIZ |   |   |   |   |   |  |  |
|          |    | rw      | rw |          |    |    |    |    |    |    |    |        |    |          |    |    |    |    |    |    |    |   |   |   |   |        |   |   |   |   |   |  |  |

Bit 31 Reserved, must be kept at reset value.

Bits 30:29 **STUPCNT**: SETUP packet count

This field specifies the number of back-to-back SETUP data packets the endpoint can receive.

01: 1 packet

10: 2 packets

11: 3 packets

Bits 28:20 Reserved, must be kept at reset value.

Bit 19 **PKTCNT**: Packet count

This field is decremented to zero after a packet is written into the RxFIFO.

Bits 18:7 Reserved, must be kept at reset value.

Bits 6:0 **XFRSIZ**: Transfer size

Indicates the transfer size in bytes for endpoint 0. The core interrupts the application only after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet.

The core decrements this field every time a packet is read from the RxFIFO and written to the external memory.

### OTG\_HS device endpoint-x transfer size register (OTG\_HS\_DIEPTSIZx) (x = 1..5, where x = Endpoint\_number)

Address offset: 0x910 + 0x20 \* x

Reset value: 0x0000 0000

The application must modify this register before enabling the endpoint. Once the endpoint is enabled using the Endpoint enable bit in the device endpoint-x control registers (EPENA bit in OTG\_HS\_DIEPCTLx), the core modifies this register. The application can only read this register once the core has cleared the Endpoint enable bit.

| 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |      |     |        |     |     |     |     |     |     |     |     |     |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|---------------------------------------------------------------------------------------|------|-----|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Reserved                                                                              | MCNT |     | PKTCNT |     |     |     |     |     |     |     |     |     | XFRSIZ |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|                                                                                       | r/w  | r/w | r/w    | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w    | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w |  |

Bit 31 **Reserved**, must be kept at reset value.

Bits 30:29 **MCNT**: Multi count

For periodic IN endpoints, this field indicates the number of packets that must be transmitted per frame on the USB. The core uses this field to calculate the data PID for isochronous IN endpoints.

01: 1 packet

10: 2 packets

11: 3 packets

Bit 28:19 **PKTCNT**: Packet count

Indicates the total number of USB packets that constitute the Transfer Size amount of data for this endpoint.

This field is decremented every time a packet (maximum size or short packet) is read from the TxFIFO.

Bits 18:0 **XFRSIZ**: Transfer size

This field contains the transfer size in bytes for the current endpoint. The core only interrupts the application after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet.

The core decrements this field every time a packet from the external memory is written to the TxFIFO.

### OTG\_HS device IN endpoint transmit FIFO status register (OTG\_HS\_DTXFSTSx) (x = 0..5, where x = Endpoint\_number)

Address offset for IN endpoints: 0x918 + 0x20 + x

This read-only register contains the free space information for the Device IN endpoint TxFIFO.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |           |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15        | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | INEPTFSAV |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | r         | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

31:16 Reserved, must be kept at reset value.

15:0 **INEPTFSAV**: IN endpoint TxFIFO space avail ()

Indicates the amount of free space available in the Endpoint TxFIFO.

Values are in terms of 32-bit words:

0x0: Endpoint TxFIFO is full

0x1: 1 word available

0x2: 2 words available

0xn: n words available (0 < n < 512)

Others: Reserved

### OTG\_HS device endpoint-x transfer size register (OTG\_HS\_DOEPTSIZx) (x = 1..5, where x = Endpoint\_number)

Address offset: 0xB10 + 0x20 \* x

Reset value: 0x0000 0000

The application must modify this register before enabling the endpoint. Once the endpoint is enabled using Endpoint Enable bit of the device endpoint-x control registers (EPENA bit in OTG\_HS\_DOEPCTLx), the core modifies this register. The application can only read this register once the core has cleared the Endpoint enable bit.

|          |                    |      |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|----------|--------------------|------|--------|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|
| 31       | 30                 | 29   | 28     | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16     | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |  |
| Reserved | RXDPID/S<br>TUPCNT |      | PKTCNT |    |    |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|          | r/rw               | r/rw | rw     | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw     | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw |  |

Bit 31 Reserved, must be kept at reset value.

Bits 30:29 **RXDPID**: Received data PID (access type is "r")

Applies to isochronous OUT endpoints only.

This is the data PID received in the last packet for this endpoint.

00: DATA0

01: DATA2

10: DATA1

11: MDATA

**STUPCNT**: SETUP packet count (access type is "rw")

Applies to control OUT Endpoints only.

This field specifies the number of back-to-back SETUP data packets the endpoint can receive.

01: 1 packet

10: 2 packets

11: 3 packets

Bit 28:19 **PKTCNT**: Packet count

Indicates the total number of USB packets that constitute the Transfer Size amount of data for this endpoint.

This field is decremented every time a packet (maximum size or short packet) is written to the RxFIFO.

Bits 18:0 **XFRSIZ**: Transfer size

This field contains the transfer size in bytes for the current endpoint. The core only interrupts the application after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet.

The core decrements this field every time a packet is read from the RxFIFO and written to the external memory.

### OTG\_HS device endpoint-x DMA address register (OTG\_HS\_DIEPDMAx / OTG\_HS\_DOEPDMAx) (x = 0..5, where x = Endpoint\_number)

Address offset for IN endpoints:  $0x914 + 0x20 * x$

Reset value: 0XXXXX XXXX

Address offset for OUT endpoints:  $0xB14 + 0x20 * x$

Reset value: 0XXXXX XXXX

|         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31      | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| DMAADDR |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| rw      | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw |

Bits 31:0 **DMAADDR**: DMA address

This bit holds the start address of the external memory for storing or fetching endpoint data.

*Note: For control endpoints, this field stores control OUT data packets as well as SETUP transaction data packets. When more than three SETUP packets are received back-to-back, the SETUP data packet in the memory is overwritten. This register is incremented on every AHB transaction. The application can give only a word-aligned address.*

### 35.12.5 OTG\_HS power and clock gating control register (OTG\_HS\_PCGCTL)

Address offset: 0xE00

Reset value: 0x0000 0000

This register is available in host and peripheral modes.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5       | 4        | 3        | 2       | 1 | 0 |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---------|----------|----------|---------|---|---|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   | PHYSUSP | Reserved | GATEHCLK | STPPCLK |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   | r/w     |          | r/w      | r/w     |   |   |

Bit 31:5 Reserved, must be kept at reset value.

Bit 4 **PHYSUSP**: PHY suspended

Indicates that the PHY has been suspended. This bit is updated once the PHY is suspended after the application has set the STPPCLK bit (bit 0).

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **GATEHCLK**: Gate HCLK

The application sets this bit to gate HCLK to modules other than the AHB Slave and Master and wake-up logic when the USB is suspended or the session is not valid. The application clears this bit when the USB is resumed or a new session starts.

Bit 0 **STPPCLK**: Stop PHY clock

The application sets this bit to stop the PHY clock when the USB is suspended, the session is not valid, or the device is disconnected. The application clears this bit when the USB is resumed or a new session starts.

### 35.12.6 OTG\_HS register map

The table below gives the USB OTG register map and reset values.

**Table 215. OTG\_HS register map and reset values**

| Offset | Register        | 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19     | 18      | 17     | 16       | 15       | 14 | 13 | 12 | 11     | 10      | 9       | 8        | 7        | 6       | 5        | 4 | 3 | 2 | 1 | 0     |      |        |      |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| 0x000  | OTG_HS_GO_TGCTL | Reserved |    |    |    |    |    |    |    |    |    |    |    | BSVLD  | ASVLD   | DBCT   | CIDSTS   | Reserved |    |    |    | DHNPEN | HSHPEN  | HNPREQ  | HNGSCS   | Reserved |         |          |   |   |   |   |       | SRQ  | SRQSCS |      |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|        | Reset value     |          |    |    |    |    |    |    |    |    |    |    |    | 0      | 0       | 0      |          |          |    |    |    | 1      | 0       | 0       | 0        |          |         |          |   |   |   |   |       | 0    | 0      | 0    | 0 |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0x004  | OTG_HS_GO_TGINT | Reserved |    |    |    |    |    |    |    |    |    |    |    | DBCDNE | ADTOCHG | HNGDET | Reserved |          |    |    |    |        | HNSSCHG | SRSSCHG | Reserved |          |         |          |   |   |   |   | SEDET | Res. |        |      |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|        | Reset value     |          |    |    |    |    |    |    |    |    |    |    |    | 0      | 0       | 0      |          |          |    |    |    |        | 0       | 0       |          |          |         |          |   |   |   |   | 0     |      | 0      | 0    | 0 | 0 | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0x008  | OTG_HS_GA_HBCFG | Reserved |    |    |    |    |    |    |    |    |    |    |    |        |         |        |          |          |    |    |    |        |         |         |          | PTXFELVL | TXFELVL | Reserved |   |   |   |   |       |      |        | GINT |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|        | Reset value     |          |    |    |    |    |    |    |    |    |    |    |    |        |         |        |          |          |    |    |    |        |         |         |          | 0        | 0       |          |   |   |   |   |       |      |        |      |   |   |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register                                   | 31       | 30     | 29       | 28       | 27       | 26 | 25     | 24     | 23      | 22       | 21      | 20         | 19                 | 18       | 17      | 16      | 15                                                      | 14      | 13                              | 12   | 11      | 10       | 9        | 8      | 7        | 6        | 5        | 4            | 3       | 2 | 1         | 0         |          |         |       |        |       |          |
|--------|--------------------------------------------|----------|--------|----------|----------|----------|----|--------|--------|---------|----------|---------|------------|--------------------|----------|---------|---------|---------------------------------------------------------|---------|---------------------------------|------|---------|----------|----------|--------|----------|----------|----------|--------------|---------|---|-----------|-----------|----------|---------|-------|--------|-------|----------|
| 0x00C  | OTG_HS_GU<br>SBCFG                         | CTXPKT   | FDMOD  | FHMOD    | Reserved |          |    |        | ULPIPD | PTCI    | PCCI     | TSDPS   | ULPIEVBUSI | ULPIEVBUSD         | ULPICSM  | ULPIAR  | ULPIFSL | Reserved                                                | PHYLPCS | Reserved                        | TRDT |         |          |          | HNPCAP | SRPCAP   | Reserved | PHYSEL   | Reserve<br>a |         |   |           | TOTAL     |          |         |       |        |       |          |
|        | Reset value                                | 0        | 0      | 0        |          |          |    |        | 0      | 0       | 0        | 0       | 0          | 0                  | 0        | 0       | 0       | 0                                                       | 0       | Reserved                        | 0    | 0 1 0 1 |          |          |        | 0        | 0        | 1        |              |         |   | 0 0 0     |           |          |         |       |        |       |          |
| 0x010  | OTG_HS_GR<br>STCTL                         | AHBIDL   | DMAREQ | Reserved |          |          |    |        |        |         |          |         |            |                    |          |         |         |                                                         |         |                                 |      |         |          |          |        |          |          | TXFNUM   |              |         |   | TXFFLSH   | RXFFLSH   | Reserved | FCRST   | HSRST | CSRST  |       |          |
|        | Reset value                                | 1        | 0      |          |          |          |    |        |        |         |          |         |            |                    |          |         |         |                                                         |         |                                 |      |         |          |          |        |          |          | 0        | 0            | 0       | 0 | 0         | 0         | 0        | 0       | 0     | 0      |       |          |
| 0x014  | OTG_HS_GIN<br>TSTS                         | WKUINT   | SRQINT | DISCINT  | CIDSCHG  | Reserved |    | PTXFE  | HCINT  | HPRINT  | Reserved |         | DATAFSUSP  | IPXFR/INCOMPISOOUT | IISOXFR  | OEPIINT | IEPIINT | Reserved                                                |         |                                 |      | EOPF    | ISOODRP  | ENUMDNE  | USBRST | USBSUSP  | ESUSP    | Reserved |              |         |   | GONAKEFF  | GINAKEFF  | NPTXFE   | RXFLVL  | SOF   | OTGINT | MMIS  | CMOD     |
|        | Reset value                                | 0        | 0      | 0        | 0        |          |    | 1      | 0      | 0       |          |         | 0          | 0                  | 0        | 0       | 0       | 0                                                       |         |                                 |      |         | 0        | 0        | 0      | 0        | 0        |          |              |         |   | 0         | 0         | 1        | 0       | 0     | 0      | 0     | 0        |
| 0x018  | OTG_HS_GIN<br>TMSK                         | WUIM     | SRQIM  | DISCINT  | CIDSCHGM | Reserved |    | PTXFEM | HCIM   | PRTIM   | Reserved |         | FSUSPM     | IPXFRM/IISOXFRM    | IISOXFRM | OEPIINT | IEPIINT | Reserved                                                |         |                                 |      | EOPFM   | ISOODRPM | ENUMDNEM | USBRST | USBSUSPM | ESUSPM   | Reserved |              |         |   | GONAKEFFM | GINAKEFFM | NPTXFEM  | RXFLVLM | SOFM  | OTGINT | MMISM | Reserved |
|        | Reset value                                | 0        | 0      | 0        | 0        |          |    | 0      | 0      | 0       |          |         | 0          | 0                  | 0        | 0       | 0       | 0                                                       |         |                                 |      |         | 0        | 0        | 0      | 0        | 0        |          |              |         |   | 0         | 0         | 0        | 0       | 0     | 0      | 0     | 0        |
| 0x01C  | OTG_HS_GR<br>XSTSR (Host<br>mode)          | Reserved |        |          |          |          |    |        |        |         |          | PKTSTS  |            |                    |          | DPID    |         | BCNT                                                    |         |                                 |      |         |          |          |        |          |          | CHNUM    |              |         |   |           |           |          |         |       |        |       |          |
|        | Reset value                                |          |        |          |          |          |    |        |        |         |          | 0 0 0 0 |            |                    |          | 0 0     |         | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0                         |         |                                 |      |         |          |          |        |          |          | 0 0 0 0  |              |         |   |           |           |          |         |       |        |       |          |
|        | OTG_HS_GR<br>XSTSR<br>(peripheral<br>mode) | Reserved |        |          |          |          |    |        |        | FRMNUM  |          |         |            | PKTSTS             |          |         |         | DPID                                                    |         | BCNT                            |      |         |          |          |        |          |          |          |              | EPNUM   |   |           |           |          |         |       |        |       |          |
|        | Reset value                                |          |        |          |          |          |    |        |        | 0 0 0 0 |          |         |            | 0 0 0 0 0          |          |         |         | 0 0                                                     |         | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |      |         |          |          |        |          |          |          |              | 0 0 0 0 |   |           |           |          |         |       |        |       |          |
| 0x020  | OTG_HS_GR<br>XSTSP (Host<br>mode)          | Reserved |        |          |          |          |    |        |        |         |          |         |            | PKTSTS             |          |         |         | DPID                                                    |         | BCNT                            |      |         |          |          |        |          |          |          |              | CHNUM   |   |           |           |          |         |       |        |       |          |
|        | Reset value                                |          |        |          |          |          |    |        |        |         |          |         |            | 0 0 0 0            |          |         |         | 0 0                                                     |         | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |      |         |          |          |        |          |          |          |              | 0 0 0 0 |   |           |           |          |         |       |        |       |          |
|        | OTG_HS_GR<br>XSTSP<br>(peripheral<br>mode) | Reserved |        |          |          |          |    |        |        | FRMNUM  |          |         |            | PKTSTS             |          |         |         | DPID                                                    |         | BCNT                            |      |         |          |          |        |          |          |          |              | EPNUM   |   |           |           |          |         |       |        |       |          |
|        | Reset value                                |          |        |          |          |          |    |        |        | 0 0 0 0 |          |         |            | 0 0 0 0 0          |          |         |         | 0 0                                                     |         | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |      |         |          |          |        |          |          |          |              | 0 0 0 0 |   |           |           |          |         |       |        |       |          |
| 0x024  | OTG_HS_GR<br>XFSIZ                         | Reserved |        |          |          |          |    |        |        |         |          |         |            |                    |          |         |         | RXFD                                                    |         |                                 |      |         |          |          |        |          |          |          |              |         |   |           |           |          |         |       |        |       |          |
|        | Reset value                                |          |        |          |          |          |    |        |        |         |          |         |            |                    |          |         |         | 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |         |                                 |      |         |          |          |        |          |          |          |              |         |   |           |           |          |         |       |        |       |          |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register                              | 31         | 30       | 29       | 28        | 27        | 26       | 25  | 24       | 23      | 22 | 21 | 20 | 19         | 18       | 17       | 16       | 15       | 14     | 13       | 12 | 11     | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1     | 0 |         |   |
|--------|---------------------------------------|------------|----------|----------|-----------|-----------|----------|-----|----------|---------|----|----|----|------------|----------|----------|----------|----------|--------|----------|----|--------|----|---|---|---|---|---|---|---|---|-------|---|---------|---|
| 0x028  | OTG_HS_GNPTXFSIZ<br>(Host mode)       | NPTXFD     |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | NPTXFSA  |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 0   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 0  | 0      | 0  | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
|        | OTG_HS_GNPTXFSIZ<br>(peripheral mode) | TX0FD      |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | TX0FSA   |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 0   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 0  | 0      | 0  | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x02C  | OTG_HS_GNPTXSTS                       | Res.       | NPTXQTOP |          |           |           |          |     | NPTQXSAV |         |    |    |    |            | NPTXFSAV |          |          |          |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           |            | 0        | 0        | 0         | 0         | 0        | 0   | 0        | 0       | 0  | 0  | 0  | 1          | 0        | 0        | 0        | 0        | 0      | 0        | 0  | 0      | 1  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x030  | OTG_HS_GI2CCTL                        | BSYDNE     | RW       | Reserved | I2CDATSE0 | I2CDEVADR | Reserved | ACK | I2CEN    | ADDR    |    |    |    |            |          | REGADDR  |          |          |        |          |    | RWDATA |    |   |   |   |   |   |   |   |   |       |   |         |   |
| 0x038  | OTG_HS_GC_CFG                         | Reserved   |          |          |           |           |          |     |          |         |    |    |    | NOVBUSSENS | SOFOUTEN | VBUSBSEN | VBUSASEN | I2CPADEN | PWRDWN | Reserved |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           |            |          |          |           |           |          |     |          |         |    |    |    | 0          | 0        | 0        | 0        | 0        | 0      |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
| 0x03C  | OTG_HS_CID                            | PRODUCT_ID |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          |          |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 0   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 1  | 0      | 0  | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x100  | OTG_HS_HPTXFSIZ                       | PTXFD      |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | PTXSA    |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 1   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 1  | 0      | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x104  | OTG_HS_DIEPTXF1                       | INEPTXFD   |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | INEPTXSA |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 1   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 0  | 1      | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x108  | OTG_HS_DIEPTXF2                       | INEPTXFD   |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | INEPTXSA |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 1   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 0  | 1      | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x10C  | OTG_HS_DIEPTXF3                       | INEPTXFD   |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | INEPTXSA |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 1   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 1  | 0      | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x110  | OTG_HS_DIEPTXF4                       | INEPTXFD   |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | INEPTXSA |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 1   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 0        | 0  | 1      | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     | 0 |         |   |
| 0x400  | OTG_HS_HCFG_G                         | Reserved   |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          |          |        |          |    |        |    |   |   |   |   |   |   |   |   | FSLSS |   | FSLSPCS |   |
|        | Reset value                           |            |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          |          |        |          |    |        |    |   |   |   |   |   |   |   |   | 0     | 0 | 0       |   |
| 0x404  | OTG_HS_HFIR                           | Reserved   |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | FRIVL    |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           |            |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | 1        | 1      | 1        | 0  | 1      | 0  | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0     | 0 | 0       | 0 |
| 0x408  | OTG_HS_HFNUM                          | FTREM      |          |          |           |           |          |     |          |         |    |    |    |            |          |          |          | FRNUM    |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 0   | 0        | 0       | 0  | 0  | 0  | 0          | 0        | 0        | 0        | 0        | 0      | 1        | 1  | 1      | 1  | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1     |   |         |   |
| 0x410  | OTG_HS_HPTXSTS                        | PTXQTOP    |          |          |           |           |          |     |          | PTXQSAV |    |    |    |            |          |          |          | PTXFSAVL |        |          |    |        |    |   |   |   |   |   |   |   |   |       |   |         |   |
|        | Reset value                           | 0          | 0        | 0        | 0         | 0         | 0        | 0   | 0        | Y       | Y  | Y  | Y  | Y          | Y        | Y        | Y        | Y        | Y      | Y        | Y  | Y      | Y  | Y | Y | Y | Y | Y | Y | Y | Y | Y     |   |         |   |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register            | 31       | 30    | 29     | 28  | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18       | 17    | 16       | 15     | 14    | 13   | 12 | 11    | 10    | 9        | 8    | 7     | 6    | 5       | 4    | 3       | 2    | 1     | 0     |   |   |   |   |   |   |   |   |   |   |
|--------|---------------------|----------|-------|--------|-----|----|----|----|----|----|----|----|----|----|----------|-------|----------|--------|-------|------|----|-------|-------|----------|------|-------|------|---------|------|---------|------|-------|-------|---|---|---|---|---|---|---|---|---|---|
| 0x414  | OTG_HS_HAI<br>NT    | Reserved |       |        |     |    |    |    |    |    |    |    |    |    |          |       |          | HAINT  |       |      |    |       |       |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         |          |       |        |     |    |    |    |    |    |    |    |    |    |          |       |          | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x418  | OTG_HS_HAI<br>NTMSK | Reserved |       |        |     |    |    |    |    |    |    |    |    |    |          |       |          | HAINTM |       |      |    |       |       |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         |          |       |        |     |    |    |    |    |    |    |    |    |    |          |       |          | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x440  | OTG_HS_HPR<br>T     | Reserved |       |        |     |    |    |    |    |    |    |    |    |    | PSP<br>D | PTCTL |          |        |       | PPWR |    | PLSTS |       | Reserved | PRST | PSUSP | PRES | POCCHNG | POCA | PENCHNG | PENA | PCDET | PCSTS |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         |          |       |        |     |    |    |    |    |    |    |    |    |    | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 |   |   |   |
| 0x500  | OTG_HS_HC<br>CHAR0  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x520  | OTG_HS_HC<br>CHAR1  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x540  | OTG_HS_HC<br>CHAR2  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x560  | OTG_HS_HC<br>CHAR3  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x580  | OTG_HS_HC<br>CHAR4  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x5A0  | OTG_HS_HC<br>CHAR5  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x5C0  | OTG_HS_HC<br>CHAR6  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x5E0  | OTG_HS_HC<br>CHAR7  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |
| 0x600  | OTG_HS_HC<br>CHAR8  | CHENA    | CHDIS | ODDFRM | DAD |    |    |    |    |    |    |    | MC |    | EPTYP    | LSDEV | Reserved | EPDIR  | EPNUM |      |    |       | MPSIZ |          |      |       |      |         |      |         |      |       |       |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0        | 0     | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0     | 0        | 0      | 0     | 0    | 0  | 0     | 0     | 0        | 0    | 0     | 0    | 0       | 0    | 0       | 0    | 0     | 0     | 0 |   |   |   |   |   |   |   |   |   |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register         | 31       | 30       | 29     | 28  | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20    | 19 | 18    | 17       | 16    | 15        | 14      | 13      | 12    | 11    | 10    | 9       | 8    | 7   | 6   | 5     | 4      | 3   | 2    | 1 | 0 |   |   |   |   |   |
|--------|------------------|----------|----------|--------|-----|----|----|----|----|----|----|----|-------|----|-------|----------|-------|-----------|---------|---------|-------|-------|-------|---------|------|-----|-----|-------|--------|-----|------|---|---|---|---|---|---|---|
| 0x620  | OTG_HS_HC_CHAR9  | CHENA    | CHDIS    | ODDFRM | DAD |    |    |    |    |    | MC |    | EPTYP |    | LSDEV | Reserved | EPDIR | EPNUM     |         |         |       | MPSIZ |       |         |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        | 0        | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0     | 0  | 0     | 0        | 0     | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 |   |   |   |   |   |
| 0x640  | OTG_HS_HC_CHAR10 | CHENA    | CHDIS    | ODDFRM | DAD |    |    |    |    |    | MC |    | EPTYP |    | LSDEV | Reserved | EPDIR | EPNUM     |         |         |       | MPSIZ |       |         |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        | 0        | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0     | 0  | 0     | 0        | 0     | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 |   |   |   |   |   |
| 0x660  | OTG_HS_HC_CHAR11 | CHENA    | CHDIS    | ODDFRM | DAD |    |    |    |    |    | MC |    | EPTYP |    | LSDEV | Reserved | EPDIR | EPNUM     |         |         |       | MPSIZ |       |         |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        | 0        | 0      | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0     | 0  | 0     | 0        | 0     | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 |   |   |   |   |   |
| 0x504  | OTG_HS_HCS_PLT0  | SPLITEN  | Reserved |        |     |    |    |    |    |    |    |    |       |    |       |          |       | COMPLSPLT | XACTPOS | HUBADDR |       |       |       | PRTADDR |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 | 0 |   |
| 0x508  | OTG_HS_HCI_NT0   | Reserved |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | DTERR | FRMOR | BBERR | TXERR   | NYET | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |   |   |   |   |   |   |
|        | Reset value      |          |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x524  | OTG_HS_HCS_PL1   | SPLITEN  | Reserved |        |     |    |    |    |    |    |    |    |       |    |       |          |       | COMPLSPLT | XACTPOS | HUBADDR |       |       |       | PRTADDR |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 |   |   |
| 0x528  | OTG_HS_HCI_NT1   | Reserved |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | DTERR | FRMOR | BBERR | TXERR   | NYET | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |   |   |   |   |   |   |
|        | Reset value      |          |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x544  | OTG_HS_HCS_PLT2  | SPLITEN  | Reserved |        |     |    |    |    |    |    |    |    |       |    |       |          |       | COMPLSPLT | XACTPOS | HUBADDR |       |       |       | PRTADDR |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 |   |   |
| 0x548  | OTG_HS_HCI_NT2   | Reserved |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | DTERR | FRMOR | BBERR | TXERR   | NYET | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |   |   |   |   |   |   |
|        | Reset value      |          |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x564  | OTG_HS_HCS_PLT3  | SPLITEN  | Reserved |        |     |    |    |    |    |    |    |    |       |    |       |          |       | COMPLSPLT | XACTPOS | HUBADDR |       |       |       | PRTADDR |      |     |     |       |        |     |      |   |   |   |   |   |   |   |
|        | Reset value      | 0        |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       | 0         | 0       | 0       | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 |   |   |
| 0x568  | OTG_HS_HCI_NT3   | Reserved |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | DTERR | FRMOR | BBERR | TXERR   | NYET | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |   |   |   |   |   |   |
|        | Reset value      |          |          |        |     |    |    |    |    |    |    |    |       |    |       |          |       |           |         |         | 0     | 0     | 0     | 0       | 0    | 0   | 0   | 0     | 0      | 0   | 0    | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register        | 31      | 30       | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16        | 15      | 14 | 13      | 12 | 11    | 10    | 9     | 8     | 7       | 6   | 5   | 4     | 3      | 2   | 1    | 0 |
|--------|-----------------|---------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|-----------|---------|----|---------|----|-------|-------|-------|-------|---------|-----|-----|-------|--------|-----|------|---|
| 0x584  | OTG_HS_HCS_PLT4 | SPLITEN | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | COMPLSPLT | XACTPOS |    | HUBADDR |    |       |       |       |       | PRTADDR |     |     |       |        |     |      |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |           | 0       | 0  | 0       | 0  | 0     | 0     | 0     | 0     | 0       | 0   | 0   | 0     | 0      | 0   | 0    | 0 |
| 0x588  | OTG_HS_HCI_NT4  |         | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |           |         |    |         |    | DTERR | FRMOR | BBERR | TXERR | NYET    | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0       | 0  | 0       | 0  |       |       |       |       |         |     |     |       |        |     |      | 0 |
| 0x5A4  | OTG_HS_HCS_PLT5 | SPLITEN | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | COMPLSPLT | XACTPOS |    | HUBADDR |    |       |       |       |       | PRTADDR |     |     |       |        |     |      |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |           | 0       | 0  | 0       | 0  | 0     | 0     | 0     | 0     | 0       | 0   | 0   | 0     | 0      | 0   | 0    | 0 |
| 0x5A8  | OTG_HS_HCI_NT5  |         | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |           |         |    |         |    | DTERR | FRMOR | BBERR | TXERR | NYET    | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0       | 0  | 0       | 0  |       |       |       |       |         |     |     |       |        |     |      | 0 |
| 0x5C4  | OTG_HS_HCS_PLT6 | SPLITEN | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | COMPLSPLT | XACTPOS |    | HUBADDR |    |       |       |       |       | PRTADDR |     |     |       |        |     |      |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |           | 0       | 0  | 0       | 0  | 0     | 0     | 0     | 0     | 0       | 0   | 0   | 0     | 0      | 0   | 0    | 0 |
| 0x5C8  | OTG_HS_HCI_NT6  |         | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |           |         |    |         |    | DTERR | FRMOR | BBERR | TXERR | NYET    | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0       | 0  | 0       | 0  |       |       |       |       |         |     |     |       |        |     |      | 0 |
| 0x5E4  | OTG_HS_HCS_PLT7 | SPLITEN | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | COMPLSPLT | XACTPOS |    | HUBADDR |    |       |       |       |       | PRTADDR |     |     |       |        |     |      |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |           | 0       | 0  | 0       | 0  | 0     | 0     | 0     | 0     | 0       | 0   | 0   | 0     | 0      | 0   | 0    | 0 |
| 0x5E8  | OTG_HS_HCI_NT7  |         | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |           |         |    |         |    | DTERR | FRMOR | BBERR | TXERR | NYET    | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0       | 0  | 0       | 0  |       |       |       |       |         |     |     |       |        |     |      | 0 |
| 0x604  | OTG_HS_HCS_PLT8 | SPLITEN | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | COMPLSPLT | XACTPOS |    | HUBADDR |    |       |       |       |       | PRTADDR |     |     |       |        |     |      |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |           | 0       | 0  | 0       | 0  | 0     | 0     | 0     | 0     | 0       | 0   | 0   | 0     | 0      | 0   | 0    | 0 |
| 0x608  | OTG_HS_HCI_NT8  |         | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |           |         |    |         |    | DTERR | FRMOR | BBERR | TXERR | NYET    | ACK | NAK | STALL | AHBERR | CHH | XFRC |   |
|        | Reset value     |         | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0       | 0  | 0       | 0  |       |       |       |       |         |     |     |       |        |     |      | 0 |

**Table 215. OTG\_HS register map and reset values (continued)**

[illegible]

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register              | 31       | 30   | 29     | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19     | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10     | 9      | 8      | 7      | 6    | 5    | 4    | 3      | 2       | 1    | 0     |
|--------|-----------------------|----------|------|--------|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|--------|--------|--------|--------|------|------|------|--------|---------|------|-------|
| 0x5AC  | OTG_HS_HCI<br>NTMSK5  | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x5CC  | OTG_HS_HCI<br>NTMSK6  | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x5EC  | OTG_HS_HCI<br>NTMSK7  | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x60C  | OTG_HS_HCI<br>NTMSK8  | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x62C  | OTG_HS_HCI<br>NTMSK9  | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x64C  | OTG_HS_HCI<br>NTMSK10 | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x66C  | OTG_HS_HCI<br>NTMSK11 | Reserved |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | DTERRM | FRMORM | BBERRM | TXERRM | NYET | ACKM | NAKM | STALLM | AHBERRM | CHHM | XFCRM |
|        | Reset value           |          |      |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x510  | OTG_HS_HCT<br>SIZ0    | DOPING   | DPID | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |        |        |        |        |      |      |      |        |         |      |       |
|        | Reset value           | 0        | 0    | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x530  | OTG_HS_HCT<br>SIZ1    | DOPING   | DPID | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |        |        |        |        |      |      |      |        |         |      |       |
|        | Reset value           | 0        | 0    | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x550  | OTG_HS_HCT<br>SIZ2    | DOPING   | DPID | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |        |        |        |        |      |      |      |        |         |      |       |
|        | Reset value           | 0        | 0    | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |
| 0x570  | OTG_HS_HCT<br>SIZ3    | DOPING   | DPID | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |        |        |        |        |      |      |      |        |         |      |       |
|        | Reset value           | 0        | 0    | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0      | 0      | 0      | 0    | 0    | 0    | 0      | 0       | 0    | 0     |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register            | 31      | 30   | 29 | 28     | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18     | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|--------|---------------------|---------|------|----|--------|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 0x590  | OTG_HS_HCT<br>SIZ4  | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x5B0  | OTG_HS_HCT<br>SIZ5  | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x5D0  | OTG_HS_HCT<br>SIZ6  | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x5F0  | OTG_HS_HCT<br>SIZ7  | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x610  | OTG_HS_HCT<br>SIZ8  | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x630  | OTG_HS_HCT<br>SIZ9  | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x650  | OTG_HS_HCT<br>SIZ10 | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x670  | OTG_HS_HCT<br>SIZ11 | DOPING  | DPID |    | PKTCNT |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x514  | OTG_HS_HC<br>DMA0   | DMAADDR |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x524  | OTG_HS_HC<br>DMA1   | DMAADDR |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x544  | OTG_HS_HC<br>DMA2   | DMAADDR |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x564  | OTG_HS_HC<br>DMA3   | DMAADDR |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x584  | OTG_HS_HC<br>DMA4   | DMAADDR |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x5A4  | OTG_HS_HC<br>DMA5   | DMAADDR |      |    |        |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|        | Reset value         | 0       | 0    | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register        | 31       | 30 | 29 | 28 | 27 | 26 | 25             | 24 | 23       | 22 | 21                        | 20 | 19 | 18 | 17            | 16 | 15           | 14 | 13          | 12 | 11             | 10 | 9               | 8 | 7             | 6 | 5              | 4 | 3               | 2 | 1               | 0 |               |  |              |  |               |   |            |   |              |   |
|--------|-----------------|----------|----|----|----|----|----|----------------|----|----------|----|---------------------------|----|----|----|---------------|----|--------------|----|-------------|----|----------------|----|-----------------|---|---------------|---|----------------|---|-----------------|---|-----------------|---|---------------|--|--------------|--|---------------|---|------------|---|--------------|---|
| 0x5C4  | OTG_HS_HC DMA6  | DMAADDR  |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x5E4  | OTG_HS_HC DMA7  | DMAADDR  |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x604  | OTG_HS_HC DMA8  | DMAADDR  |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x624  | OTG_HS_HC DMA9  | DMAADDR  |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x644  | OTG_HS_HC DMA10 | DMAADDR  |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x664  | OTG_HS_HC DMA11 | DMAADDR  |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x800  | OTG_HS_DCFG     | Reserved |    |    |    |    |    | PERSCHVL<br>10 |    | Reserved |    | Reserved                  |    |    |    |               |    |              |    |             |    | PFIVL<br>00    |    | DAD<br>00000000 |   |               |   |                |   | Reserved<br>0   |   | NZLSOHSK<br>00  |   | DSPD<br>00    |  |              |  |               |   |            |   |              |   |
|        |                 |          |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
| 0x804  | OTG_HS_DCTL     | Reserved |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    | POPRGDNE<br>00 |    | CGONAK<br>00    |   | SGONAK<br>00  |   | CGINAK<br>00   |   | SGINAK<br>00    |   | TCTL<br>0000    |   |               |  | GONSTS<br>00 |  | GINSTS<br>00  |   | SDIS<br>00 |   | RWUSIG<br>00 |   |
|        |                 |          |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
| 0x808  | OTG_HS_DSTS     | Reserved |    |    |    |    |    |                |    |          |    | FNSOF<br>0000000000000000 |    |    |    |               |    |              |    |             |    | Reserved       |    |                 |   |               |   | EERR<br>00     |   | ENUMSPD<br>00   |   | SUSPSTS<br>00   |   |               |  |              |  |               |   |            |   |              |   |
|        |                 |          |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
| 0x810  | OTG_HS_DIEPMSK  | Reserved |    |    |    |    |    |                |    |          |    |                           |    |    |    | NAKMSK<br>0   |    | Reserved     |    |             |    | TXFURM<br>00   |    | Reserved<br>00  |   | INEPNEM<br>00 |   | INEPNMM<br>00  |   | ITTXFEMSK<br>00 |   | TOM<br>00       |   | AHBERRM<br>00 |  | EPDM<br>00   |  | XFRM<br>00    |   |            |   |              |   |
|        |                 |          |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
| 0x814  | OTG_HS_DOEPMSK  | Reserved |    |    |    |    |    |                |    |          |    |                           |    |    |    | NYETMSK<br>00 |    | NAKMSK<br>00 |    | BERRM<br>00 |    | Reserved       |    |                 |   | OPEM<br>00    |   | Reserved<br>00 |   | B2BSTUP<br>00   |   | STSPHSRXM<br>00 |   | OTEPDM<br>00  |  | STUPM<br>00  |  | AHBERRM<br>00 |   | EPDM<br>00 |   | XFRM<br>00   |   |
|        |                 |          |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
| 0x818  | OTG_HS_DAINT    | OEPINT   |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    | IEPINT       |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x81C  | OTG_HS_DAINTMSK | OEPM     |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    | IEPM         |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     | 0        | 0  | 0  | 0  | 0  | 0  | 0              | 0  | 0        | 0  | 0                         | 0  | 0  | 0  | 0             | 0  | 0            | 0  | 0           | 0  | 0              | 0  | 0               | 0 | 0             | 0 | 0              | 0 | 0               | 0 | 0               | 0 |               |  |              |  |               |   |            |   |              |   |
| 0x828  | OTG_HS_DVBUSDIS | Reserved |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    | VBUSDT       |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  |               |   |            |   |              |   |
|        | Reset value     |          |    |    |    |    |    |                |    |          |    |                           |    |    |    |               |    |              |    |             |    |                |    |                 |   |               |   |                |   |                 |   |                 |   |               |  |              |  | 0             | 0 | 0          | 1 | 0            | 1 |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register                    | 31       | 30    | 29      | 28             | 27    | 26       | 25      | 24 | 23 | 22 | 21    | 20       | 19    | 18     | 17         | 16       | 15        | 14   | 13    | 12           | 11     | 10     | 9        | 8       | 7       | 6         | 5   | 4       | 3    | 2     | 1 | 0        |             |
|--------|-----------------------------|----------|-------|---------|----------------|-------|----------|---------|----|----|----|-------|----------|-------|--------|------------|----------|-----------|------|-------|--------------|--------|--------|----------|---------|---------|-----------|-----|---------|------|-------|---|----------|-------------|
| 0x82C  | OTG_HS_DVB<br>USPULSE       | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              | DVBUSP |        |          |         |         |           |     |         |      |       |   |          |             |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              | 0      | 1      | 0        | 1       | 1       | 0         | 1   | 1       | 0    | 0     | 0 |          |             |
| 0x830  | OTG_HS_DTH<br>RCTL          | Reserved |       |         |                | ARPEN | Reserved | RXTHRLN |    |    |    |       |          |       |        | RXTHREN    | Reserved |           |      |       |              |        |        |          | TXTHRLN |         |           |     |         |      |       |   | ISOThREN | NONISOThREN |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         |      |       |   |          |             |
| 0x834  | OTG_HS_DIE<br>PEMPMSK       | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | INEPTXFEM |      |       |              |        |        |          |         |         |           |     |         |      |       |   |          |             |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | 0         | 0    | 0     | 0            | 0      | 0      | 0        | 0       | 0       | 0         | 0   | 0       | 0    | 0     | 0 | 0        | 0           |
| 0x838  | OTG_HS_DEA<br>CHINT         | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        | 0          | Reserved |           |      |       |              |        |        |          |         |         |           |     |         |      |       |   | 0        | Reserved    |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         |      |       |   |          |             |
| 0x83C  | OTG_HS_DEA<br>CHINTMSK      | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        | 0          | Reserved |           |      |       |              |        |        |          |         |         |           |     |         |      |       |   | 0        | Reserved    |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         |      |       |   |          |             |
| 0x844  | OTG_HS_DIE<br>PEACHMSK1     | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      | NAKM  | Reserve<br>d | BIM    | TXFURM | Reserved | INEPNEM | INEPNMM | ITTXFEMSK | TOM | AHBERRM | EPDM | XFCRM |   |          |             |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         |      |       | 0 | 0        | 0           |
| 0x884  | OTG_HS_DO<br>EPEACHMSK<br>1 | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | NYETM     | NAKM | BERRM | Reserved     | BIM    | TXFURM | Reserved | INEPNEM | INEPNMM | ITTXFEMSK | TOM | AHBERRM | EPDM | XFCRM |   |          |             |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         |      |       | 0 | 0        | 0           |
| 0x900  | OTG_HS_DIE<br>PCTL0         | EPENA    | EPDIS | SODDFRM | SD0PID/SEVNFRM | SNAK  | CNAK     | TXFNUM  |    |    |    | STALL | Reserved | EPTYP | NAKSTS | EONUM/DPID | USBAEP   | Reserved  |      |       |              | MPSIZ  |        |          |         |         |           |     |         |      |       |   |          |             |
|        | Reset value                 | 0        | 0     | 0       | 0              | 0     | 0        |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         | 0    | 0     | 0 | 0        | 0           |
| 0x918  | OTG_HS_DTX<br>FSTS0         | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | INEPTFSAV |      |       |              |        |        |          |         |         |           |     |         |      |       |   |          |             |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | 0         | 0    | 0     | 0            | 0      | 0      | 0        | 1       | 0       | 0         | 0   | 0       | 0    | 0     | 0 | 0        | 0           |
| 0x920  | OTG_HS_DIE<br>PCTL1         | EPENA    | EPDIS | SODDFRM | SD0PID/SEVNFRM | SNAK  | CNAK     | TXFNUM  |    |    |    | Stall | Reserved | EPTYP | NAKSTS | EONUM/DPID | USBAEP   | Reserved  |      |       |              | MPSIZ  |        |          |         |         |           |     |         |      |       |   |          |             |
|        | Reset value                 | 0        | 0     | 0       | 0              | 0     | 0        |         |    |    |    |       |          |       |        |            |          |           |      |       |              |        |        |          |         |         |           |     |         | 0    | 0     | 0 | 0        | 0           |
| 0x938  | OTG_HS_DTX<br>FSTS1         | Reserved |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | INEPTFSAV |      |       |              |        |        |          |         |         |           |     |         |      |       |   |          |             |
|        | Reset value                 |          |       |         |                |       |          |         |    |    |    |       |          |       |        |            |          | 0         | 0    | 0     | 0            | 0      | 0      | 0        | 0       | 1       | 0         | 0   | 0       | 0    | 0     | 0 | 0        | 0           |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register         | 31       | 30    | 29       | 28              | 27   | 26       | 25     | 24 | 23 | 22    | 21    | 20       | 19     | 18       | 17         | 16       | 15        | 14 | 13 | 12 | 11    | 10 | 9 | 8 | 7 | 6 | 5 | 4     | 3 | 2 | 1 | 0 |   |   |
|--------|------------------|----------|-------|----------|-----------------|------|----------|--------|----|----|-------|-------|----------|--------|----------|------------|----------|-----------|----|----|----|-------|----|---|---|---|---|---|-------|---|---|---|---|---|---|
| 0x940  | OTG_HS_DIE_PCTL2 | EPENA    | EPDIS | SODDFRM  | SD0PID/SEVNFIRM | SNAK | CNAK     | TXFNUM |    |    |       | Stall | Reserved | EPTYP  | NAKSTS   | EONUM/DPID | USBAEP   | Reserved  |    |    |    | MPSIZ |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     | 0        | 0               | 0    | 0        | 0      | 0  | 0  | 0     | 0     |          | 0      | 0        | 0          | 0        | 0         |    |    |    |       |    | 0 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 |   |
| 0x958  | OTG_HS_DTX_FSTS2 | Reserved |       |          |                 |      |          |        |    |    |       |       |          |        |          |            |          | INEPTFSAV |    |    |    |       |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      |          |       |          |                 |      |          |        |    |    |       |       |          |        |          |            |          | 0         | 0  | 0  | 0  | 0     | 0  | 1 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x960  | OTG_HS_DIE_PCTL3 | EPENA    | EPDIS | SODDFRM  | SD0PID/SEVNFIRM | SNAK | CNAK     | TXFNUM |    |    |       | Stall | Reserved | EPTYP  | NAKSTS   | EONUM/DPID | USBAEP   | Reserved  |    |    |    | MPSIZ |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     | 0        | 0               | 0    | 0        | 0      | 0  | 0  | 0     | 0     |          | 0      | 0        | 0          | 0        | 0         |    |    |    |       |    | 0 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 |   |
| 0x978  | OTG_HS_DTX_FSTS3 | Reserved |       |          |                 |      |          |        |    |    |       |       |          |        |          |            |          | INEPTFSAV |    |    |    |       |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      |          |       |          |                 |      |          |        |    |    |       |       |          |        |          |            |          | 0         | 0  | 0  | 0  | 0     | 0  | 1 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 | 0 |
| 0x980  | OTG_HS_DIE_PCTL4 | EPENA    | EPDIS | SODDFRM  | SD0PID/SEVNFIRM | SNAK | CNAK     | TXFNUM |    |    |       | Stall | Reserved | EPTYP  | NAKSTS   | EONUM/DPID | USBAEP   | Reserved  |    |    |    | MPSIZ |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     | 0        | 0               | 0    | 0        | 0      | 0  | 0  | 0     | 0     |          | 0      | 0        | 0          | 0        | 0         |    |    |    |       |    | 0 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 |   |
| 0x9A0  | OTG_HS_DIE_PCTL5 | EPENA    | EPDIS | SODDFRM  | SD0PID/SEVNFIRM | SNAK | CNAK     | TXFNUM |    |    |       | STALL | Reserved | EPTYP  | NAKSTS   | EONUM/DPID | USBAEP   | Reserved  |    |    |    | MPSIZ |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     | 0        | 0               | 0    | 0        | 0      | 0  | 0  | 0     | 0     |          | 0      | 0        | 0          | 0        | 0         |    |    |    |       |    | 0 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 |   |
| 0x9C0  | OTG_HS_DIE_PCTL6 | EPENA    | EPDIS | SODDFRM  | SD0PID/SEVNFIRM | SNAK | CNAK     | TXFNUM |    |    |       | STALL | Reserved | EPTYP  | NAKSTS   | EONUM/DPID | USBAEP   | Reserved  |    |    |    | MPSIZ |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     | 0        | 0               | 0    | 0        | 0      | 0  | 0  | 0     | 0     |          | 0      | 0        | 0          | 0        | 0         |    |    |    |       |    | 0 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 |   |
| 0x9E0  | OTG_HS_DIE_PCTL7 | EPENA    | EPDIS | SODDFRM  | SD0PID/SEVNFIRM | SNAK | CNAK     | TXFNUM |    |    |       | STALL | Reserved | EPTYP  | NAKSTS   | EONUM/DPID | USBAEP   | Reserved  |    |    |    | MPSIZ |    |   |   |   |   |   |       |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     | 0        | 0               | 0    | 0        | 0      | 0  | 0  | 0     | 0     |          | 0      | 0        | 0          | 0        | 0         |    |    |    |       |    | 0 | 0 | 0 | 0 | 0 | 0     | 0 | 0 | 0 | 0 | 0 |   |
| 0xB00  | OTG_HS_DO_EPCTL0 | EPENA    | EPDIS | Reserved | SNAK            | CNAK | Reserved |        |    |    | STALL | SNPM  | EPTYP    | NAKSTS | Reserved | USBAEP     | Reserved |           |    |    |    |       |    |   |   |   |   |   | MPSIZ |   |   |   |   |   |   |
|        | Reset value      | 0        | 0     |          | 0               | 0    |          |        |    |    | 0     | 0     | 0        | 0      | 0        | 1          |          |           |    |    |    |       |    |   |   |   |   |   | 0     | 0 |   |   |   |   |   |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register            | 31       | 30    | 29      | 28              | 27   | 26   | 25       | 24 | 23 | 22 | 21    | 20   | 19    | 18 | 17     | 16         | 15     | 14       | 13  | 12       | 11 | 10       | 9          | 8    | 7      | 6      | 5      | 4   | 3      | 2      | 1    | 0 |   |
|--------|---------------------|----------|-------|---------|-----------------|------|------|----------|----|----|----|-------|------|-------|----|--------|------------|--------|----------|-----|----------|----|----------|------------|------|--------|--------|--------|-----|--------|--------|------|---|---|
| 0xB20  | OTG_HS_DO<br>EPCTL1 | EPENA    | EPDIS | SODDFRM | SD0PID/SEVNFIRM | SNAK | CNAK | Reserved |    |    |    | STALL | SNPM | EPTYP |    | NAKSTS | EONUM/DPID | USBAEP | Reserved |     |          |    | MPSIZ    |            |      |        |        |        |     |        |        |      |   |   |
|        | Reset value         | 0        | 0     | 0       | 0               | 0    | 0    |          |    |    |    | 0     | 0    | 0     | 0  | 0      | 0          | 0      |          |     |          |    | 0        | 0          | 0    | 0      | 0      | 0      | 0   | 0      | 0      | 0    | 0 | 0 |
| 0xB40  | OTG_HS_DO<br>EPCTL2 | EPENA    | EPDIS | SODDFRM | SD0PID/SEVNFIRM | SNAK | CNAK | Reserved |    |    |    | Stall | SNPM | EPTYP |    | NAKSTS | EONUM/DPID | USBAEP | Reserved |     |          |    | MPSIZ    |            |      |        |        |        |     |        |        |      |   |   |
|        | Reset value         | 0        | 0     | 0       | 0               | 0    | 0    |          |    |    |    | 0     | 0    | 0     | 0  | 0      | 0          | 0      | 0        |     |          |    |          | 0          | 0    | 0      | 0      | 0      | 0   | 0      | 0      | 0    | 0 | 0 |
| 0xB60  | OTG_HS_DO<br>EPCTL3 | EPENA    | EPDIS | SODDFRM | SD0PID/SEVNFIRM | SNAK | CNAK | Reserved |    |    |    | Stall | SNPM | EPTYP |    | NAKSTS | EONUM/DPID | USBAEP | Reserved |     |          |    | MPSIZ    |            |      |        |        |        |     |        |        |      |   |   |
|        | Reset value         | 0        | 0     | 0       | 0               | 0    | 0    |          |    |    |    | 0     | 0    | 0     | 0  | 0      | 0          | 0      | 0        |     |          |    |          | 0          | 0    | 0      | 0      | 0      | 0   | 0      | 0      | 0    | 0 | 0 |
| 0x908  | OTG_HS_DIE<br>PINT0 | Reserved |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | NAK | Reserved |    | Reserved | TXFIFOUDRN | TXFE | INEPNE | INEPNM | ITTXFE | TOC | AHBERR | EPDISD | XFRC |   |   |
|        | Reset value         |          |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | 0   | 0        |    |          |            |      |        |        |        |     |        |        |      |   |   |
| 0x928  | OTG_HS_DIE<br>PINT1 | Reserved |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | NAK | Reserved |    | Reserved | TXFIFOUDRN | TXFE | INEPNE | INEPNM | ITTXFE | TOC | AHBERR | EPDISD | XFRC |   |   |
|        | Reset value         |          |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | 0   | 0        |    |          |            |      |        |        |        |     |        |        |      |   |   |
| 0x948  | OTG_HS_DIE<br>PINT2 | Reserved |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | NAK | BERR     |    | Reserved | TXFIFOUDRN | TXFE | INEPNE | INEPNM | ITTXFE | TOC | AHBERR | EPDISD | XFRC |   |   |
|        | Reset value         |          |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | 0   | 0        |    |          |            |      |        |        |        |     |        |        |      |   |   |
| 0x968  | OTG_HS_DIE<br>PINT3 | Reserved |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | NAK | Reserved |    | Reserved | TXFIFOUDRN | TXFE | INEPNE | INEPNM | ITTXFE | TOC | AHBERR | EPDISD | XFRC |   |   |
|        | Reset value         |          |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | 0   | 0        | 0  |          |            |      |        |        |        |     |        |        |      |   |   |
| 0x988  | OTG_HS_DIE<br>PINT4 | Reserved |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | NAK | Reserved |    | Reserved | TXFIFOUDRN | TXFE | INEPNE | INEPNM | ITTXFE | TOC | AHBERR | EPDISD | XFRC |   |   |
|        | Reset value         |          |       |         |                 |      |      |          |    |    |    |       |      |       |    |        |            |        |          | 0   | 0        |    |          |            |      |        |        |        |     |        |        |      |   |   |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register            | 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12   | 11       | 10       | 9        | 8          | 7        | 6       | 5        | 4       | 3    | 2      | 1      | 0    |
|--------|---------------------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|----------|----------|----------|------------|----------|---------|----------|---------|------|--------|--------|------|
| 0x9A8  | OTG_HS_DIE<br>PINT5 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NAK  | Reserved | Reserved | Reserved | TXFIFOUDRN | TXFE     | INEPNE  | INENPM   | ITTXFE  | TOC  | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 1        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0x9C8  | OTG_HS_DIE<br>PINT6 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NAK  | Reserved | Reserved | Reserved | TXFIFOUDRN | TXFE     | INEPNE  | INENPM   | ITTXFE  | TOC  | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 1        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0x9E8  | OTG_HS_DIE<br>PINT7 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NAK  | Reserved | Reserved | Reserved | TXFIFOUDRN | TXFE     | INEPNE  | INENPM   | ITTXFE  | TOC  | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 1        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0xB08  | OTG_HS_DO<br>EPINT0 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET | NAK      | BERR     | Reserved | OUTPKTERR  | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 0        | 0       | 0        | 0       | 1    | 0      | 0      | 0    |
| 0xB28  | OTG_HS_DO<br>EPINT1 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET | NAK      | BERR     | Reserved | OUTPKTERR  | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0xB48  | OTG_HS_DO<br>EPINT2 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET | NAK      | BERR     | Reserved | OUTPKTERR  | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0xB68  | OTG_HS_DO<br>EPINT3 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET | NAK      | BERR     | Reserved | OUTPKTERR  | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0xB88  | OTG_HS_DO<br>EPINT4 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET | NAK      | BERR     | Reserved | OUTPKTERR  | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |
| 0xBA8  | OTG_HS_DO<br>EPINT5 | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | NYET | NAK      | BERR     | Reserved | OUTPKTERR  | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |
|        | Reset value         |          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0    | 0        | 0        | 0        | 0          | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |

Table 215. OTG\_HS register map and reset values (continued)

| Offset | Register          | 31       | 30              | 29 | 28       | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19      | 18      | 17       | 16       | 15 | 14 | 13   | 12  | 11   | 10       | 9 | 8         | 7        | 6       | 5        | 4       | 3    | 2      | 1      | 0    |   |
|--------|-------------------|----------|-----------------|----|----------|----|----|----|----|----|----|----|----|---------|---------|----------|----------|----|----|------|-----|------|----------|---|-----------|----------|---------|----------|---------|------|--------|--------|------|---|
| 0xBC8  | OTG_HS_DO EPINT6  | Reserved |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    | NYET | NAK | BERR | Reserved |   | OUTPKTERR | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |   |
|        | Reset value       |          |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    | 0    | 0   | 0    |          |   | 0         |          | 0       |          | 0       | 0    | 0      | 0      | 0    | 0 |
| 0xBE8  | OTG_HS_DO EPINT7  | Reserved |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    | NYET | NAK | BERR | Reserved |   | OUTPKTERR | Reserved | B2BSTUP | Reserved | OTEPDIS | STUP | AHBERR | EPDISD | XFRC |   |
|        | Reset value       |          |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    | 0    | 0   | 0    |          |   | 0         |          | 0       |          | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x910  | OTG_HS_DIE PTSIZ0 | Reserved |                 |    |          |    |    |    |    |    |    |    |    | PKT CNT |         | Reserved |          |    |    |      |     |      |          |   |           | XFRSIZ   |         |          |         |      |        |        |      |   |
|        | Reset value       |          |                 |    |          |    |    |    |    |    |    |    |    | 0       | 0       |          |          |    |    |      |     |      |          |   |           | 0        | 0       | 0        | 0       | 0    | 0      | 0      |      |   |
| 0x930  | OTG_HS_DIE PTSIZ1 | Reserved | MCN T           |    | PKTCNT   |    |    |    |    |    |    |    |    |         | XFRSIZ  |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       |          | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      |      |   |
| 0x934  | OTG_HS_DIE PDMA1  | DMAADDR  |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    |   |
| 0x93C  | OTG_HS_DIE PDMAB1 | DMABADDR |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x950  | OTG_HS_DIE PTSIZ2 | Reserved | MCN T           |    | PKTCNT   |    |    |    |    |    |    |    |    |         | XFRSIZ  |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       |          | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x954  | OTG_HS_DIE PDMA2  | DMAADDR  |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x95C  | OTG_HS_DIE PDMAB2 | DMABADDR |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x970  | OTG_HS_DIE PTSIZ3 | Reserved | MCN T           |    | PKTCNT   |    |    |    |    |    |    |    |    |         | XFRSIZ  |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       |          | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x974  | OTG_HS_DIE PDMA3  | DMAADDR  |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0x97C  | OTG_HS_DIE PDMAB3 | DMABADDR |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0xB10  | OTG_HS_DO EPTSIZ0 | Reserved | STU PCN T       |    | Reserved |    |    |    |    |    |    |    |    |         | PKT CNT |          | Reserved |    |    |      |     |      |          |   |           |          | XFRSIZ  |          |         |      |        |        |      |   |
|        | Reset value       |          | 0               | 0  |          |    |    |    |    |    |    |    |    |         | 0       |          |          |    |    |      |     |      |          |   |           | 0        | 0       | 0        | 0       | 0    | 0      | 0      |      |   |
| 0xB30  | OTG_HS_DO EPTSIZ1 | Reserved | RXDPID/ STUPCNT |    | PKTCNT   |    |    |    |    |    |    |    |    |         | XFRSIZ  |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       |          | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |
| 0xB34  | OTG_HS_DO EPDMA1  | DMAADDR  |                 |    |          |    |    |    |    |    |    |    |    |         |         |          |          |    |    |      |     |      |          |   |           |          |         |          |         |      |        |        |      |   |
|        | Reset value       | 0        | 0               | 0  | 0        | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0       | 0       | 0        | 0        | 0  | 0  | 0    | 0   | 0    | 0        | 0 | 0         | 0        | 0       | 0        | 0       | 0    | 0      | 0      | 0    | 0 |

**Table 215. OTG\_HS register map and reset values (continued)**

| Offset | Register             | 31       | 30                 | 29     | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17     | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3       | 2 | 1        | 0 |          |  |         |  |
|--------|----------------------|----------|--------------------|--------|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|---|---|---|---|---|---|---------|---|----------|---|----------|--|---------|--|
| 0xB3C  | OTG_HS_DO<br>EPDMAB1 | DMABADDR |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          | 0        | 0                  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xB50  | OTG_HS_DO<br>EPTSIZ2 | Reserved | RXDPID/<br>STUPCNT | PKTCNT |    |    |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          |          |                    | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xB54  | OTG_HS_DO<br>EPDMA2  | DMAADDR  |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          | 0        | 0                  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xB5C  | OTG_HS_DO<br>EPDMAB2 | DMABADDR |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          | 0        | 0                  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xB70  | OTG_HS_DO<br>EPTSIZ3 | Reserved | RXDPID/<br>STUPCNT | PKTCNT |    |    |    |    |    |    |    |    |    |    |    | XFRSIZ |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          |          |                    | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xB74  | OTG_HS_DO<br>EPDMA3  | DMAADDR  |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          | 0        | 0                  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xB7C  | OTG_HS_DO<br>EPDMAB3 | DMABADDR |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |
|        | Reset value          | 0        | 0                  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0       | 0 | 0        | 0 |          |  |         |  |
| 0xE00  | OTG_HS_PC<br>GCCTL   | Reserved |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   | PHYSUSP |   | Reserved |   | GATEHCLK |  | STPPCLK |  |
|        | Reset value          |          |                    |        |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |   |   |   |   |   |   |         |   |          |   |          |  |         |  |

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

### 35.13 OTG\_HS programming model

### 35.13.1 Core initialization

The application must perform the core initialization sequence. If the cable is connected during power-up, the current mode of operation bit in the Core interrupt register (CMOD bit in OTG\_HS\_GINTSTS) reflects the mode. The OTG\_HS controller enters host mode when an “A” plug is connected or peripheral mode when a “B” plug is connected.

This section explains the initialization of the OTG\_HS controller after power-on. The application must follow the initialization sequence irrespective of host or peripheral mode operation. All core global registers are initialized according to the core's configuration:

1. Program the following fields in the Global AHB configuration (OTG\_HS\_GAHBCFG) register:
  - DMA mode bit
  - AHB burst length field
  - Global interrupt mask bit GINT = 1
  - RxFIFO nonempty (RXFLVL bit in OTG\_HS\_GINTSTS)
  - Periodic TxFIFO empty level
2. Program the following fields in OTG\_HS\_GUSBCFG register:
  - HNP capable bit
  - SRP capable bit
  - FS timeout calibration field
  - USB turnaround time field
3. The software must unmask the following bits in the GINTMSK register:  
OTG interrupt mask  
Mode mismatch interrupt mask
4. The software can read the CMOD bit in OTG\_HS\_GINTSTS to determine whether the OTG\_HS controller is operating in host or peripheral mode.

### 35.13.2 Host initialization

To initialize the core as host, the application must perform the following steps:

1. Program the HPRTINT in GINTMSK to unmask
2. Program the OTG\_HS\_HCFG register to select full-speed host
3. Program the PPWR bit in OTG\_HS\_HPRT to 1. This drives  $V_{BUS}$  on the USB.
4. Wait for the PCDET interrupt in OTG\_HS\_HPRT0. This indicates that a device is connecting to the port.
5. Program the PRST bit in OTG\_HS\_HPRT to 1. This starts the reset process.
6. Wait at least 10 ms for the reset process to complete.
7. Program the PRST bit in OTG\_HS\_HPRT to 0.
8. Wait for the PENCHNG interrupt in OTG\_HS\_HPRT.
9. Read the PSPD bit in OTG\_HS\_HPRT to get the enumerated speed.
10. Program the HFIR register with a value corresponding to the selected PHY clock 1.
11. Program the FLSPCS field in OTG\_HS\_HCFG register according to the speed of the detected device read in step 9. If FLSPCS has been changed, reset the port.
12. Program the OTG\_HS\_GRXFSIZ register to select the size of the receive FIFO.
13. Program the OTG\_HS\_GNPTXFSIZ register to select the size and the start address of the nonperiodic transmit FIFO for nonperiodic transactions.
14. Program the OTG\_HS\_HPTXFSIZ register to select the size and start address of the periodic transmit FIFO for periodic transactions.

To communicate with devices, the system software must initialize and enable at least one channel.

### 35.13.3 Device initialization

The application must perform the following steps to initialize the core as a device on power-up or after a mode change from host to device.

1. Program the following fields in the OTG\_HS\_DCFG register:
  - Device speed
  - Nonzero-length status OUT handshake
2. Program the OTG\_HS\_GINTMSK register to unmask the following interrupts:
  - USB reset
  - Enumeration done
  - Early suspend
  - USB suspend
  - SOF
3. Program the VBUSSEN bit in the OTG\_HS\_GCCFG register to enable  $V_{BUS}$  sensing in “B” peripheral mode and supply the 5 volts across the pull-up resistor on the DP line.
4. Wait for the USBRST interrupt in OTG\_HS\_GINTSTS. It indicates that a reset has been detected on the USB that lasts for about 10 ms on receiving this interrupt.

Wait for the ENUMDNE interrupt in OTG\_HS\_GINTSTS. This interrupt indicates the end of reset on the USB. On receiving this interrupt, the application must read the OTG\_HS\_DSTS register to determine the enumeration speed and perform the steps listed in [Endpoint initialization on enumeration completion on page 1504](#).

At this point, the device is ready to accept SOF packets and perform control transfers on control endpoint 0.

### 35.13.4 DMA mode

The OTG host uses the AHB master interface to fetch the transmit packet data (AHB to USB) and receive the data update (USB to AHB). The AHB master uses the programmed DMA address (HCDMAx register in host mode and DIEPDMAx/DOEPDMAx register in peripheral mode) to access the data buffers.

### 35.13.5 Host programming model

#### Channel initialization

The application must initialize one or more channels before it can communicate with connected devices. To initialize and enable a channel, the application must perform the following steps:

1. Program the GINTMSK register to unmask the following:
2. Channel interrupt
  - Nonperiodic transmit FIFO empty for OUT transactions (applicable for Slave mode that operates in pipelined transaction-level with the packet count field programmed with more than one).
  - Nonperiodic transmit FIFO half-empty for OUT transactions (applicable for Slave mode that operates in pipelined transaction-level with the packet count field programmed with more than one).
3. Program the OTG\_HS\_HAINTMSK register to unmask the selected channels' interrupts.
4. Program the OTG\_HS\_HCINTMSK register to unmask the transaction-related interrupts of interest given in the host channel interrupt register.
5. Program the selected channel's OTG\_HS\_HCTSIZx register with the total transfer size, in bytes, and the expected number of packets, including short packets. The application must program the PID field with the initial data PID (to be used on the first OUT transaction or to be expected from the first IN transaction).
6. Program the selected channels in the OTG\_HS\_HCSPLTx register(s) with the hub and port addresses (split transactions only).
7. Program the selected channels in the HCDMAx register(s) with the buffer start address.
8. Program the OTG\_HS\_HCCHARx register of the selected channel with the device's endpoint characteristics, such as type, speed, direction, and so forth. (The channel can be enabled by setting the channel enable bit to 1 only when the application is ready to transmit or receive any packet).

### Halting a channel

The application can disable any channel by programming the OTG\_HS\_HCCHARx register with the CHDIS and CHENA bits set to 1. This enables the OTG\_HS host to flush the posted requests (if any) and generates a channel halted interrupt. The application must wait for the CHH interrupt in OTG\_HS\_HCINTx before reallocating the channel for other transactions. The OTG\_HS host does not interrupt the transaction that has already been started on the USB.

To disable a channel in DMA mode operation, the application does not need to check for space in the request queue. The OTG\_HS host checks for space to write the disable request on the disabled channel's turn during arbitration. Meanwhile, all posted requests are dropped from the request queue when the CHDIS bit in HCCHARx is set to 1.

Before disabling a channel, the application must ensure that there is at least one free space available in the nonperiodic request queue (when disabling a nonperiodic channel) or the periodic request queue (when disabling a periodic channel). The application can simply flush the posted requests when the Request queue is full (before disabling the channel), by programming the OTG\_HS\_HCCHARx register with the CHDIS bit set to 1, and the CHENA bit cleared to 0.

The application is expected to disable a channel on any of the following conditions:

1. When an XFRC interrupt in OTG\_HS\_HCINTx is received during a nonperiodic IN transfer or high-bandwidth interrupt IN transfer (Slave mode only)
2. When an STALL, TXERR, BBERR or DTERR interrupt in OTG\_HS\_HCINTx is received for an IN or OUT channel (Slave mode only). For high-bandwidth interrupt INs in Slave mode, once the application has received a DTERR interrupt it must disable the

channel and wait for a channel halted interrupt. The application must be able to receive other interrupts (DTERR, NAK, Data, TXERR) for the same channel before receiving the halt.

3. When a DISCINT (Disconnect Device) interrupt in OTG\_HS\_GINTSTS is received. (The application is expected to disable all enabled channels)
4. When the application aborts a transfer before normal completion.

### Ping protocol

When the OTG\_HS host operates in high speed, the application must initiate the ping protocol when communicating with high-speed bulk or control (data and status stage) OUT endpoints.

The application must initiate the ping protocol when it receives a NAK/NYET/TXERR interrupt. When the HS\_OTG host receives one of the above responses, it does not continue any transaction for a specific endpoint, drops all posted or fetched OUT requests (from the request queue), and flushes the corresponding data (from the transmit FIFO).

This is valid in slave mode only. In Slave mode, the application can send a ping token either by setting the DOPING bit in HCTSIZx before enabling the channel or by just writing the HCTSIZx register with the DOPING bit set when the channel is already enabled. This enables the HS\_OTG host to write a ping request entry to the request queue. The application must wait for the response to the ping token (a NAK, ACK, or TXERR interrupt) before continuing the transaction or sending another ping token. The application can continue the data transaction only after receiving an ACK from the OUT endpoint for the requested ping. In DMA mode operation, the application does not need to set the DOPING bit in HCTSIZx for a NAK/NYET response in case of Bulk/Control OUT. The OTG\_HS host automatically sets the DOPING bit in HCTSIZx, and issues the ping tokens for Bulk/Control OUT. The HS\_OTG host continues sending ping tokens until it receives an ACK, and then switches automatically to the data transaction.

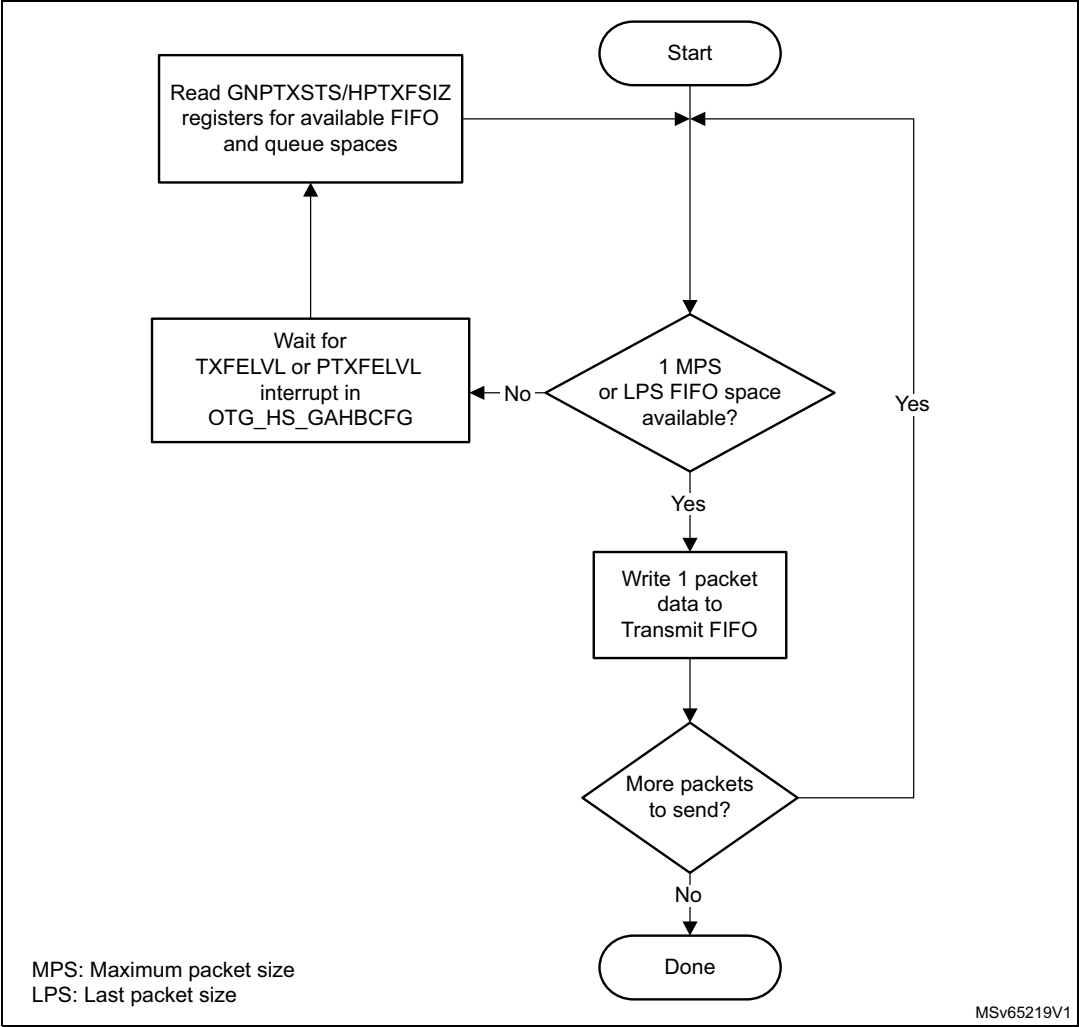
### Operational model

The application must initialize a channel before communicating to the connected device. This section explains the sequence of operation to be performed for different types of USB transactions.

- **Writing the transmit FIFO**

The OTG\_HS host automatically writes an entry (OUT request) to the periodic/nonperiodic request queue, along with the last word write of a packet. The application must ensure that at least one free space is available in the periodic/nonperiodic request queue before starting to write to the transmit FIFO. The application must always write to the transmit FIFO in words. If the packet size is non-word-aligned, the application must use padding. The OTG\_HS host determines the actual packet size based on the programmed maximum packet size and transfer size.

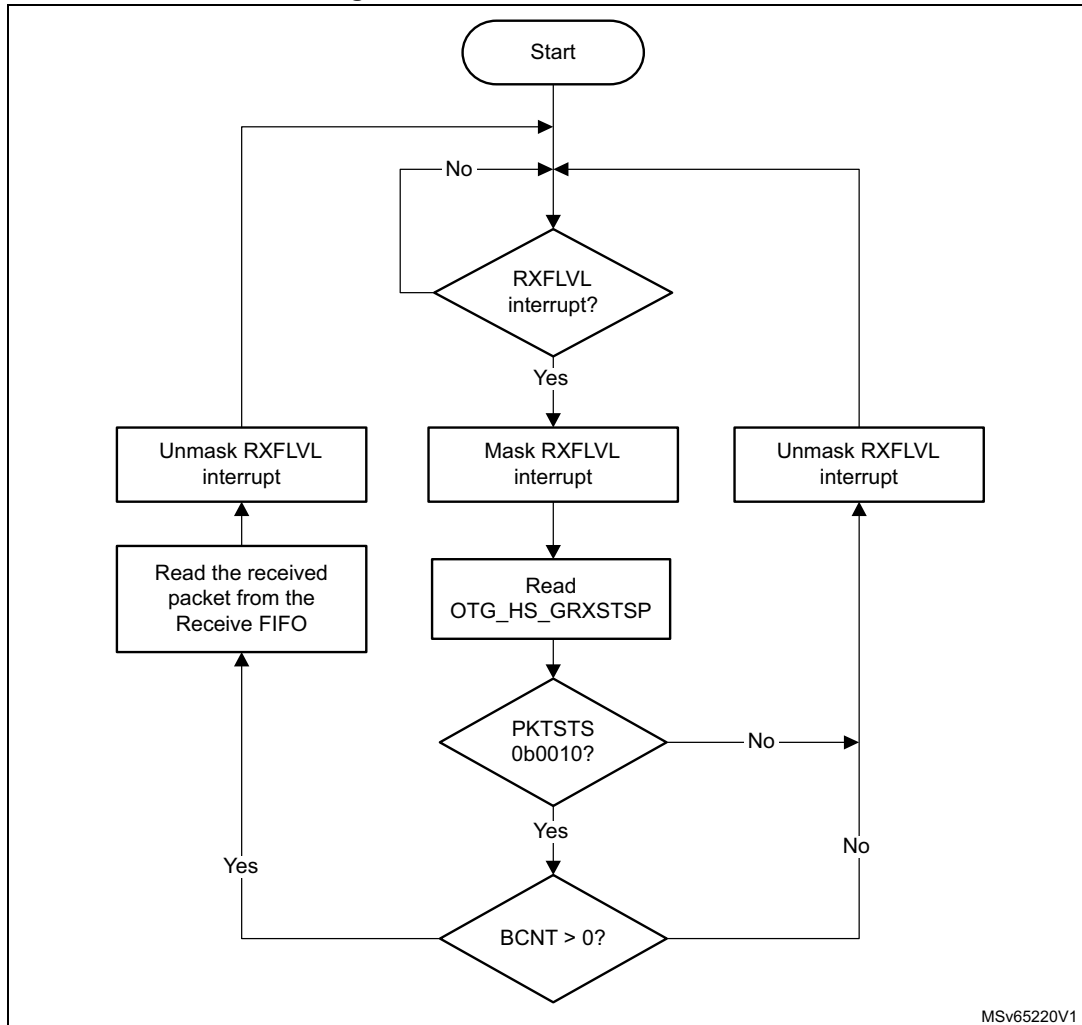
Figure 416. Transmit FIFO write task



- **Reading the receive FIFO**

The application must ignore all packet statuses other than IN data packet (bx0010).

**Figure 417. Receive FIFO read task**



- **Bulk and control OUT/SETUP transactions**

A typical bulk or control OUT/SETUP pipelined transaction-level operation is shown in [Figure 418](#). See channel 1 (ch\_1). Two bulk OUT packets are transmitted. A control SETUP transaction operates in the same way but has only one packet. The assumptions are:

- The application is attempting to send two maximum-packet-size packets (transfer size = 1,024 bytes).
- The nonperiodic transmit FIFO can hold two packets (128 bytes for FS).
- The nonperiodic request queue depth = 4.

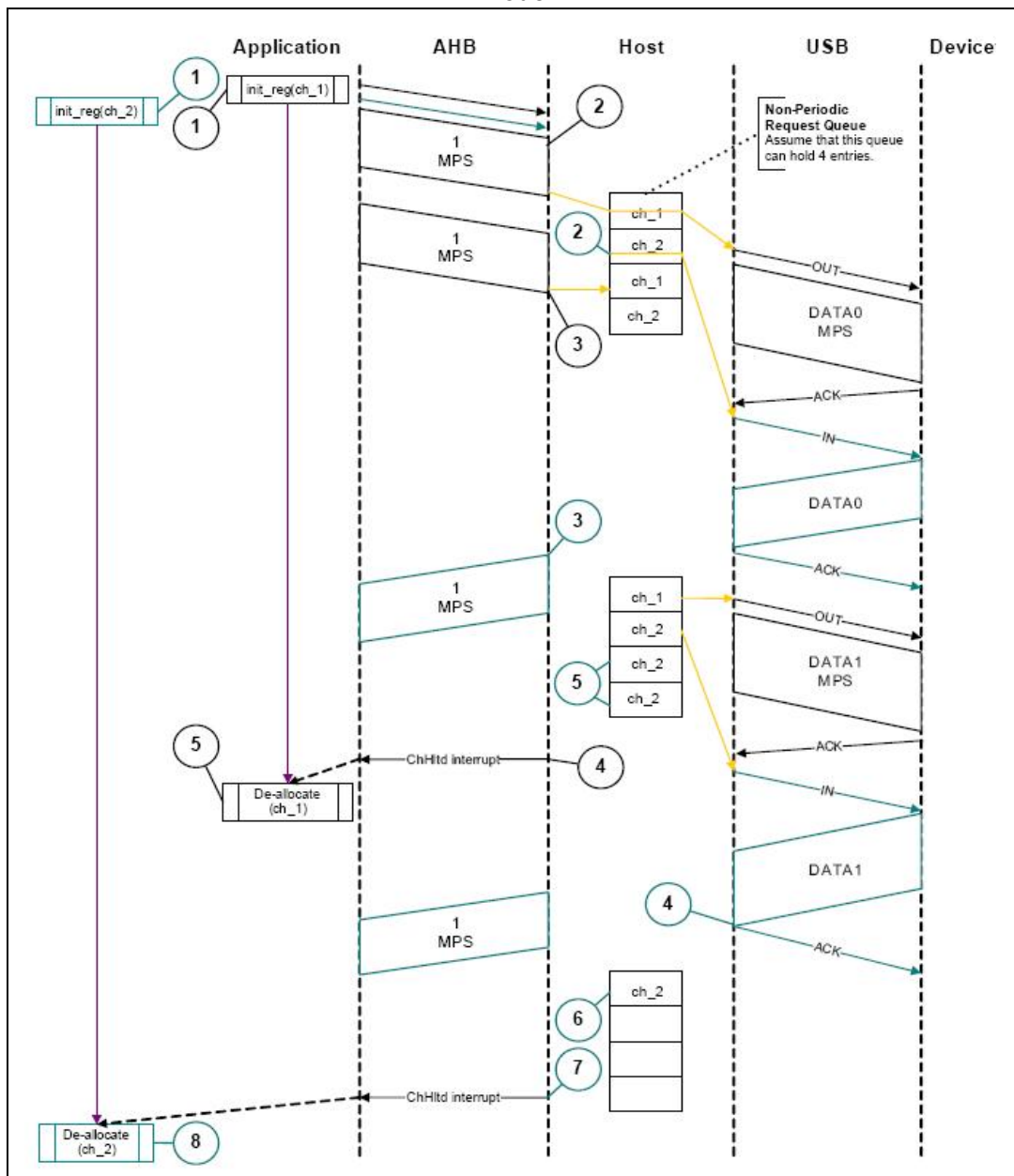
- **Normal bulk and control OUT/SETUP operations**

The sequence of operations for channel 1 is as follows:

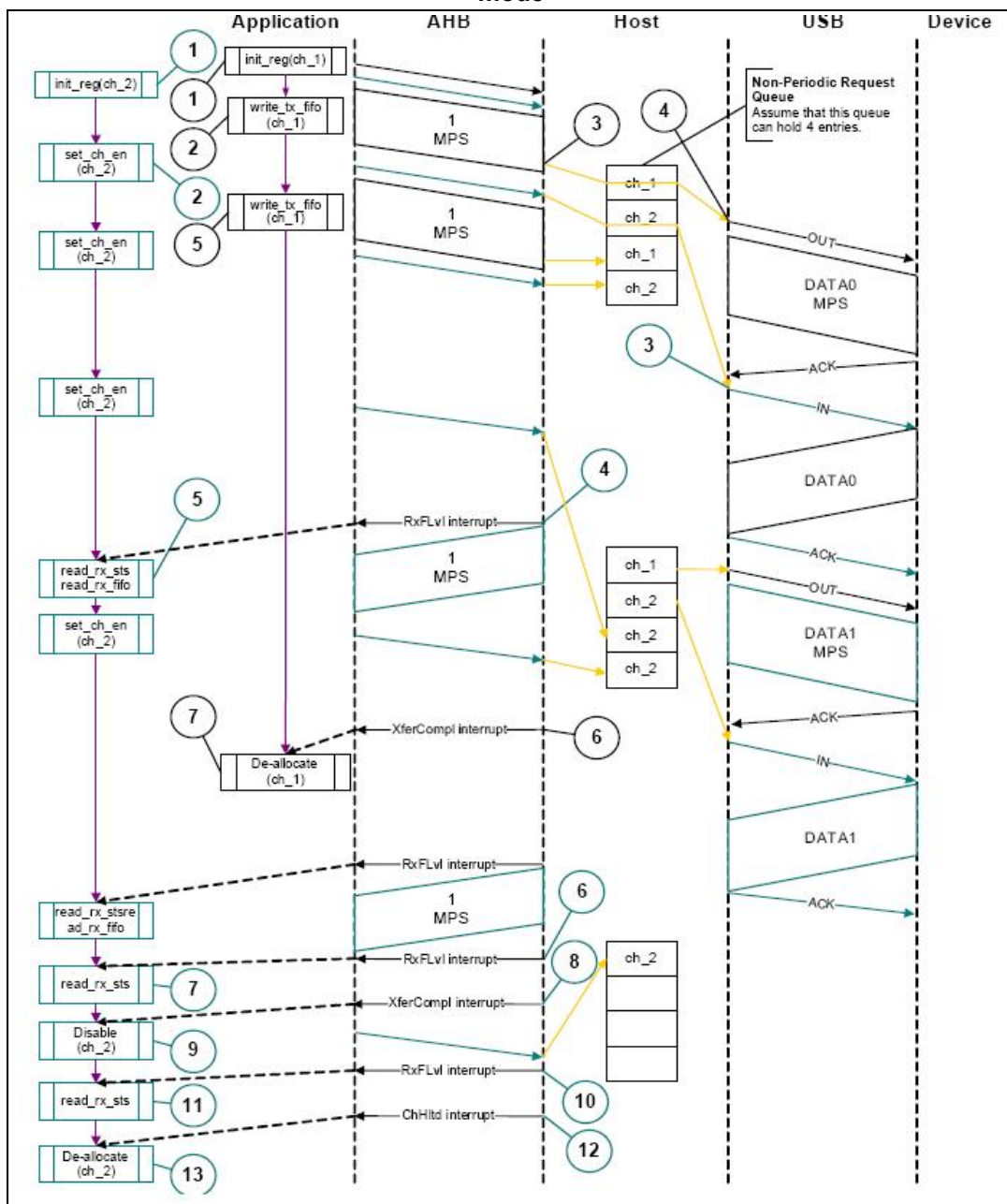
- Initialize channel 1
- Write the first packet for channel 1

- c) Along with the last word write, the core writes an entry to the nonperiodic request queue
- d) As soon as the nonperiodic queue becomes nonempty, the core attempts to send an OUT token in the current frame
- e) Write the second (last) packet for channel 1
- f) The core generates the XFRC interrupt as soon as the last transaction is completed successfully
- g) In response to the XFRC interrupt, de-allocate the channel for other transfers
- h) Handling nonACK responses

**Figure 418. Normal bulk/control OUT/SETUP and bulk/control IN transactions - DMA mode**



**Figure 419. Normal bulk/control OUT/SETUP and bulk/control IN transactions - Slave mode**



The channel-specific interrupt service routine for bulk and control OUT/SETUP transactions in Slave mode is shown in the following code samples.

- Interrupt service routine for bulk/control OUT/SETUP and bulk/control IN transactions**

a) Bulk/Control OUT/SETUP

```
Unmask (NAK/TXERR/STALL/XFRC)
if (XFRC)
{
    Reset Error Count
}
```

```

    Mask ACK
    De-allocate Channel
}
else if (STALL)
{
    Transfer Done = 1
    Unmask CHH
    Disable Channel
}
else if (NAK or TXERR )
{
    Rewind Buffer Pointers
    Unmask CHH
    Disable Channel
    if (TXERR)
    {
        Increment Error Count
        Unmask ACK
    }
    else
    {
        Reset Error Count
    }
}
else if (CHH)
{
    Mask CHH
    if (Transfer Done or (Error_count == 3))
    {
        De-allocate Channel
    }
    else
    {
        Re-initialize Channel
    }
}
else if (ACK)
{
    Reset Error Count
    Mask ACK
}

```

The application is expected to write the data packets into the transmit FIFO as and when the space is available in the transmit FIFO and the Request queue. The application can make use of the NPTXFE interrupt in OTG\_HS\_GINTSTS to find the transmit FIFO space.

b) Bulk/Control IN

```

Unmask (TXERR/XFRC/BBERR/STALL/DTERR)
if (XFRC)
{
    Reset Error Count
    Unmask CHH
    Disable Channel
}

```

```

    Reset Error Count
    Mask ACK
}
else if (TXERR or BBERR or STALL)
{
    Unmask CHH
    Disable Channel
    if (TXERR)
    {
        Increment Error Count
        Unmask ACK
    }
}
else if (CHH)
{
    Mask CHH
    if (Transfer Done or (Error_count == 3))
    {
        De-allocate Channel
    }
    else
    {
        Re-initialize Channel
    }
}
else if (ACK)
{
    Reset Error Count
    Mask ACK
}
else if (DTERR)
{
    Reset Error Count
}

```

The application is expected to write the requests as and when the Request queue space is available and until the XFRC interrupt is received.

- **Bulk and control IN transactions**

A typical bulk or control IN pipelined transaction-level operation is shown in [Figure 420](#). See channel 2 (ch\_2). The assumptions are:

- The application is attempting to receive two maximum-packet-size packets (transfer size = 1 024 bytes).
- The receive FIFO can contain at least one maximum-packet-size packet and two status words per packet (72 bytes for FS).
- The nonperiodic request queue depth = 4.

Figure 420. Bulk/control IN transactions - DMA mode

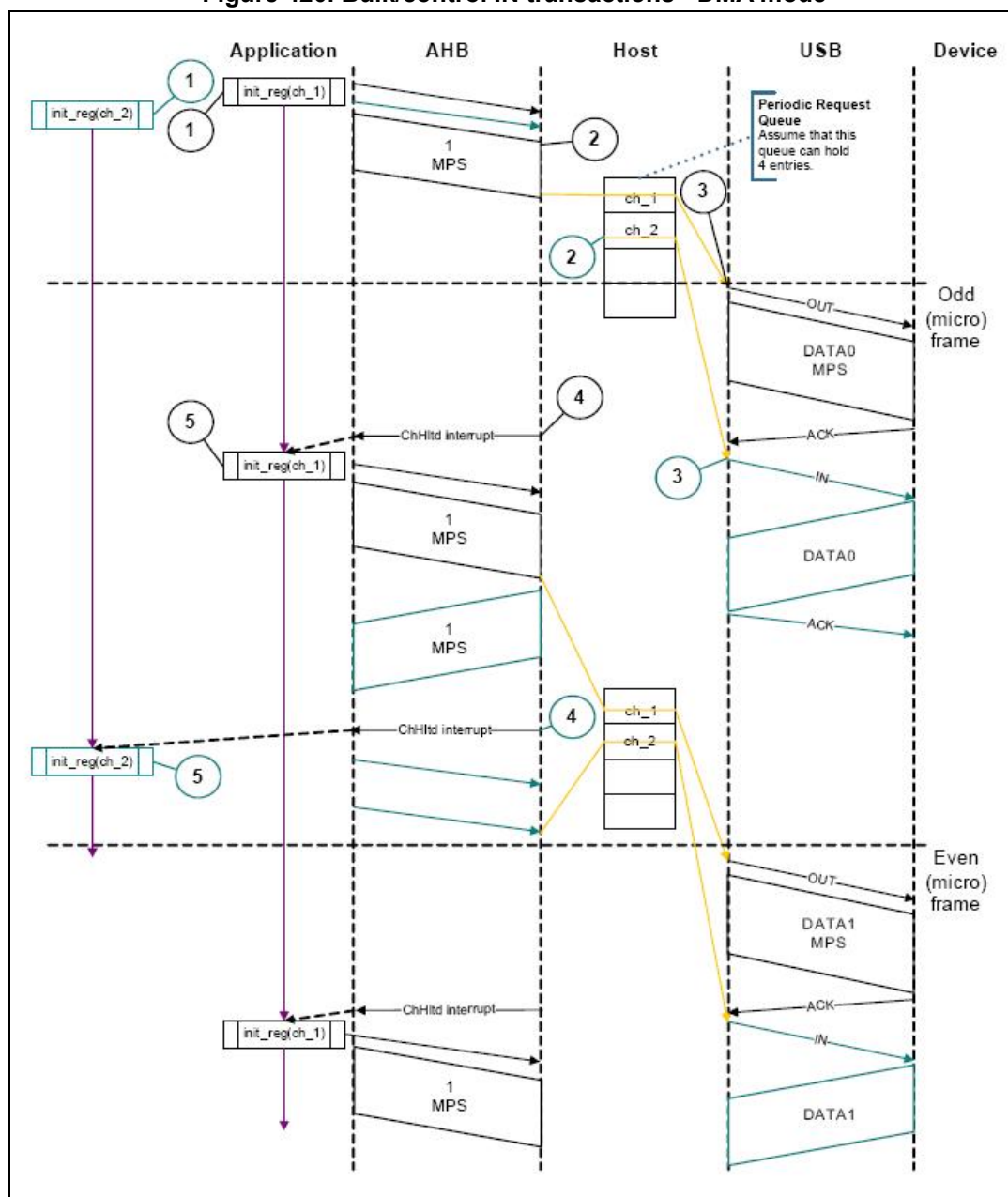
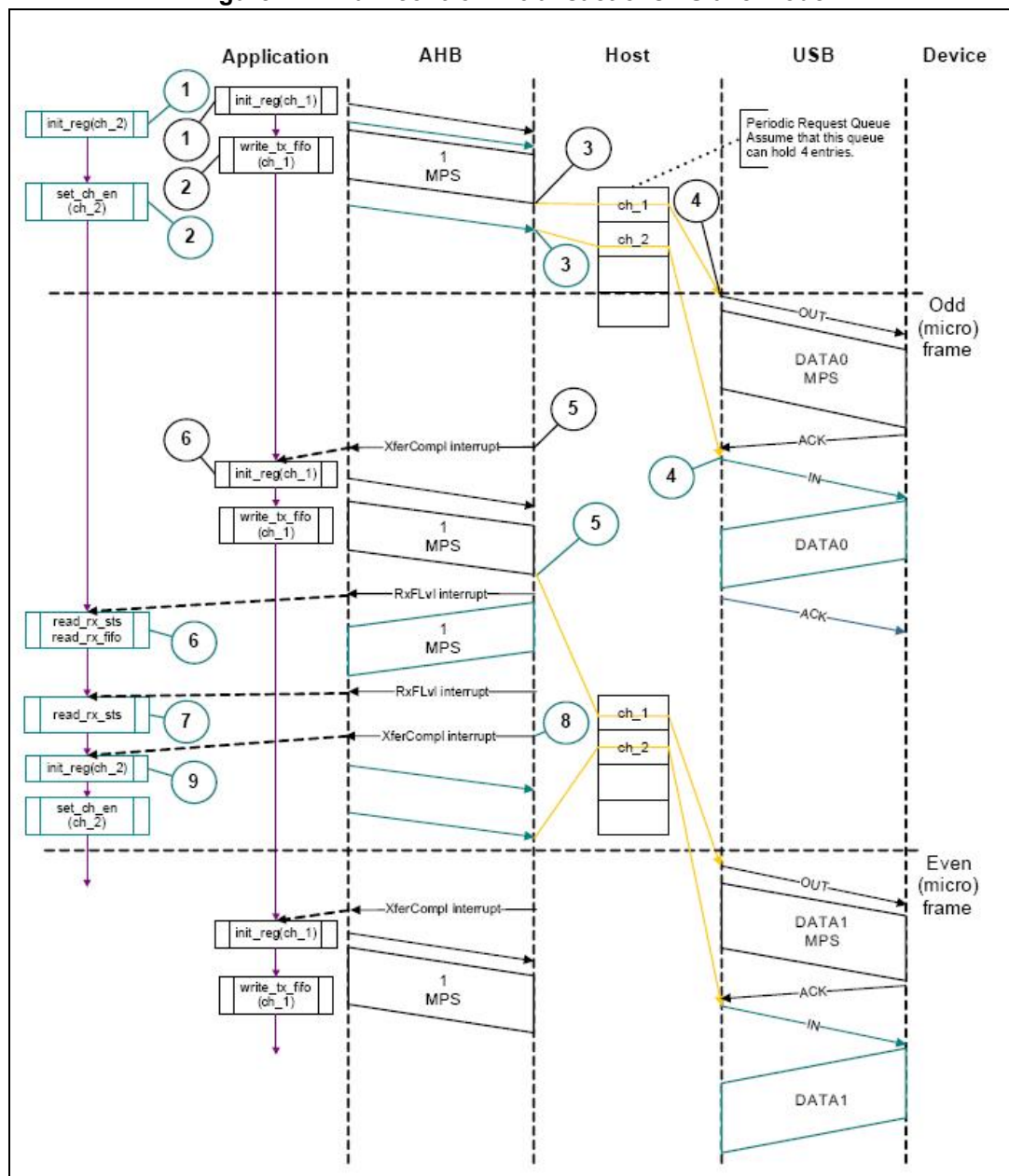


Figure 421. Bulk/control IN transactions - Slave mode



The sequence of operations is as follows:

- Initialize channel 2.
- Set the CHENA bit in HCCHAR2 to write an IN request to the nonperiodic request queue.
- The core attempts to send an IN token after completing the current OUT transaction.
- The core generates an RXFLVL interrupt as soon as the received packet is written to the receive FIFO.
- In response to the RXFLVL interrupt, mask the RXFLVL interrupt and read the received packet status to determine the number of bytes received, then read the

receive FIFO accordingly. Following this, unmask the RXFLVL interrupt.

- f) The core generates the RXFLVL interrupt for the transfer completion status entry in the receive FIFO.
- g) The application must read and ignore the receive packet status when the receive packet status is not an IN data packet (PKTSTS in GRXSTSR  $\neq$  0b0010).
- h) The core generates the XFRC interrupt as soon as the receive packet status is read.
- i) In response to the XFRC interrupt, disable the channel and stop writing the OTG\_HS\_HCCHAR2 register for further requests. The core writes a channel disable request to the nonperiodic request queue as soon as the OTG\_HS\_HCCHAR2 register is written.
- j) The core generates the RXFLVL interrupt as soon as the halt status is written to the receive FIFO.
- k) Read and ignore the receive packet status.
- l) The core generates a CHH interrupt as soon as the halt status is popped from the receive FIFO.
- m) In response to the CHH interrupt, de-allocate the channel for other transfers.
- n) Handling nonACK responses
- **Control transactions in slave mode**  
Setup, Data, and Status stages of a control transfer must be performed as three separate transfers. Setup-, Data- or Status-stage OUT transactions are performed similarly to the bulk OUT transactions explained previously. Data- or Status-stage IN transactions are performed similarly to the bulk IN transactions explained previously. For all three stages, the application is expected to set the EPTYP field in OTG\_HS\_HCCHAR1 to Control. During the Setup stage, the application is expected to set the PID field in OTG\_HS\_HCTSIZ1 to SETUP.
- **Interrupt OUT transactions**  
A typical interrupt OUT operation in Slave mode is shown in [Figure 422](#). The assumptions are:
  - The application is attempting to send one packet in every frame (up to 1 maximum packet size), starting with the odd frame (transfer size = 1 024 bytes)
  - The periodic transmit FIFO can hold one packet (1 KB)
  - Periodic request queue depth = 4
 The sequence of operations is as follows:
  - a) Initialize and enable channel 1. The application must set the ODDFRM bit in OTG\_HS\_HCCHAR1.
  - b) Write the first packet for channel 1. For a high-bandwidth interrupt transfer, the application must write the subsequent packets up to MCNT (maximum number of packets to be transmitted in the next frame times) before switching to another channel.
  - c) Along with the last word write of each packet, the OTG\_HS host writes an entry to the periodic request queue.
  - d) The OTG\_HS host attempts to send an OUT token in the next (odd) frame.
  - e) The OTG\_HS host generates an XFRC interrupt as soon as the last packet is transmitted successfully.
  - f) In response to the XFRC interrupt, reinitialize the channel for the next transfer.

Figure 422. Normal interrupt OUT/IN transactions - DMA mode

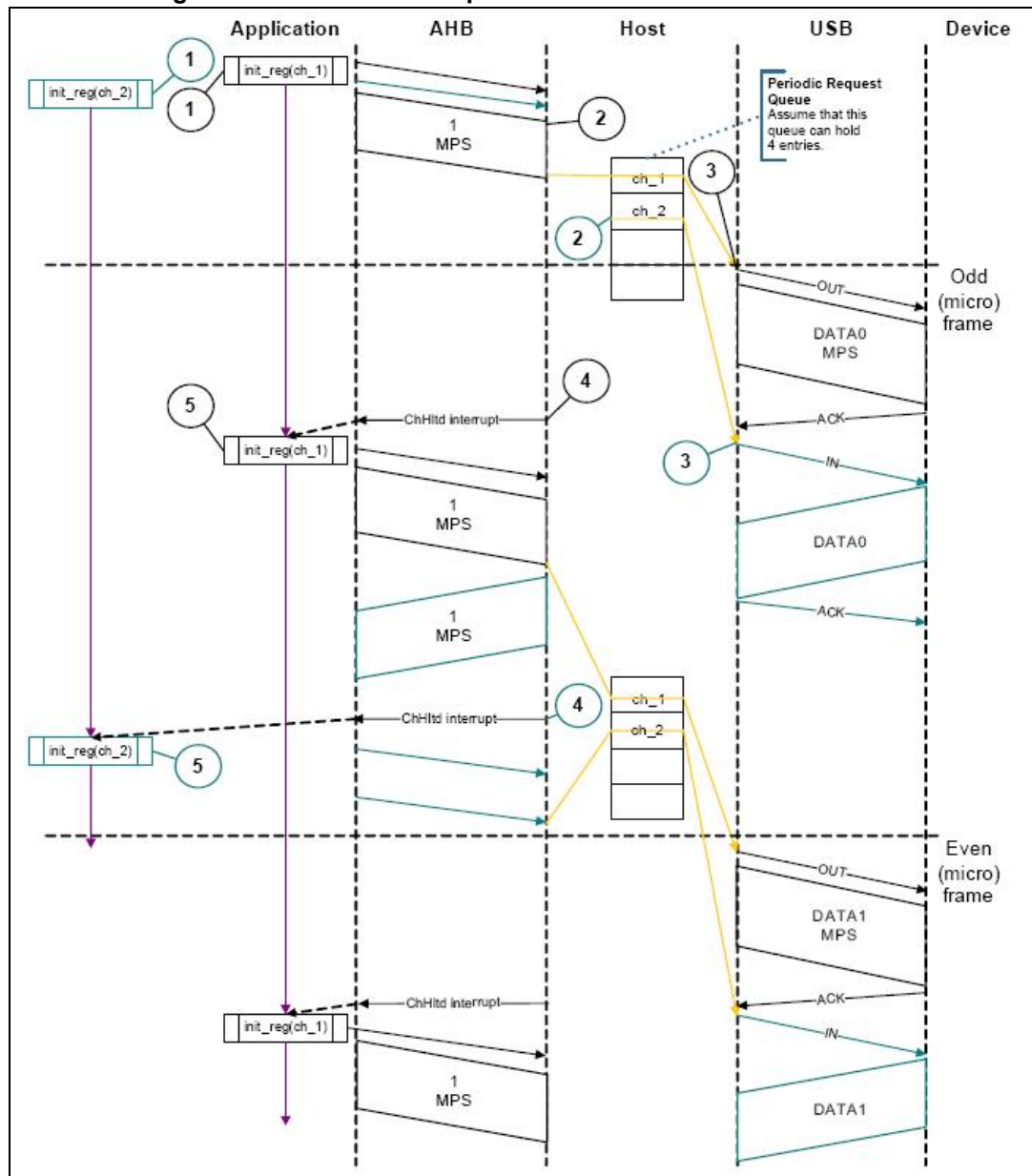
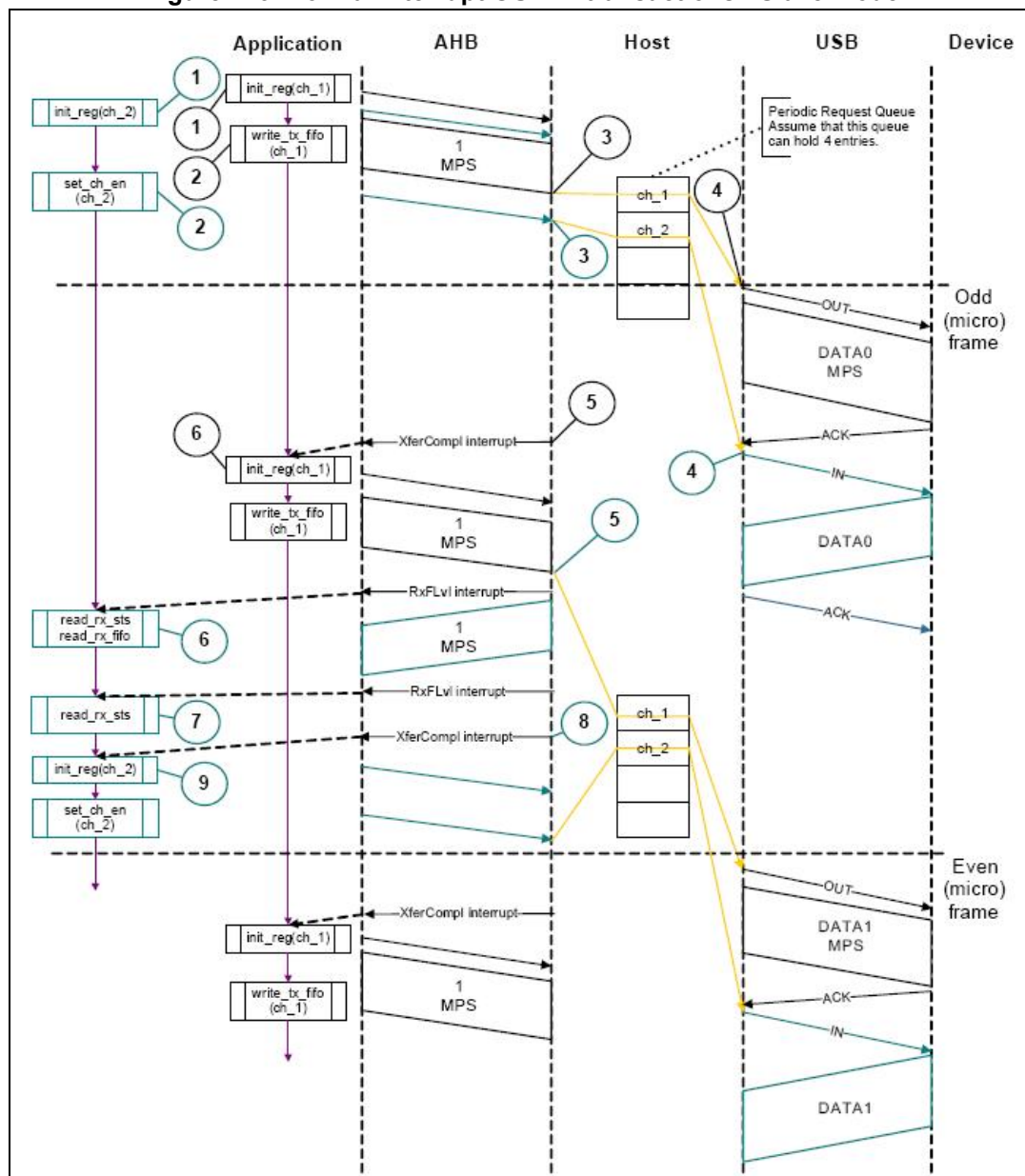


Figure 423. Normal interrupt OUT/IN transactions - Slave mode



- Interrupt service routine for interrupt OUT/IN transactions

- a) Interrupt OUT

**Unmask (NAK/TXERR/STALL/XFRC/FRMOR)**

**if (XFRC)**

```
{
  Reset Error Count
  Mask ACK
  De-allocate Channel
}
```

**else**

```
  if (STALL or FRMOR)
  {
```

```

Mask ACK
Unmask CHH
Disable Channel
if (STALL)
{
    Transfer Done = 1
}
}
else
if (NAK or TXERR)
{
    Rewind Buffer Pointers
    Reset Error Count
    Mask ACK
    Unmask CHH
    Disable Channel
}
else
if (CHH)
{
    Mask CHH
    if (Transfer Done or (Error_count == 3))
    {
        De-allocate Channel
    }
    else
    {
        Re-initialize Channel (in next b_interval - 1 Frame)
    }
}
else
if (ACK)
{
    Reset Error Count
    Mask ACK
}

```

The application is expected to write the data packets into the transmit FIFO when the space is available in the transmit FIFO and the Request queue up to the count specified in the MCNT field before switching to another channel. The application uses the NPTXFE interrupt in OTG\_HS\_GINTSTS to find the transmit FIFO space.

b) Interrupt IN

```

Unmask (NAK/TXERR/XFRC/BBERR/STALL/FRMOR/DTERR)
if (XFRC)
{
    Reset Error Count
    Mask ACK
    if (OTG_HS_HCTSIZx.PKTCNT == 0)
    {
        De-allocate Channel
    }
    else
    {

```

```
        Transfer Done = 1
        Unmask CHH
        Disable Channel
    }
}
else
    if (STALL or FRMOR or NAK or DTERR or BBERR)
    {
        Mask ACK
        Unmask CHH
        Disable Channel
        if (STALL or BBERR)
        {
            Reset Error Count
            Transfer Done = 1
        }
        else
            if (!FRMOR)
            {
                Reset Error Count
            }
    }
else
    if (TXERR)
    {
        Increment Error Count
        Unmask ACK
        Unmask CHH
        Disable Channel
    }
else
    if (CHH)
    {
        Mask CHH
        if (Transfer Done or (Error_count == 3))
        {
            De-allocate Channel
        }
        else
            Re-initialize Channel (in next b_interval - 1 /Frame)
    }
}
else
    if (ACK)
    {
        Reset Error Count
        Mask ACK
    }
```

}

The application is expected to write the requests for the same channel when the Request queue space is available up to the count specified in the MCNT field before switching to another channel (if any).

- **Interrupt IN transactions**

The assumptions are:

- The application is attempting to receive one packet (up to 1 maximum packet size) in every frame, starting with odd (transfer size = 1 024 bytes).
- The receive FIFO can hold at least one maximum-packet-size packet and two status words per packet (1 031 bytes).
- Periodic request queue depth = 4.

- **Normal interrupt IN operation**

The sequence of operations is as follows:

- Initialize channel 2. The application must set the ODDFRM bit in OTG\_HS\_HCCHAR2.
- Set the CHENA bit in OTG\_HS\_HCCHAR2 to write an IN request to the periodic request queue. For a high-bandwidth interrupt transfer, the application must write the OTG\_HS\_HCCHAR2 register MCNT (maximum number of expected packets in the next frame times) before switching to another channel.
- The OTG\_HS host writes an IN request to the periodic request queue for each OTG\_HS\_HCCHAR2 register write with the CHENA bit set.
- The OTG\_HS host attempts to send an IN token in the next (odd) frame.
- As soon as the IN packet is received and written to the receive FIFO, the OTG\_HS host generates an RXFLVL interrupt.
- In response to the RXFLVL interrupt, read the received packet status to determine the number of bytes received, then read the receive FIFO accordingly. The application must mask the RXFLVL interrupt before reading the receive FIFO, and unmask after reading the entire packet.
- The core generates the RXFLVL interrupt for the transfer completion status entry in the receive FIFO. The application must read and ignore the receive packet status when the receive packet status is not an IN data packet (PKTSTS in GRXSTSR ≠ 0b0010).
- The core generates an XFRC interrupt as soon as the receive packet status is read.
- In response to the XFRC interrupt, read the PKTCNT field in OTG\_HS\_HCTSIZ2. If the PKTCNT bit in OTG\_HS\_HCTSIZ2 is not equal to 0, disable the channel before re-initializing the channel for the next transfer, if any). If PKTCNT bit in

OTG\_HS\_HCTSIZ2 = 0, reinitialize the channel for the next transfer. This time, the application must reset the ODDFRM bit in OTG\_HS\_HCCHAR2.

- **Isochronous OUT transactions**

A typical isochronous OUT operation in Slave mode is shown in [Figure 424](#). The assumptions are:

- The application is attempting to send one packet every frame (up to 1 maximum packet size), starting with an odd frame. (transfer size = 1 024 bytes).
- The periodic transmit FIFO can hold one packet (1 KB).
- Periodic request queue depth = 4.

The sequence of operations is as follows:

- a) Initialize and enable channel 1. The application must set the ODDFRM bit in OTG\_HS\_HCCHAR1.
- b) Write the first packet for channel 1. For a high-bandwidth isochronous transfer, the application must write the subsequent packets up to MCNT (maximum number of packets to be transmitted in the next frame times before switching to another channel).
- c) Along with the last word write of each packet, the OTG\_HS host writes an entry to the periodic request queue.
- d) The OTG\_HS host attempts to send the OUT token in the next frame (odd).
- e) The OTG\_HS host generates the XFRC interrupt as soon as the last packet is transmitted successfully.
- f) In response to the XFRC interrupt, reinitialize the channel for the next transfer.
- g) Handling nonACK responses

Figure 424. Normal isochronous OUT/IN transactions - DMA mode

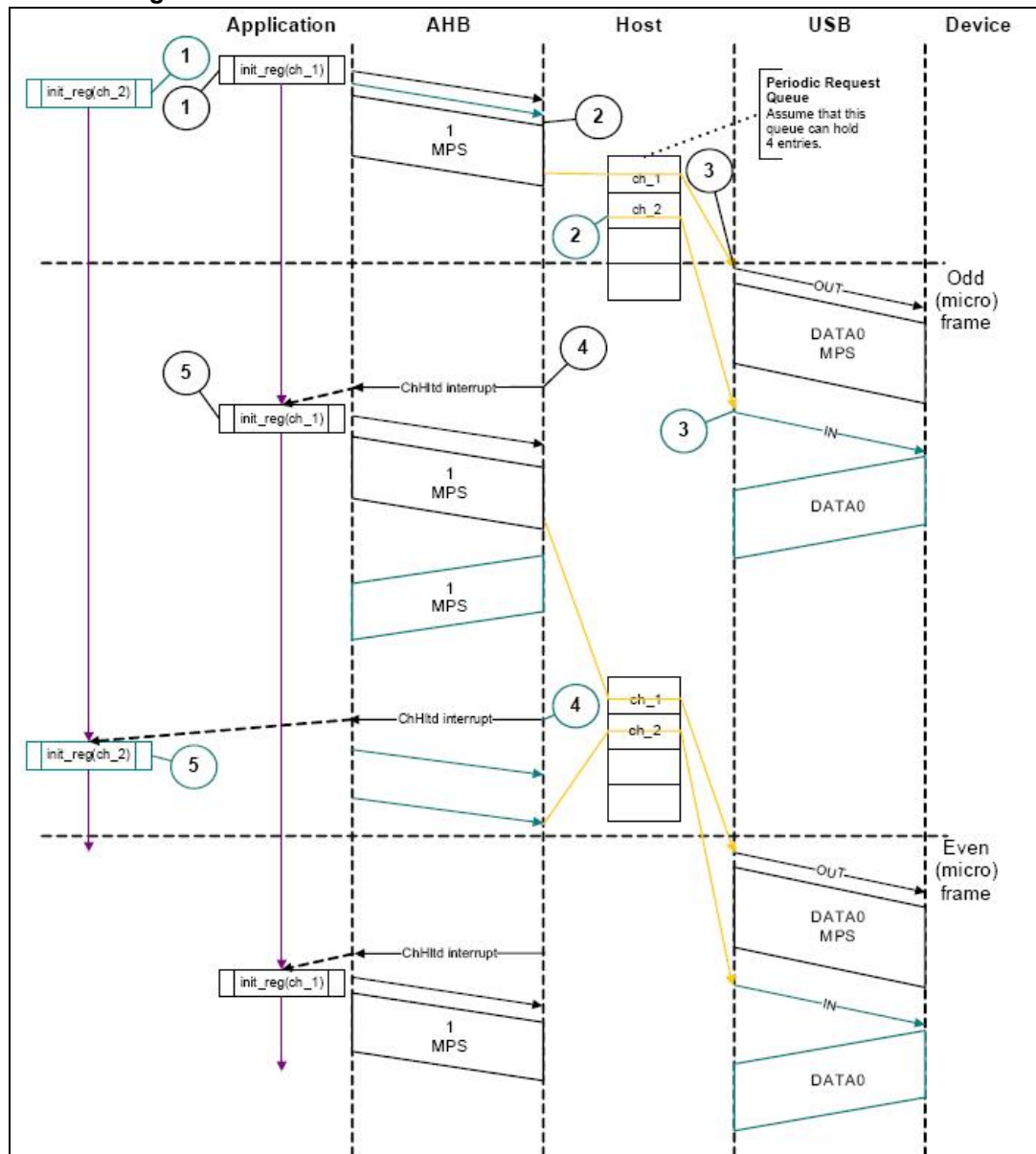
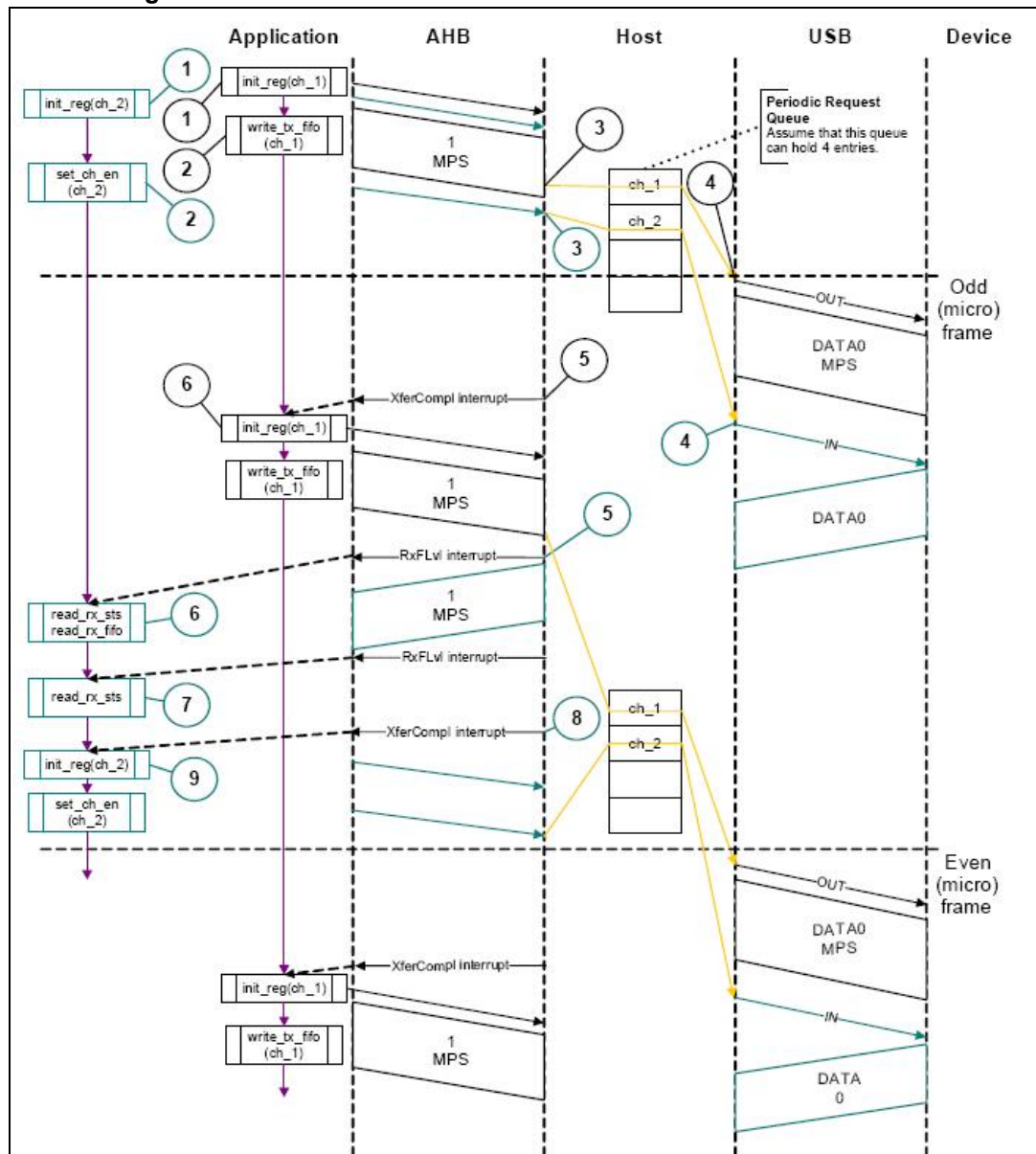


Figure 425. Normal isochronous OUT/IN transactions - Slave mode



- Interrupt service routine for isochronous OUT/IN transactions

Code sample: Isochronous OUT

```

Unmask (FRMOR/XFRC)
if (XFRC)
{
    De-allocate Channel
}
else
{
    if (FRMOR)
    {
        Unmask CHH
        Disable Channel
    }
}

```

```
else
if (CHH)
{
Mask CHH
De-allocate Channel
}
Code sample: Isochronous IN
Unmask (TXERR/XFRC/FRMOR/BBERR)
if (XFRC or FRMOR)
{
if (XFRC and (OTG_HS_HCTSIZx.PKTCNT == 0))
{
Reset Error Count
De-allocate Channel
}
else
{
Unmask CHH
Disable Channel
}
}
else
if (TXERR or BBERR)
{
Increment Error Count
Unmask CHH
Disable Channel
}
else
if (CHH)
{
Mask CHH
if (Transfer Done or (Error_count == 3))
{
De-allocate Channel
}
else
{
Re-initialize Channel
}
}
}
```

- **Isochronous IN transactions**

The assumptions are:

- The application is attempting to receive one packet (up to 1 maximum packet size) in every frame starting with the next odd frame (transfer size = 1 024 bytes).
- The receive FIFO can hold at least one maximum-packet-size packet and two status words per packet (1 031 bytes).
- Periodic request queue depth = 4.

The sequence of operations is as follows:

- Initialize channel 2. The application must set the ODDFRM bit in OTG\_HS\_HCCHAR2.
- Set the CHENA bit in OTG\_HS\_HCCHAR2 to write an IN request to the periodic request queue. For a high-bandwidth isochronous transfer, the application must write the OTG\_HS\_HCCHAR2 register MCNT (maximum number of expected packets in the next frame times) before switching to another channel.
- The OTG\_HS host writes an IN request to the periodic request queue for each OTG\_HS\_HCCHAR2 register write with the CHENA bit set.
- The OTG\_HS host attempts to send an IN token in the next odd frame.
- As soon as the IN packet is received and written to the receive FIFO, the OTG\_HS host generates an RXFLVL interrupt.
- In response to the RXFLVL interrupt, read the received packet status to determine the number of bytes received, then read the receive FIFO accordingly. The application must mask the RXFLVL interrupt before reading the receive FIFO, and unmask it after reading the entire packet.
- The core generates an RXFLVL interrupt for the transfer completion status entry in the receive FIFO. This time, the application must read and ignore the receive packet status when the receive packet status is not an IN data packet (PKTSTS bit in OTG\_HS\_GRXSTSR ≠ 0b0010).
- The core generates an XFRC interrupt as soon as the receive packet status is read.
- In response to the XFRC interrupt, read the PKTCNT field in OTG\_HS\_HCTSIZ2. If PKTCNT ≠ 0 in OTG\_HS\_HCTSIZ2, disable the channel before re-initializing the channel for the next transfer, if any. If PKTCNT = 0 in OTG\_HS\_HCTSIZ2, reinitialize the channel for the next transfer. This time, the application must reset the ODDFRM bit in OTG\_HS\_HCCHAR2.

- **Selecting the queue depth**

Choose the periodic and nonperiodic request queue depths carefully to match the number of periodic/nonperiodic endpoints accessed.

The nonperiodic request queue depth affects the performance of nonperiodic transfers. The deeper the queue (along with sufficient FIFO size), the more often the core is able to pipeline nonperiodic transfers. If the queue size is small, the core is able to put in new requests only when the queue space is freed up.

The core's periodic request queue depth is critical to perform periodic transfers as scheduled. Select the periodic queue depth, based on the number of periodic transfers scheduled in a micro-frame. In Slave mode, however, the application must also take into account the disable entry that must be put into the queue. So, if there are two nonhigh-bandwidth periodic endpoints, the periodic request queue depth must be at least 4. If at least one high-bandwidth endpoint is supported, the queue depth must be

8. If the periodic request queue depth is smaller than the periodic transfers scheduled in a micro-frame, a frame overrun condition occurs.

- **Handling babble conditions**

OTG\_HS controller handles two cases of babble: packet babble and port babble.

Packet babble occurs if the device sends more data than the maximum packet size for the channel. Port babble occurs if the core continues to receive data from the device at EOF2 (the end of frame 2, which is very close to SOF).

When OTG\_HS controller detects a packet babble, it stops writing data into the Rx buffer and waits for the end of packet (EOP). When it detects an EOP, it flushes already written data in the Rx buffer and generates a Babble interrupt to the application.

When OTG\_HS controller detects a port babble, it flushes the RxFIFO and disables the port. The core then generates a Port disabled interrupt (HPRTINT in OTG\_HS\_GINTSTS, PENCHNG in OTG\_HS\_HPRT). On receiving this interrupt, the application must determine that this is not due to an overcurrent condition (another cause of the Port Disabled interrupt) by checking POCA in OTG\_HS\_HPRT, then perform a soft reset. The core does not send any more tokens after it has detected a port babble condition.

- **Bulk and control OUT/SETUP transactions in DMA mode**

The sequence of operations is as follows:

- Initialize and enable channel 1 as explained in [Section : Channel initialization](#).
- The HS\_OTG host starts fetching the first packet as soon as the channel is enabled. For internal DMA mode, the OTG\_HS host uses the programmed DMA address to fetch the packet.
- After fetching the last word of the second (last) packet, the OTG\_HS host masks channel 1 internally for further arbitration.
- The HS\_OTG host generates a CHH interrupt as soon as the last packet is sent.
- In response to the CHH interrupt, de-allocate the channel for other transfers.

- **NAK and NYET handling with internal DMA**

- The OTG\_HS host sends a bulk OUT transaction.
- The device responds with NAK or NYET.
- If the application has unmasked NAK or NYET, the core generates the corresponding interrupt(s) to the application. The application is not required to service these interrupts, since the core takes care of rewinding the buffer pointers and re-initializing the Channel without application intervention.
- The core automatically issues a ping token.
- When the device returns an ACK, the core continues with the transfer. Optionally, the application can utilize these interrupts, in which case the NAK or NYET interrupt is masked by the application.

The core does not generate a separate interrupt when NAK or NYET is received by the host functionality.

- **Bulk and control IN transactions in DMA mode**

The sequence of operations is as follows:

- Initialize and enable the used channel (channel x) as explained in [Section : Channel initialization](#).
- The OTG\_HS host writes an IN request to the request queue as soon as the channel receives the grant from the arbiter (arbitration is performed in a round-robin fashion).

- c) The OTG\_HS host starts writing the received data to the system memory as soon as the last byte is received with no errors.
- d) When the last packet is received, the OTG\_HS host sets an internal flag to remove any extra IN requests from the request queue.
- e) The OTG\_HS host flushes the extra requests.
- f) The final request to disable channel x is written to the request queue. At this point, channel 2 is internally masked for further arbitration.
- g) The OTG\_HS host generates the CHH interrupt as soon as the disable request comes to the top of the queue.
- h) In response to the CHH interrupt, de-allocate the channel for other transfers.
- **Interrupt OUT transactions in DMA mode**
  - a) Initialize and enable channel x as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host starts fetching the first packet as soon the channel is enabled and writes the OUT request along with the last word fetch. In high-bandwidth transfers, the HS\_OTG host continues fetching the next packet (up to the value specified in the MC field) before switching to the next channel.
  - c) The OTG\_HS host attempts to send the OUT token at the beginning of the next odd frame/micro-frame.
  - d) After successfully transmitting the packet, the OTG\_HS host generates a CHH interrupt.
  - e) In response to the CHH interrupt, reinitialize the channel for the next transfer.
- **Interrupt IN transactions in DMA mode**

The sequence of operations (channelx) is as follows:

  - a) Initialize and enable channel x as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host writes an IN request to the request queue as soon as the channel x gets the grant from the arbiter (round-robin with fairness). In high-bandwidth transfers, the OTG\_HS host writes consecutive writes up to MC times.
  - c) The OTG\_HS host attempts to send an IN token at the beginning of the next (odd) frame/micro-frame.
  - d) As soon the packet is received and written to the receive FIFO, the OTG\_HS host generates a CHH interrupt.
  - e) In response to the CHH interrupt, reinitialize the channel for the next transfer.
- **Isochronous OUT transactions in DMA mode**
  - a) Initialize and enable channel x as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host starts fetching the first packet as soon as the channel is enabled, and writes the OUT request along with the last word fetch. In high-bandwidth transfers, the OTG\_HS host continues fetching the next packet (up to the value specified in the MC field) before switching to the next channel.
  - c) The OTG\_HS host attempts to send an OUT token at the beginning of the next (odd) frame/micro-frame.
  - d) After successfully transmitting the packet, the HS\_OTG host generates a CHH interrupt.
  - e) In response to the CHH interrupt, reinitialize the channel for the next transfer.
- **Isochronous IN transactions in DMA mode**

The sequence of operations ((channel x) is as follows:

- a) Initialize and enable channel x as explained in [Section : Channel initialization](#).
- b) The OTG\_HS host writes an IN request to the request queue as soon as the channel x gets the grant from the arbiter (round-robin with fairness). In high-bandwidth transfers, the OTG\_HS host performs consecutive write operations up to MC times.
- c) The OTG\_HS host attempts to send an IN token at the beginning of the next (odd) frame/micro-frame.
- d) As soon the packet is received and written to the receive FIFO, the OTG\_HS host generates a CHH interrupt.
- e) In response to the CHH interrupt, reinitialize the channel for the next transfer.
- **Bulk and control OUT/SETUP split transactions in DMA mode**  
The sequence of operations in (channel x) is as follows:
  - a) Initialize and enable channel x for start split as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host starts fetching the first packet as soon the channel is enabled and writes the OUT request along with the last word fetch.
  - c) After successfully transmitting start split, the OTG\_HS host generates the CHH interrupt.
  - d) In response to the CHH interrupt, set the COMPLSPLT bit in HCSPLT1 to send the complete split.
  - e) After successfully transmitting complete split, the OTG\_HS host generates the CHH interrupt.
  - f) In response to the CHH interrupt, de-allocate the channel.
- **Bulk/Control IN split transactions in DMA mode**  
The sequence of operations (channel x) is as follows:
  - a) Initialize and enable channel x as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host writes the start split request to the nonperiodic request after getting the grant from the arbiter. The OTG\_HS host masks the channel x internally for the arbitration after writing the request.
  - c) As soon as the IN token is transmitted, the OTG\_HS host generates the CHH interrupt.
  - d) In response to the CHH interrupt, set the COMPLSPLT bit in HCSPLT2 and re-enable the channel to send the complete split token. This unmask channel x for arbitration.
  - e) The OTG\_HS host writes the complete split request to the nonperiodic request after receiving the grant from the arbiter.
  - f) The OTG\_HS host starts writing the packet to the system memory after receiving the packet successfully.
  - g) As soon as the received packet is written to the system memory, the OTG\_HS host generates a CHH interrupt.
  - h) In response to the CHH interrupt, de-allocate the channel.
- **Interrupt OUT split transactions in DMA mode**  
The sequence of operations in (channel x) is as follows:
  - a) Initialize and enable channel 1 for start split as explained in [Section : Channel initialization](#). The application must set the ODDFRM bit in HCCHAR1.
  - b) The HS\_OTG host starts reading the packet.

- c) The HS\_OTG host attempts to send the start split transaction.
- d) After successfully transmitting the start split, the OTG\_HS host generates the CHH interrupt.
- e) In response to the CHH interrupt, set the COMPLSPLT bit in HCSPLT1 to send the complete split.
- f) After successfully completing the complete split transaction, the OTG\_HS host generates the CHH interrupt.
- g) In response to CHH interrupt, de-allocate the channel.
- **Interrupt IN split transactions in DMA mode**

The sequence of operations in (channel x) is as follows:

  - a) Initialize and enable channel x for start split as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host writes an IN request to the request queue as soon as channel x receives the grant from the arbiter.
  - c) The OTG\_HS host attempts to send the start split IN token at the beginning of the next odd micro-frame.
  - d) The OTG\_HS host generates the CHH interrupt after successfully transmitting the start split IN token.
  - e) In response to the CHH interrupt, set the COMPLSPLT bit in HCSPLT2 to send the complete split.
  - f) As soon as the packet is received successfully, the OTG\_HS host starts writing the data to the system memory.
  - g) The OTG\_HS host generates the CHH interrupt after transferring the received data to the system memory.
  - h) In response to the CHH interrupt, de-allocate or reinitialize the channel for the next start split.
- **Isochronous OUT split transactions in DMA mode**

The sequence of operations (channel x) is as follows:

  - a) Initialize and enable channel x for start split (begin) as explained in [Section : Channel initialization](#). The application must set the ODDFRM bit in HCCHAR1. Program the MPS field.
  - b) The HS\_OTG host starts reading the packet.
  - c) After successfully transmitting the start split (begin), the HS\_OTG host generates the CHH interrupt.
  - d) In response to the CHH interrupt, reinitialize the registers to send the start split (end).
  - e) After successfully transmitting the start split (end), the OTG\_HS host generates a CHH interrupt.
  - f) In response to the CHH interrupt, de-allocate the channel.
- **Isochronous IN split transactions in DMA mode**

The sequence of operations (channel x) is as follows:

  - a) Initialize and enable channel x for start split as explained in [Section : Channel initialization](#).
  - b) The OTG\_HS host writes an IN request to the request queue as soon as channel x receives the grant from the arbiter.

- c) The OTG\_HS host attempts to send the start split IN token at the beginning of the next odd micro-frame.
- d) The OTG\_HS host generates the CHH interrupt after successfully transmitting the start split IN token.
- e) In response to the CHH interrupt, set the COMPLSPLT bit in HCSPLT2 to send the complete split.
- f) As soon as the packet is received successfully, the OTG\_HS host starts writing the data to the system memory.
- g) The OTG\_HS host generates the CHH interrupt after transferring the received data to the system memory. In response to the CHH interrupt, de-allocate the channel or reinitialize the channel for the next start split.

### 35.13.6 Device programming model

#### Endpoint initialization on USB reset

1. Set the NAK bit for all OUT endpoints
  - SNAK = 1 in OTG\_HS\_DOEPCTLx (for all OUT endpoints)
2. Unmask the following interrupt bits
  - INEP0 = 1 in OTG\_HS\_DAINMSK (control 0 IN endpoint)
  - OUTEP0 = 1 in OTG\_HS\_DAINMSK (control 0 OUT endpoint)
  - STUP = 1 in DOEPMSK
  - XFRC = 1 in DOEPMSK
  - XFRC = 1 in DIEPMSK
  - TOC = 1 in DIEPMSK
3. Set up the Data FIFO RAM for each of the FIFOs
  - Program the OTG\_HS\_GRXFSIZ register, to be able to receive control OUT data and setup data. If thresholding is not enabled, at a minimum, this must be equal to 1 max packet size of control endpoint 0 + 2 words (for the status of the control OUT data packet) + 10 words (for setup packets).
  - Program the OTG\_HS\_TX0FSIZ register (depending on the FIFO number chosen) to be able to transmit control IN data. At a minimum, this must be equal to 1 max packet size of control endpoint 0.
4. Program the following fields in the endpoint-specific registers for control OUT endpoint 0 to receive a SETUP packet
  - STUPCNT = 3 in OTG\_HS\_DOEPTSIZ0 (to receive up to 3 back-to-back SETUP packets)
5. In DMA mode, the DOEPDMA0 register should have a valid memory address to store any SETUP packets received.

At this point, all initialization required to receive SETUP packets is done.

**Endpoint initialization on enumeration completion**

1. On the Enumeration Done interrupt (ENUMDNE in OTG\_HS\_GINTSTS), read the OTG\_HS\_DSTS register to determine the enumeration speed.
2. Program the MPSIZ field in OTG\_HS\_DIEPCTL0 to set the maximum packet size. This step configures control endpoint 0. The maximum packet size for a control endpoint depends on the enumeration speed.
3. In DMA mode, program the DOEPTL0 register to enable control OUT endpoint 0, to receive a SETUP packet.
  - EPENA bit in DOEPTL0 = 1

At this point, the device is ready to receive SOF packets and is configured to perform control transfers on control endpoint 0.

**Endpoint initialization on SetAddress command**

This section describes what the application must do when it receives a SetAddress command in a SETUP packet.

1. Program the OTG\_HS\_DCFG register with the device address received in the SetAddress command
1. Program the core to send out a status IN packet

**Endpoint initialization on SetConfiguration/SetInterface command**

This section describes what the application must do when it receives a SetConfiguration or SetInterface command in a SETUP packet.

1. When a SetConfiguration command is received, the application must program the endpoint registers to configure them with the characteristics of the valid endpoints in the new configuration.
2. When a SetInterface command is received, the application must program the endpoint registers of the endpoints affected by this command.
3. Some endpoints that were active in the prior configuration or alternate setting are not valid in the new configuration or alternate setting. These invalid endpoints must be deactivated.
4. Unmask the interrupt for each active endpoint and mask the interrupts for all inactive endpoints in the OTG\_HS\_DAINMSK register.
5. Set up the Data FIFO RAM for each FIFO.
6. After all required endpoints are configured; the application must program the core to send a status IN packet.

At this point, the device core is configured to receive and transmit any type of data packet.

### Endpoint activation

This section describes the steps required to activate a device endpoint or to configure an existing device endpoint to a new type.

1. Program the characteristics of the required endpoint into the following fields of the OTG\_HS\_DIEPCTLx register (for IN or bidirectional endpoints) or the OTG\_HS\_DOEPCTLx register (for OUT or bidirectional endpoints).
  - Maximum packet size
  - USB active endpoint = 1
  - Endpoint start data toggle (for interrupt and bulk endpoints)
  - Endpoint type
  - TxFIFO number
2. Once the endpoint is activated, the core starts decoding the tokens addressed to that endpoint and sends out a valid handshake for each valid token received for the endpoint.

### Endpoint deactivation

This section describes the steps required to deactivate an existing endpoint.

1. In the endpoint to be deactivated, clear the USB active endpoint bit in the OTG\_HS\_DIEPCTLx register (for IN or bidirectional endpoints) or the OTG\_HS\_DOEPCTLx register (for OUT or bidirectional endpoints).
2. Once the endpoint is deactivated, the core ignores tokens addressed to that endpoint, which results in a timeout on the USB.

*Note:* The application must meet the following conditions to set up the device core to handle traffic:  
NPTXFEM and RXFLVLM in GINTMSK must be cleared.

## 35.13.7 Operational model

### SETUP and OUT data transfers

This section describes the internal data flow and application-level operations during data OUT transfers and SETUP transactions.

- **Packet read**

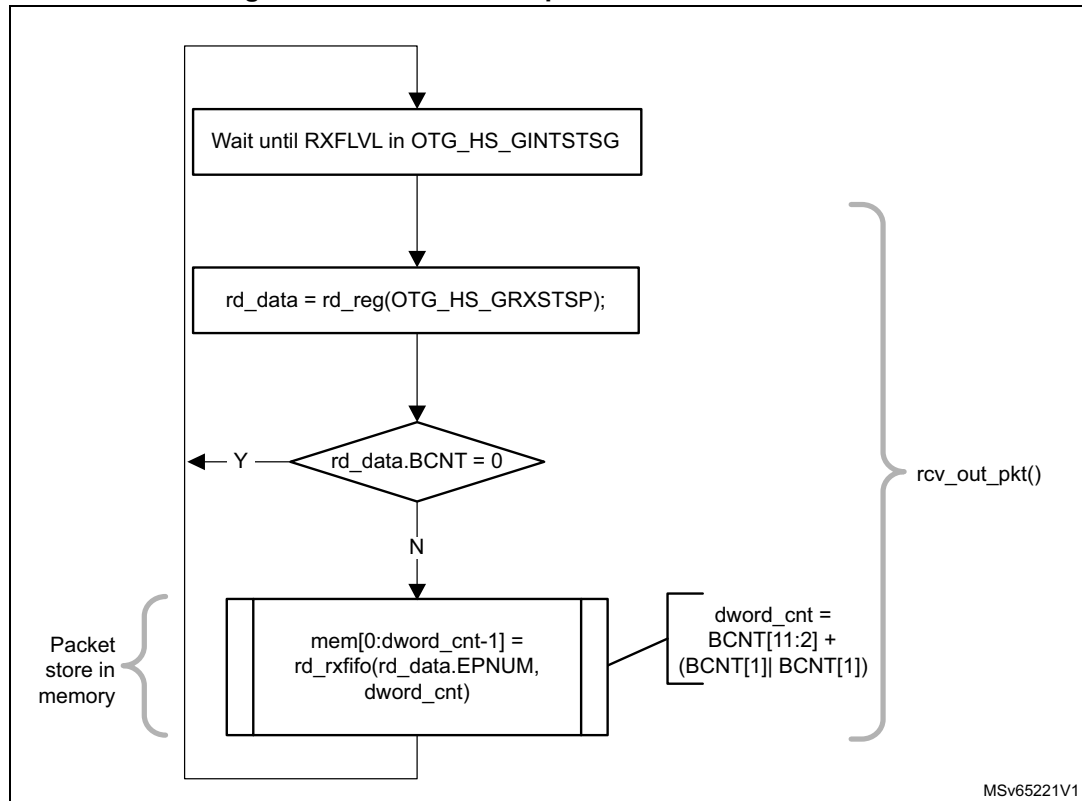
This section describes how to read packets (OUT data and SETUP packets) from the receive FIFO in Slave mode.

1. On catching an RXFLVL interrupt (OTG\_HS\_GINTSTS register), the application must read the Receive status pop register (OTG\_HS\_GRXSTSP).
2. The application can mask the RXFLVL interrupt (in OTG\_HS\_GINTSTS) by writing to RXFLVL = 0 (in GINTMSK), until it has read the packet from the receive FIFO.
3. If the received packet's byte count is not 0, the byte count amount of data is popped from the receive Data FIFO and stored in memory. If the received packet byte count is 0, no data is popped from the receive data FIFO.

4. The receive FIFO packet status readout indicates one of the following:
  - a) Global OUT NAK pattern:  
PKTSTS = Global OUT NAK, BCNT = 0x000, EPNUM = Don't Care (0x0),  
DPID = Don't Care (0b00).  
These data indicate that the global OUT NAK bit has taken effect.
  - b) SETUP packet pattern:  
PKTSTS = SETUP, BCNT = 0x008, EPNUM = Control EP Num, DPID = D0.  
These data indicate that a SETUP packet for the specified endpoint is now available for reading from the receive FIFO.
  - c) Setup stage done pattern:  
PKTSTS = Setup Stage Done, BCNT = 0x0, EPNUM = Control EP Num,  
DPID = Don't Care (0b00).  
These data indicate that the Setup stage for the specified endpoint has completed and the Data stage has started. After this entry is popped from the receive FIFO, the core asserts a Setup interrupt on the specified control OUT endpoint.
  - d) Data OUT packet pattern:  
PKTSTS = DataOUT, BCNT = size of the received data OUT packet  
( $0 \leq \text{BCNT} \leq 1024$ ), EPNUM = EPNUM on which the packet was received,  
DPID = Actual Data PID.
  - e) Data transfer completed pattern:  
PKTSTS = Data OUT Transfer Done, BCNT = 0x0, EPNUM = OUT EP Num  
on which the data transfer is complete, DPID = Don't Care (0b00).  
These data indicate that an OUT data transfer for the specified OUT endpoint has completed. After this entry is popped from the receive FIFO, the core asserts a Transfer Completed interrupt on the specified OUT endpoint.
5. After the data payload is popped from the receive FIFO, the RXFLVL interrupt (OTG\_HS\_GINTSTS) must be unmasked.
6. Steps 1–5 are repeated every time the application detects assertion of the interrupt line due to RXFLVL in OTG\_HS\_GINTSTS. Reading an empty receive FIFO can result in undefined core behavior.

*Figure 426* provides a flowchart of the above procedure.

Figure 426. Receive FIFO packet read in slave mode



#### • SETUP transactions

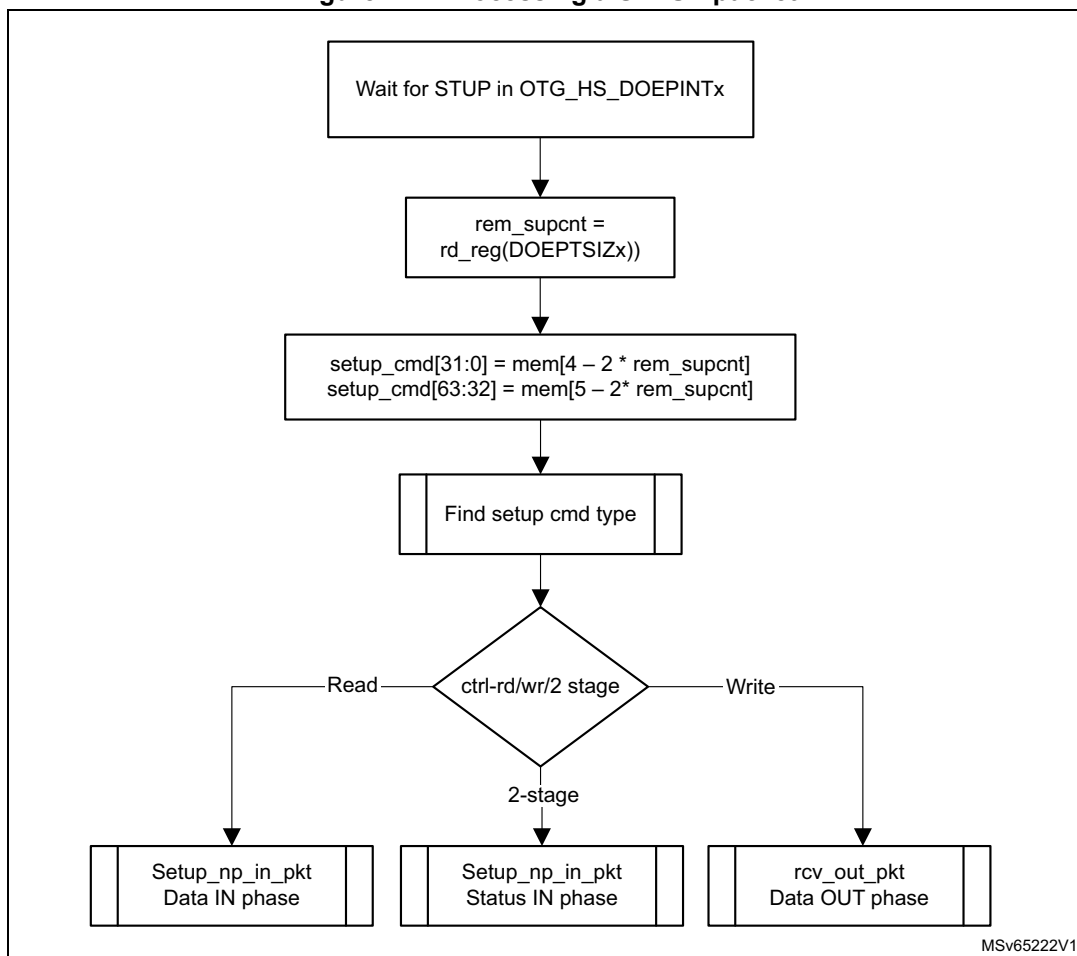
This section describes how the core handles SETUP packets and the application's sequence for handling SETUP transactions.

#### • Application requirements

- To receive a SETUP packet, the STUPCNT field (OTG\_HS\_DOEPTSIZx) in a control OUT endpoint must be programmed to a nonzero value. When the application programs the STUPCNT field to a nonzero value, the core receives SETUP packets and writes them to the receive FIFO, irrespective of the NAK status and EPENA bit setting in OTG\_HS\_DOEPCTLx. The STUPCNT field is decremented every time the control endpoint receives a SETUP packet. If the STUPCNT field is not programmed to a proper value before receiving a SETUP packet, the core still receives the SETUP packet and decrements the STUPCNT field, but the application may not be able to determine the correct number of SETUP packets received in the Setup stage of a control transfer.
  - STUPCNT = 3 in OTG\_HS\_DOEPTSIZx
- The application must always allocate some extra space in the Receive data FIFO, to be able to receive up to three SETUP packets on a control endpoint.
  - The space to be reserved is 10 words. Three words are required for the first SETUP packet, 1 word is required for the Setup stage done word and 6 words are required to store two extra SETUP packets among all control endpoints.
  - 3 words per SETUP packet are required to store 8 bytes of SETUP data and 4 bytes of SETUP status (Setup packet pattern). The core reserves this space in the receive data.
  - FIFO to write SETUP data only, and never uses this space for data packets.

3. The application must read the 2 words of the SETUP packet from the receive FIFO.
4. The application must read and discard the Setup stage done word from the receive FIFO.
- **Internal data flow**
5. When a SETUP packet is received, the core writes the received data to the receive FIFO, without checking for available space in the receive FIFO and irrespective of the endpoint's NAK and STALL bit settings.
  - The core internally sets the IN NAK and OUT NAK bits for the control IN/OUT endpoints on which the SETUP packet was received.
6. For every SETUP packet received on the USB, 3 words of data are written to the receive FIFO, and the STUPCNT field is decremented by 1.
  - The first word contains control information used internally by the core
  - The second word contains the first 4 bytes of the SETUP command
  - The third word contains the last 4 bytes of the SETUP command
7. When the Setup stage changes to a Data IN/OUT stage, the core writes an entry (Setup stage done word) to the receive FIFO, indicating the completion of the Setup stage.
8. On the AHB side, SETUP packets are emptied by the application.
9. When the application pops the Setup stage done word from the receive FIFO, the core interrupts the application with an STUP interrupt (OTG\_HS\_DOEPINTx), indicating it can process the received SETUP packet.
  - The core clears the endpoint enable bit for control OUT endpoints.
- **Application programming sequence**
1. Program the OTG\_HS\_DOEPTSIz register.
  - STUPCNT = 3
2. Wait for the RXFLVL interrupt (OTG\_HS\_GINTSTS) and empty the data packets from the receive FIFO.
3. Assertion of the STUP interrupt (OTG\_HS\_DOEPINTx) marks a successful completion of the SETUP Data Transfer.
  - On this interrupt, the application must read the OTG\_HS\_DOEPTSIz register to determine the number of SETUP packets received and process the last received SETUP packet.

Figure 427. Processing a SETUP packet



- **Handling more than three back-to-back SETUP packets**

Per the USB 2.0 specification, normally, during a SETUP packet error, a host does not send more than three back-to-back SETUP packets to the same endpoint. However, the USB 2.0 specification does not limit the number of back-to-back SETUP packets a host can send to the same endpoint. When this condition occurs, the OTG\_HS controller generates an interrupt (B2BSTUP in OTG\_HS\_DOEPINTx).

- **Setting the global OUT NAK**

Internal data flow:

1. When the application sets the Global OUT NAK (SGONAK bit in OTG\_HS\_DCTL), the core stops writing data, except SETUP packets, to the receive FIFO. Irrespective of the space availability in the receive FIFO, nonisochronous OUT tokens receive a NAK handshake response, and the core ignores isochronous OUT data packets
2. The core writes the Global OUT NAK pattern to the receive FIFO. The application must reserve enough receive FIFO space to write this data pattern.
3. When the application pops the Global OUT NAK pattern word from the receive FIFO, the core sets the GONAKEFF interrupt (OTG\_HS\_GINTSTS).
4. Once the application detects this interrupt, it can assume that the core is in Global OUT NAK mode. The application can clear this interrupt by clearing the SGONAK bit in OTG\_HS\_DCTL.

Application programming sequence:

1. To stop receiving any kind of data in the receive FIFO, the application must set the Global OUT NAK bit by programming the following field:
  - SGONAK = 1 in OTG\_HS\_DCTL
2. Wait for the assertion of the GONAKEFF interrupt in OTG\_HS\_GINTSTS. When asserted, this interrupt indicates that the core has stopped receiving any type of data except SETUP packets.
3. The application can receive valid OUT packets after it has set SGONAK in OTG\_HS\_DCTL and before the core asserts the GONAKEFF interrupt (OTG\_HS\_GINTSTS).
4. The application can temporarily mask this interrupt by writing to the GINAKEFFM bit in GINTMSK.
  - GINAKEFFM = 0 in GINTMSK
5. Whenever the application is ready to exit the Global OUT NAK mode, it must clear the SGONAK bit in OTG\_HS\_DCTL. This also clears the GONAKEFF interrupt (OTG\_HS\_GINTSTS).
  - OTG\_HS\_DCTL = 1 in CGONAK
6. If the application has masked this interrupt earlier, it must be unmasked as follows:
  - GINAKEFFM = 1 in GINTMSK

- **Disabling an OUT endpoint**

The application must use this sequence to disable an OUT endpoint that it has enabled.

Application programming sequence:

1. Before disabling any OUT endpoint, the application must enable Global OUT NAK mode in the core.
  - SGONAK = 1 in OTG\_HS\_DCTL
2. Wait for the GONAKEFF interrupt (OTG\_HS\_GINTSTS)
3. Disable the required OUT endpoint by programming the following fields:
  - EPDIS = 1 in OTG\_HS\_DOEPCTLx
  - SNAK = 1 in OTG\_HS\_DOEPCTLx
4. Wait for the EPDISD interrupt (OTG\_HS\_DOEPINTx), which indicates that the OUT endpoint is completely disabled. When the EPDISD interrupt is asserted, the core also clears the following bits:
  - EPDIS = 0 in OTG\_HS\_DOEPCTLx
  - EPENA = 0 in OTG\_HS\_DOEPCTLx
5. The application must clear the Global OUT NAK bit to start receiving data from other nondisabled OUT endpoints.
  - SGONAK = 0 in OTG\_HS\_DCTL

- **Transfer Stop Programming for OUT endpoints**

The application must use the following programming sequence to stop any transfers (because of an interrupt from the host, typically a reset).

**Sequence of operations:**

1. Enable all OUT endpoints by setting
  - EPENA = 1 in all OTG\_HS\_DOEPCTLx registers.
2. Flush the RxFIFO as follows
  - Poll OTG\_HS\_GRSTCTL.AHBIDL until it is 1. This indicates that AHB master is idle.
  - Perform read modify write operation on OTG\_HS\_GRSTCTL.RXFFLSH = 1
  - Poll OTG\_HS\_GRSTCTL.RXFFLSH until it is 0, but also using a timeout of less than 10 milli-seconds (corresponds to minimum reset signaling duration). If 0 is seen before the timeout, then the RxFIFO flush is successful. If at the moment the timeout occurs, there is still a 1, (this may be due to a packet on EP0 coming from the host) then go back (once only) to the previous step (“Perform read modify write operation”).
3. Before disabling any OUT endpoint, the application must enable Global OUT NAK mode in the core, according to the instructions in [“Setting the global OUT NAK on page 1509”](#). This ensures that data in the RxFIFO is sent to the application successfully. Set SGONAK = 1 in OTG\_HS\_DCTL
4. Wait for the GONAKEFF interrupt (OTG\_HS\_GINTSTS)
5. Disable all active OUT endpoints by programming the following register bits:
  - EPDIS = 1 in registers OTG\_HS\_DOEPCTLx
  - SNAK = 1 in registers OTG\_HS\_DOEPCTLx
6. Wait for the EPDIS interrupt in OTG\_HS\_DOEPINTx for each OUT endpoint programmed in the previous step. The EPDIS interrupt in OTG\_HS\_DOEPINTx indicates that the corresponding OUT endpoint is completely disabled. When the EPDIS interrupt is asserted, the following bits are cleared:
  - EPENA = 0 in registers OTG\_HS\_DOEPCTLx
  - EPDIS = 0 in registers OTG\_HS\_DOEPCTLx
  - SNAK = 0 in registers OTG\_HS\_DOEPCTLx

• **Generic non-isochronous OUT data transfers**

This section describes a regular nonisochronous OUT data transfer (control, bulk, or interrupt).

Application requirements:

1. Before setting up an OUT transfer, the application must allocate a buffer in the memory to accommodate all data to be received as part of the OUT transfer.
2. For OUT transfers, the transfer size field in the endpoint’s transfer size register must be a multiple of the maximum packet size of the endpoint, adjusted to the word boundary.
  - $\text{transfer size}[\text{EPNUM}] = n \times (\text{MPSIZ}[\text{EPNUM}] + 4 - (\text{MPSIZ}[\text{EPNUM}] \bmod 4))$
  - $\text{packet count}[\text{EPNUM}] = n$
  - $n > 0$
3. On any OUT endpoint interrupt, the application must read the endpoint’s transfer size register to calculate the size of the payload in the memory. The received payload size can be less than the programmed transfer size.
  - $\text{Payload size in memory} = \text{application programmed initial transfer size} - \text{core updated final transfer size}$
  - $\text{Number of USB packets in which this payload was received} = \text{application programmed initial packet count} - \text{core updated final packet count}$

Internal data flow:

1. The application must set the transfer size and packet count fields in the endpoint-specific registers, clear the NAK bit, and enable the endpoint to receive the data.
2. Once the NAK bit is cleared, the core starts receiving data and writes it to the receive FIFO, as long as there is space in the receive FIFO. For every data packet received on the USB, the data packet and its status are written to the receive FIFO. Every packet (maximum packet size or short packet) written to the receive FIFO decrements the packet count field for that endpoint by 1.
  - OUT data packets received with bad data CRC are flushed from the receive FIFO automatically.
  - After sending an ACK for the packet on the USB, the core discards nonisochronous OUT data packets that the host, which cannot detect the ACK, re-sends. The application does not detect multiple back-to-back data OUT packets on the same endpoint with the same data PID. In this case the packet count is not decremented.
  - If there is no space in the receive FIFO, isochronous or nonisochronous data packets are ignored and not written to the receive FIFO. Additionally, nonisochronous OUT tokens receive a NAK handshake reply.
  - In all the above three cases, the packet count is not decremented because no data are written to the receive FIFO.
3. When the packet count becomes 0 or when a short packet is received on the endpoint, the NAK bit for that endpoint is set. Once the NAK bit is set, the isochronous or nonisochronous data packets are ignored and not written to the receive FIFO, and nonisochronous OUT tokens receive a NAK handshake reply.
4. After the data are written to the receive FIFO, the application reads the data from the receive FIFO and writes it to external memory, one packet at a time per endpoint.
5. At the end of every packet write on the AHB to external memory, the transfer size for the endpoint is decremented by the size of the written packet.
6. The OUT data transfer completed pattern for an OUT endpoint is written to the receive FIFO on one of the following conditions:
  - The transfer size is 0 and the packet count is 0
  - The last OUT data packet written to the receive FIFO is a short packet ( $0 \leq \text{packet size} < \text{maximum packet size}$ )
7. When either the application pops this entry (OUT data transfer completed), a transfer completed interrupt is generated for the endpoint and the endpoint enable is cleared.

Application programming sequence:

1. Program the OTG\_HS\_DOEPTSIZE register for the transfer size and the corresponding packet count.
2. Program the OTG\_HS\_DOEPCTL register with the endpoint characteristics, and set the EPENA and CNAK bits.
  - EPENA = 1 in OTG\_HS\_DOEPCTL
  - CNAK = 1 in OTG\_HS\_DOEPCTL
3. Wait for the RXFLVL interrupt (in OTG\_HS\_GINTSTS) and empty the data packets from the receive FIFO.
  - This step can be repeated many times, depending on the transfer size.
4. Asserting the XFRC interrupt (OTG\_HS\_DOEPINT) marks a successful completion of the nonisochronous OUT data transfer.
5. Read the OTG\_HS\_DOEPTSIZE register to determine the size of the received data payload.

- **Generic isochronous OUT data transfer**

This section describes a regular isochronous OUT data transfer.

Application requirements:

1. All the application requirements for nonisochronous OUT data transfers also apply to isochronous OUT data transfers.
2. For isochronous OUT data transfers, the transfer size and packet count fields must always be set to the number of maximum-packet-size packets that can be received in a single frame and no more. Isochronous OUT data transfers cannot span more than 1 frame.
3. The application must read all isochronous OUT data packets from the receive FIFO (data and status) before the end of the periodic frame (EOPF interrupt in OTG\_HS\_GINTSTS).
4. To receive data in the following frame, an isochronous OUT endpoint must be enabled after the EOPF (OTG\_HS\_GINTSTS) and before the SOF (OTG\_HS\_GINTSTS).

Internal data flow:

1. The internal data flow for isochronous OUT endpoints is the same as that for nonisochronous OUT endpoints, but for a few differences.
2. When an isochronous OUT endpoint is enabled by setting the Endpoint Enable and clearing the NAK bits, the Even/Odd frame bit must also be set appropriately. The core receives data on an isochronous OUT endpoint in a particular frame only if the following condition is met:
  - EONUM (in OTG\_HS\_DOEPCTL) = SOFFN[0] (in OTG\_HS\_DSTS)
3. When the application completely reads an isochronous OUT data packet (data and status) from the receive FIFO, the core updates the RXDPID field in OTG\_HS\_DOEPTSIZE with the data PID of the last isochronous OUT data packet read from the receive FIFO.

Application programming sequence:

1. Program the OTG\_HS\_DOEPTSLx register for the transfer size and the corresponding packet count
2. Program the OTG\_HS\_DOEPCTLx register with the endpoint characteristics and set the Endpoint Enable, ClearNAK, and Even/Odd frame bits.
  - EPENA = 1
  - CNAK = 1
  - EONUM = (0: Even/1: Odd)
3. In Slave mode, wait for the RXFLVL interrupt (in OTG\_HS\_GINTSTS) and empty the data packets from the receive FIFO
  - This step can be repeated many times, depending on the transfer size.
4. The assertion of the XFRC interrupt (in OTG\_HS\_DOEPINTx) marks the completion of the isochronous OUT data transfer. This interrupt does not necessarily mean that the data in memory are good.
5. This interrupt cannot always be detected for isochronous OUT transfers. Instead, the application can detect the IISOXFRM interrupt in OTG\_HS\_GINTSTS.
6. Read the OTG\_HS\_DOEPTSLx register to determine the size of the received transfer and to determine the validity of the data received in the frame. The application must treat the data received in memory as valid only if one of the following conditions is met:
  - RXDPID = D0 (in OTG\_HS\_DOEPTSLx) and the number of USB packets in which this payload was received = 1
  - RXDPID = D1 (in OTG\_HS\_DOEPTSLx) and the number of USB packets in which this payload was received = 2
  - RXDPID = D2 (in OTG\_HS\_DOEPTSLx) and the number of USB packets in which this payload was received = 3

The number of USB packets in which this payload was received =  
Application programmed initial packet count – Core updated final packet count

The application can discard invalid data packets.

• **Incomplete isochronous OUT data transfers**

This section describes the application programming sequence when isochronous OUT data packets are dropped inside the core.

Internal data flow:

1. For isochronous OUT endpoints, the XFRC interrupt (in OTG\_HS\_DOEPINTx) may not always be asserted. If the core drops isochronous OUT data packets, the application could fail to detect the XFRC interrupt (OTG\_HS\_DOEPINTx) under the following circumstances:
  - When the receive FIFO cannot accommodate the complete ISO OUT data packet, the core drops the received ISO OUT data
  - When the isochronous OUT data packet is received with CRC errors
  - When the isochronous OUT token received by the core is corrupted
  - When the application is very slow in reading the data from the receive FIFO
2. When the core detects an end of periodic frame before transfer completion to all isochronous OUT endpoints, it asserts the incomplete Isochronous OUT data interrupt (IISOXFRM in OTG\_HS\_GINTSTS), indicating that an XFRC interrupt (in OTG\_HS\_DOEPINTx) is not asserted on at least one of the isochronous OUT endpoints. At this point, the endpoint with the incomplete transfer remains enabled, but no active transfers remain in progress on this endpoint on the USB.

Application programming sequence:

1. Asserting the IISOXFRM interrupt (OTG\_HS\_GINTSTS) indicates that in the current frame, at least one isochronous OUT endpoint has an incomplete transfer.
2. If this occurs because isochronous OUT data is not completely emptied from the endpoint, the application must ensure that the application empties all isochronous OUT data (data and status) from the receive FIFO before proceeding.
  - When all data are emptied from the receive FIFO, the application can detect the XFRC interrupt (OTG\_HS\_DOEPINTx). In this case, the application must re-enable the endpoint to receive isochronous OUT data in the next frame.
3. When it receives an IISOXFRM interrupt (in OTG\_HS\_GINTSTS), the application must read the control registers of all isochronous OUT endpoints (OTG\_HS\_DOEPCTLx) to determine which endpoints had an incomplete transfer in the current micro-frame. An endpoint transfer is incomplete if both the following conditions are met:
  - EONUM bit (in OTG\_HS\_DOEPCTLx) = SOFFN[0] (in OTG\_HS\_DSTS)
  - EPENA = 1 (in OTG\_HS\_DOEPCTLx)
4. The previous step must be performed before the SOF interrupt (in OTG\_HS\_GINTSTS) is detected, to ensure that the current frame number is not changed.
5. For isochronous OUT endpoints with incomplete transfers, the application must discard the data in the memory and disable the endpoint by setting the EPDIS bit in OTG\_HS\_DOEPCTLx.
6. Wait for the EPDIS interrupt (in OTG\_HS\_DOEPINTx) and enable the endpoint to receive new data in the next frame.
  - Because the core can take some time to disable the endpoint, the application may not be able to receive the data in the next frame after receiving bad isochronous data.

- **Stalling a nonisochronous OUT endpoint**

This section describes how the application can stall a nonisochronous endpoint.

1. Put the core in the Global OUT NAK mode.
2. Disable the required endpoint
  - When disabling the endpoint, instead of setting the SNAK bit in OTG\_HS\_DOEPCTL, set STALL = 1 (in OTG\_HS\_DOEPCTL).  
The STALL bit always takes precedence over the NAK bit.
3. When the application is ready to end the STALL handshake for the endpoint, the STALL bit (in OTG\_HS\_DOEPCTLx) must be cleared.
4. If the application is setting or clearing a STALL for an endpoint due to a SetFeature.Endpoint Halt or ClearFeature.Endpoint Halt command, the STALL bit must be set or cleared before the application sets up the Status stage transfer on the control endpoint.

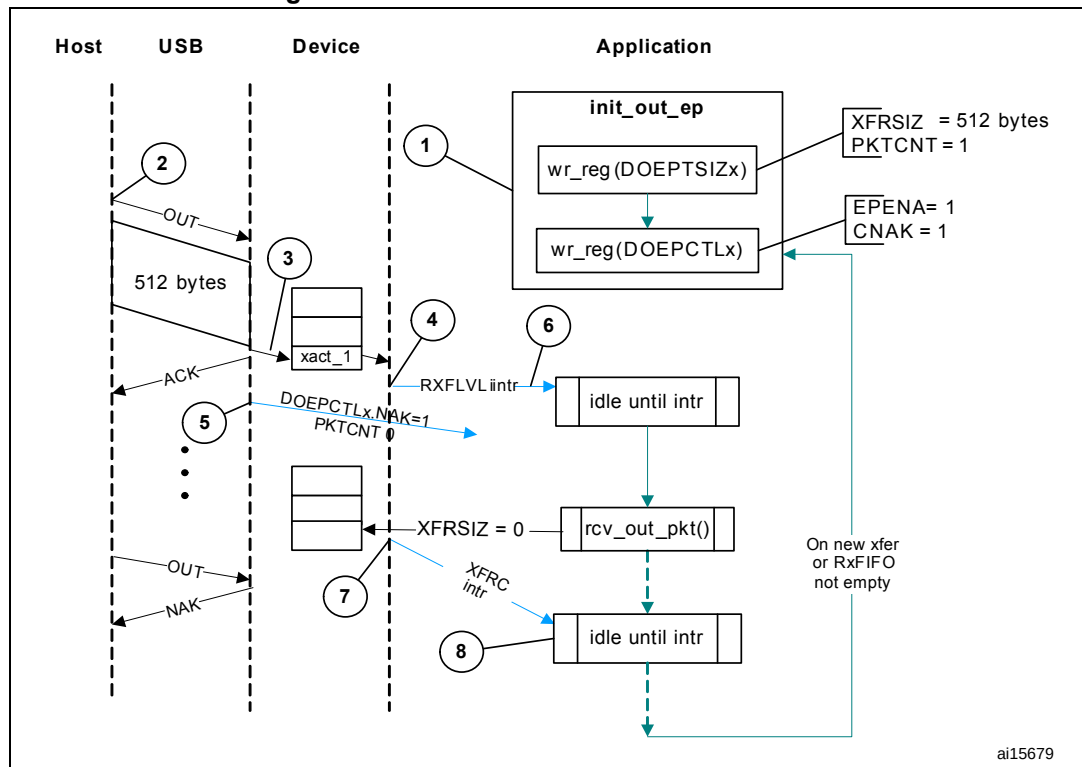
## Examples

This section describes and depicts some fundamental transfer types and scenarios.

- Slave mode bulk OUT transaction

[Figure 428](#) depicts the reception of a single Bulk OUT Data packet from the USB to the AHB and describes the events involved in the process.

Figure 428. Slave mode bulk OUT transaction



After a SetConfiguration/SetInterface command, the application initializes all OUT endpoints by setting CNAK = 1 and EPENA = 1 (in OTG\_HS\_DOEPCTLx), and setting a suitable XFRSIZ and PKTCNT in the OTG\_HS\_DOEPTSIzX register.

1. Host attempts to send data (OUT token) to an endpoint.
2. When the core receives the OUT token on the USB, it stores the packet in the RxFIFO because space is available there.
3. After writing the complete packet in the RxFIFO, the core then asserts the RXFLVL interrupt (in OTG\_HS\_GINTSTS).
4. On receiving the PKTCNT number of USB packets, the core internally sets the NAK bit for this endpoint to prevent it from receiving any more packets.
5. The application processes the interrupt and reads the data from the RxFIFO.
6. When the application has read all the data (equivalent to XFRSIZ), the core generates an XFRSIZ interrupt (in OTG\_HS\_DOEPINTx).
7. The application processes the interrupt and uses the setting of the XFRSIZ interrupt bit (in OTG\_HS\_DOEPINTx) to determine that the intended transfer is complete.

## IN data transfers

### • Packet write

This section describes how the application writes data packets to the endpoint FIFO in Slave mode when dedicated transmit FIFOs are enabled.

1. The application can either choose the polling or the interrupt mode.
  - In polling mode, the application monitors the status of the endpoint transmit data FIFO by reading the OTG\_HS\_DTXFSTSx register, to determine if there is enough space in the data FIFO.
  - In interrupt mode, the application waits for the TXFE interrupt (in OTG\_HS\_DIEPINTx) and then reads the OTG\_HS\_DTXFSTSx register, to determine if there is enough space in the data FIFO.
  - To write a single nonzero length data packet, there must be space to write the entire packet in the data FIFO.
  - To write zero length packet, the application must not look at the FIFO space.
2. Using one of the above mentioned methods, when the application determines that there is enough space to write a transmit packet, the application must first write into the endpoint control register, before writing the data into the data FIFO. Typically, the application, must do a read modify write on the OTG\_HS\_DIEPCTLx register to avoid modifying the contents of the register, except for setting the Endpoint Enable bit.

The application can write multiple packets for the same endpoint into the transmit FIFO, if space is available. For periodic IN endpoints, the application must write packets only for one micro-frame. It can write packets for the next periodic transaction only after getting transfer complete for the previous transaction.

- **Setting IN endpoint NAK**

Internal data flow:

1. When the application sets the IN NAK for a particular endpoint, the core stops transmitting data on the endpoint, irrespective of data availability in the endpoint's transmit FIFO.
2. Nonisochronous IN tokens receive a NAK handshake reply
  - Isochronous IN tokens receive a zero-data-length packet reply
3. The core asserts the INEPNE (IN endpoint NAK effective) interrupt in OTG\_HS\_DIEPINTx in response to the SNAK bit in OTG\_HS\_DIEPCTLx.
4. Once this interrupt is seen by the application, the application can assume that the endpoint is in IN NAK mode. This interrupt can be cleared by the application by setting the CNAK bit in OTG\_HS\_DIEPCTLx.

Application programming sequence:

1. To stop transmitting any data on a particular IN endpoint, the application must set the IN NAK bit. To set this bit, the following field must be programmed.
  - SNAK = 1 in OTG\_HS\_DIEPCTLx
2. Wait for assertion of the INEPNE interrupt in OTG\_HS\_DIEPINTx. This interrupt indicates that the core has stopped transmitting data on the endpoint.
3. The core can transmit valid IN data on the endpoint after the application has set the NAK bit, but before the assertion of the NAK Effective interrupt.
4. The application can mask this interrupt temporarily by writing to the INEPNEM bit in DIEPMSK.
  - INEPNEM = 0 in DIEPMSK
5. To exit Endpoint NAK mode, the application must clear the NAK status bit (NAKSTS) in OTG\_HS\_DIEPCTLx. This also clears the INEPNE interrupt (in OTG\_HS\_DIEPINTx).
  - CNAK = 1 in OTG\_HS\_DIEPCTLx
6. If the application masked this interrupt earlier, it must be unmasked as follows:
  - INEPNEM = 1 in DIEPMSK

- **IN endpoint disable**

Use the following sequence to disable a specific IN endpoint that has been previously enabled.

Application programming sequence:

1. The application must stop writing data on the AHB for the IN endpoint to be disabled.
2. The application must set the endpoint in NAK mode.
  - SNAK = 1 in OTG\_HS\_DIEPCTLx
3. Wait for the INEPNE interrupt in OTG\_HS\_DIEPINTx.
4. Set the following bits in the OTG\_HS\_DIEPCTLx register for the endpoint that must be disabled.
  - EPDIS = 1 in OTG\_HS\_DIEPCTLx
  - SNAK = 1 in OTG\_HS\_DIEPCTLx
5. Assertion of the EPDISD interrupt in OTG\_HS\_DIEPINTx indicates that the core has completely disabled the specified endpoint. Along with the assertion of the interrupt, the core also clears the following bits:
  - EPENA = 0 in OTG\_HS\_DIEPCTLx
  - EPDIS = 0 in OTG\_HS\_DIEPCTLx
6. The application must read the OTG\_HS\_DIEPTSIZx register for the periodic IN EP, to calculate how much data on the endpoint were transmitted on the USB.
7. The application must flush the data in the Endpoint transmit FIFO, by setting the following fields in the OTG\_HS\_GRSTCTL register:
  - TXFNUM (in OTG\_HS\_GRSTCTL) = Endpoint transmit FIFO number
  - TXFFLSH in (OTG\_HS\_GRSTCTL) = 1

The application must poll the OTG\_HS\_GRSTCTL register, until the TXFFLSH bit is cleared by the core, which indicates the end of flush operation. To transmit new data on this endpoint, the application can re-enable the endpoint at a later point.

- **Transfer Stop Programming for IN endpoints**

The application must use the following programming sequence to stop any transfers (because of an interrupt from the host, typically a reset).

**Sequence of operations:**

1. Disable the IN endpoint by setting:
  - EPDIS = 1 in all OTG\_HS\_DIEPCTLx registers
2. Wait for the EPDIS interrupt in OTG\_HS\_DIEPINTx, which indicates that the IN endpoint is completely disabled. When the EPDIS interrupt is asserted the following bits are cleared:
  - EPDIS = 0 in OTG\_HS\_DIEPCTLx
  - EPENA = 0 in OTG\_HS\_DIEPCTLx
3. Flush the Tx FIFO by programming the following bits:
  - TXFFLSH = 1 in OTG\_HS\_GRSTCTL
  - TXFNUM = “FIFO number specific to endpoint” in OTG\_HS\_GRSTCTL
4. The application can start polling till TXFFLSH in OTG\_HS\_GRSTCTL is cleared. When this bit is cleared, it ensures that there is no data left in the Tx FIFO.

- **Generic non-periodic IN data transfers**

**Application requirements:**

1. Before setting up an IN transfer, the application must ensure that all data to be transmitted as part of the IN transfer are part of a single buffer.
2. For IN transfers, the Transfer Size field in the Endpoint Transfer Size register denotes a payload that constitutes multiple maximum-packet-size packets and a single short packet. This short packet is transmitted at the end of the transfer.
  - To transmit a few maximum-packet-size packets and a short packet at the end of the transfer:
 
$$\text{Transfer size}[\text{EPNUM}] = x \times \text{MPSIZ}[\text{EPNUM}] + \text{sp}$$
 If ( $\text{sp} > 0$ ), then  $\text{packet count}[\text{EPNUM}] = x + 1$ .  
 Otherwise,  $\text{packet count}[\text{EPNUM}] = x$
  - To transmit a single zero-length data packet:
 
$$\text{Transfer size}[\text{EPNUM}] = 0$$

$$\text{Packet count}[\text{EPNUM}] = 1$$
  - To transmit a few maximum-packet-size packets and a zero-length data packet at the end of the transfer, the application must split the transfer into two parts. The first sends maximum-packet-size data packets and the second sends the zero-length data packet alone.
 
$$\text{First transfer: transfer size}[\text{EPNUM}] = x \times \text{MPSIZ}[\text{epnum}]; \text{ packet count} = n;$$

$$\text{Second transfer: transfer size}[\text{EPNUM}] = 0; \text{ packet count} = 1;$$
3. Once an endpoint is enabled for data transfers, the core updates the Transfer size register. At the end of the IN transfer, the application must read the Transfer size register to determine how much data posted in the transmit FIFO have already been sent on the USB.
4. Data fetched into transmit FIFO = Application-programmed initial transfer size – core-updated final transfer size
  - Data transmitted on USB = (application-programmed initial packet count – Core updated final packet count)  $\times$  MPSIZ[EPNUM]
  - Data yet to be transmitted on USB = (Application-programmed initial transfer size – data transmitted on USB)

**Internal data flow:**

1. The application must set the transfer size and packet count fields in the endpoint-specific registers and enable the endpoint to transmit the data.
2. The application must also write the required data to the transmit FIFO for the endpoint.
3. Every time a packet is written into the transmit FIFO by the application, the transfer size for that endpoint is decremented by the packet size. The data is fetched from the memory by the application, until the transfer size for the endpoint becomes 0. After writing the data into the FIFO, the “number of packets in FIFO” count is incremented (this is a 3-bit count, internally maintained by the core for each IN endpoint transmit FIFO. The maximum number of packets maintained by the core at any time in an IN endpoint FIFO is eight). For zero-length packets, a separate flag is set for each FIFO, without any data in the FIFO.
4. Once the data are written to the transmit FIFO, the core reads them out upon receiving an IN token. For every nonisochronous IN data packet transmitted with an ACK handshake, the packet count for the endpoint is decremented by one, until the packet count is zero. The packet count is not decremented on a timeout.
5. For zero length packets (indicated by an internal zero length flag), the core sends out a zero-length packet for the IN token and decrements the packet count field.
6. If there are no data in the FIFO for a received IN token and the packet count field for that endpoint is zero, the core generates an “IN token received when TxFIFO is empty” (ITTXFE) Interrupt for the endpoint, provided that the endpoint NAK bit is not set. The core responds with a NAK handshake for nonisochronous endpoints on the USB.
7. The core internally rewinds the FIFO pointers and no timeout interrupt is generated.
8. When the transfer size is 0 and the packet count is 0, the transfer complete (XFRC) interrupt for the endpoint is generated and the endpoint enable is cleared.

Application programming sequence:

1. Program the OTG\_HS\_DIEPTSIz register with the transfer size and corresponding packet count.
2. Program the OTG\_HS\_DIEPCTLx register with the endpoint characteristics and set the CNAK and EPENA (Endpoint Enable) bits.
3. When transmitting nonzero length data packet, the application must poll the OTG\_HS\_DTXFSTSx register (where x is the FIFO number associated with that endpoint) to determine whether there is enough space in the data FIFO. The application can optionally use TXFE (in OTG\_HS\_DIEPINTx) before writing the data.

- **Generic periodic IN data transfers**

This section describes a typical periodic IN data transfer.

Application requirements:

1. Application requirements 1, 2, 3, and 4 of [Generic non-periodic IN data transfers on page 1519](#) also apply to periodic IN data transfers, except for a slight modification of requirement 2.
  - The application can only transmit multiples of maximum-packet-size data packets or multiples of maximum-packet-size packets, plus a short packet at the end. To

transmit a few maximum-packet-size packets and a short packet at the end of the transfer, the following conditions must be met:

$\text{transfer size}[\text{EPNUM}] = x \times \text{MPSIZ}[\text{EPNUM}] + \text{sp}$

(where  $x$  is an integer  $\geq 0$ , and  $0 \leq \text{sp} < \text{MPSIZ}[\text{EPNUM}]$ )

If ( $\text{sp} > 0$ ),  $\text{packet count}[\text{EPNUM}] = x + 1$

Otherwise,  $\text{packet count}[\text{EPNUM}] = x$ ;

$\text{MCNT}[\text{EPNUM}] = \text{packet count}[\text{EPNUM}]$

- The application cannot transmit a zero-length data packet at the end of a transfer. It can transmit a single zero-length data packet by itself. To transmit a single zero-length data packet:
  - $\text{transfer size}[\text{EPNUM}] = 0$
  - $\text{packet count}[\text{EPNUM}] = 1$
  - $\text{MCNT}[\text{EPNUM}] = \text{packet count}[\text{EPNUM}]$
- 2. The application can only schedule data transfers one frame at a time.
  - $(\text{MCNT} - 1) \times \text{MPSIZ} \leq \text{XFERSIZ} \leq \text{MCNT} \times \text{MPSIZ}$
  - $\text{PKTCNT} = \text{MCNT}$  (in OTG\_HS\_DIEPTSIZE<sub>x</sub>)
  - If  $\text{XFERSIZ} < \text{MCNT} \times \text{MPSIZ}$ , the last data packet of the transfer is a short packet.
  - Note that: MCNT is in OTG\_HS\_DIEPTSIZE<sub>x</sub>, MPSIZ is in OTG\_HS\_DIEPCTL<sub>x</sub>, PKTCNT is in OTG\_HS\_DIEPTSIZE<sub>x</sub> and XFERSIZ is in OTG\_HS\_DIEPTSIZE<sub>x</sub>
- 3. The complete data to be transmitted in the frame must be written into the transmit FIFO by the application, before the IN token is received. Even when 1 word of the data to be transmitted per frame is missing in the transmit FIFO when the IN token is received, the core behaves as when the FIFO is empty. When the transmit FIFO is empty:
  - A zero data length packet would be transmitted on the USB for isochronous IN endpoints
  - A NAK handshake would be transmitted on the USB for interrupt IN endpoints
- 4. For a high-bandwidth IN endpoint with three packets in a frame, the application can program the endpoint FIFO size to be  $2 \times \text{max\_pkt\_size}$  and have the third packet loaded in after the first packet has been transmitted on the USB.

Internal data flow:

1. The application must set the transfer size and packet count fields in the endpoint-specific registers and enable the endpoint to transmit the data.
2. The application must also write the required data to the associated transmit FIFO for the endpoint.
3. Every time the application writes a packet to the transmit FIFO, the transfer size for that endpoint is decremented by the packet size. The data are fetched from application memory until the transfer size for the endpoint becomes 0.
4. When an IN token is received for a periodic endpoint, the core transmits the data in the FIFO, if available. If the complete data payload (complete packet, in dedicated FIFO

- mode) for the frame is not present in the FIFO, then the core generates an IN token received when TxFIFO empty interrupt for the endpoint.
- A zero-length data packet is transmitted on the USB for isochronous IN endpoints
  - A NAK handshake is transmitted on the USB for interrupt IN endpoints
5. The packet count for the endpoint is decremented by 1 under the following conditions:
    - For isochronous endpoints, when a zero- or nonzero-length data packet is transmitted
    - For interrupt endpoints, when an ACK handshake is transmitted
    - When the transfer size and packet count are both 0, the transfer completed interrupt for the endpoint is generated and the endpoint enable is cleared.
  6. At the “Periodic frame Interval” (controlled by PFIVL in OTG\_HS\_DCFG), when the core finds nonempty any of the isochronous IN endpoint FIFOs scheduled for the current frame nonempty, the core generates an IISOIXFR interrupt in OTG\_HS\_GINTSTS.

Application programming sequence:

1. Program the OTG\_HS\_DIEPCTLx register with the endpoint characteristics and set the CNAK and EPENA bits.
2. Write the data to be transmitted in the next frame to the transmit FIFO.
3. Asserting the ITTXFE interrupt (in OTG\_HS\_DIEPINTx) indicates that the application has not yet written all data to be transmitted to the transmit FIFO.
4. If the interrupt endpoint is already enabled when this interrupt is detected, ignore the interrupt. If it is not enabled, enable the endpoint so that the data can be transmitted on the next IN token attempt.
5. Asserting the XFRC interrupt (in OTG\_HS\_DIEPINTx) with no ITTXFE interrupt in OTG\_HS\_DIEPINTx indicates the successful completion of an isochronous IN transfer. A read to the OTG\_HS\_DIEPTSIZx register must give transfer size = 0 and packet count = 0, indicating all data were transmitted on the USB.
6. Asserting the XFRC interrupt (in OTG\_HS\_DIEPINTx), with or without the ITTXFE interrupt (in OTG\_HS\_DIEPINTx), indicates the successful completion of an interrupt IN transfer. A read to the OTG\_HS\_DIEPTSIZx register must give transfer size = 0 and packet count = 0, indicating all data were transmitted on the USB.
7. Asserting the incomplete isochronous IN transfer (IISOIXFR) interrupt in OTG\_HS\_GINTSTS with none of the aforementioned interrupts indicates the core did not receive at least 1 periodic IN token in the current frame.

- **Incomplete isochronous IN data transfers**

This section describes what the application must do on an incomplete isochronous IN data transfer.

Internal data flow:

1. An isochronous IN transfer is treated as incomplete in one of the following conditions:
  - a) The core receives a corrupted isochronous IN token on at least one isochronous IN endpoint. In this case, the application detects an incomplete isochronous IN transfer interrupt (IISOIXFR in OTG\_HS\_GINTSTS).
  - b) The application is slow to write the complete data payload to the transmit FIFO and an IN token is received before the complete data payload is written to the FIFO. In this case, the application detects an IN token received when TxFIFO empty interrupt in OTG\_HS\_DIEPINTx. The application can ignore this interrupt,

as it eventually results in an incomplete isochronous IN transfer interrupt (IISOIXFR in OTG\_HS\_GINTSTS) at the end of periodic frame.

The core transmits a zero-length data packet on the USB in response to the received IN token.

2. The application must stop writing the data payload to the transmit FIFO as soon as possible.
3. The application must set the NAK bit and the disable bit for the endpoint.
4. The core disables the endpoint, clears the disable bit, and asserts the Endpoint Disable interrupt for the endpoint.

#### Application programming sequence

1. The application can ignore the IN token received when TxFIFO empty interrupt in OTG\_HS\_DIEPINTx on any isochronous IN endpoint, as it eventually results in an incomplete isochronous IN transfer interrupt (in OTG\_HS\_GINTSTS).
2. Assertion of the incomplete isochronous IN transfer interrupt (in OTG\_HS\_GINTSTS) indicates an incomplete isochronous IN transfer on at least one of the isochronous IN endpoints.
3. The application must read the Endpoint Control register for all isochronous IN endpoints to detect endpoints with incomplete IN data transfers.
4. The application must stop writing data to the Periodic Transmit FIFOs associated with these endpoints on the AHB.
5. Program the following fields in the OTG\_HS\_DIEPCTLx register to disable the endpoint:
  - SNAK = 1 in OTG\_HS\_DIEPCTLx
  - EPDIS = 1 in OTG\_HS\_DIEPCTLx
6. The assertion of the Endpoint Disabled interrupt in OTG\_HS\_DIEPINTx indicates that the core has disabled the endpoint.
  - At this point, the application must flush the data in the associated transmit FIFO or overwrite the existing data in the FIFO by enabling the endpoint for a new transfer in the next micro-frame. To flush the data, the application must use the OTG\_HS\_GRSTCTL register.

- **Stalling nonisochronous IN endpoints**

This section describes how the application can stall a nonisochronous endpoint.

Application programming sequence:

1. Disable the IN endpoint to be stalled. Set the STALL bit as well.
2. EPDIS = 1 in OTG\_HS\_DIEPCTLx, when the endpoint is already enabled
  - STALL = 1 in OTG\_HS\_DIEPCTLx
  - The STALL bit always takes precedence over the NAK bit
3. Assertion of the Endpoint Disabled interrupt (in OTG\_HS\_DIEPINTx) indicates to the application that the core has disabled the specified endpoint.
4. The application must flush the nonperiodic or periodic transmit FIFO, depending on the endpoint type. In case of a nonperiodic endpoint, the application must re-enable the other nonperiodic endpoints that do not need to be stalled, to transmit data.
5. Whenever the application is ready to end the STALL handshake for the endpoint, the STALL bit must be cleared in OTG\_HS\_DIEPCTLx.
6. If the application sets or clears a STALL bit for an endpoint due to a SetFeature.Endpoint Halt command or ClearFeature.Endpoint Halt command, the STALL bit must be set or cleared before the application sets up the Status stage transfer on the control endpoint.

Special case: stalling the control OUT endpoint

The core must stall IN/OUT tokens if, during the data stage of a control transfer, the host sends more IN/OUT tokens than are specified in the SETUP packet. In this case, the application must enable the ITTXFE interrupt in OTG\_HS\_DIEPINTx and the OTEPDIS interrupt in OTG\_HS\_DOEPINTx during the data stage of the control transfer, after the core has transferred the amount of data specified in the SETUP packet. Then, when the application receives this interrupt, it must set the STALL bit in the corresponding endpoint control register, and clear this interrupt.

### 35.13.8 Worst case response time

When the OTG\_HS controller acts as a device, there is a worst case response time for any tokens that follow an isochronous OUT. This worst case response time depends on the AHB clock frequency.

The core registers are in the AHB domain, and the core does not accept another token before updating these register values. The worst case is for any token following an isochronous OUT, because for an isochronous transaction, there is no handshake and the next token could come sooner. This worst case value is 7 PHY clocks when the AHB clock is the same as the PHY clock. When the AHB clock is faster, this value is smaller.

If this worst case condition occurs, the core responds to bulk/interrupt tokens with a NAK and drops isochronous and SETUP tokens. The host interprets this as a timeout condition for SETUP and retries the SETUP packet. For isochronous transfers, the Incomplete isochronous IN transfer interrupt (IISOIXFR) and Incomplete isochronous OUT transfer interrupt (IISOOXFR) inform the application that isochronous IN/OUT packets were dropped.

#### Choosing the value of TRDT in OTG\_HS\_GUSBCFG

The value in TRDT (OTG\_HS\_GUSBCFG) is the time it takes for the MAC, in terms of PHY clocks after it has received an IN token, to get the FIFO status, and thus the first data from the PFC block. This time involves the synchronization delay between the PHY and AHB clocks. The worst case delay for this is when the AHB clock is the same as the PHY clock. In this case, the delay is 5 clocks.

Once the MAC receives an IN token, this information (token received) is synchronized to the AHB clock by the PFC (the PFC runs on the AHB clock). The PFC then reads the data from the SPRAM and writes them into the dual clock source buffer. The MAC then reads the data out of the source buffer (4 deep).

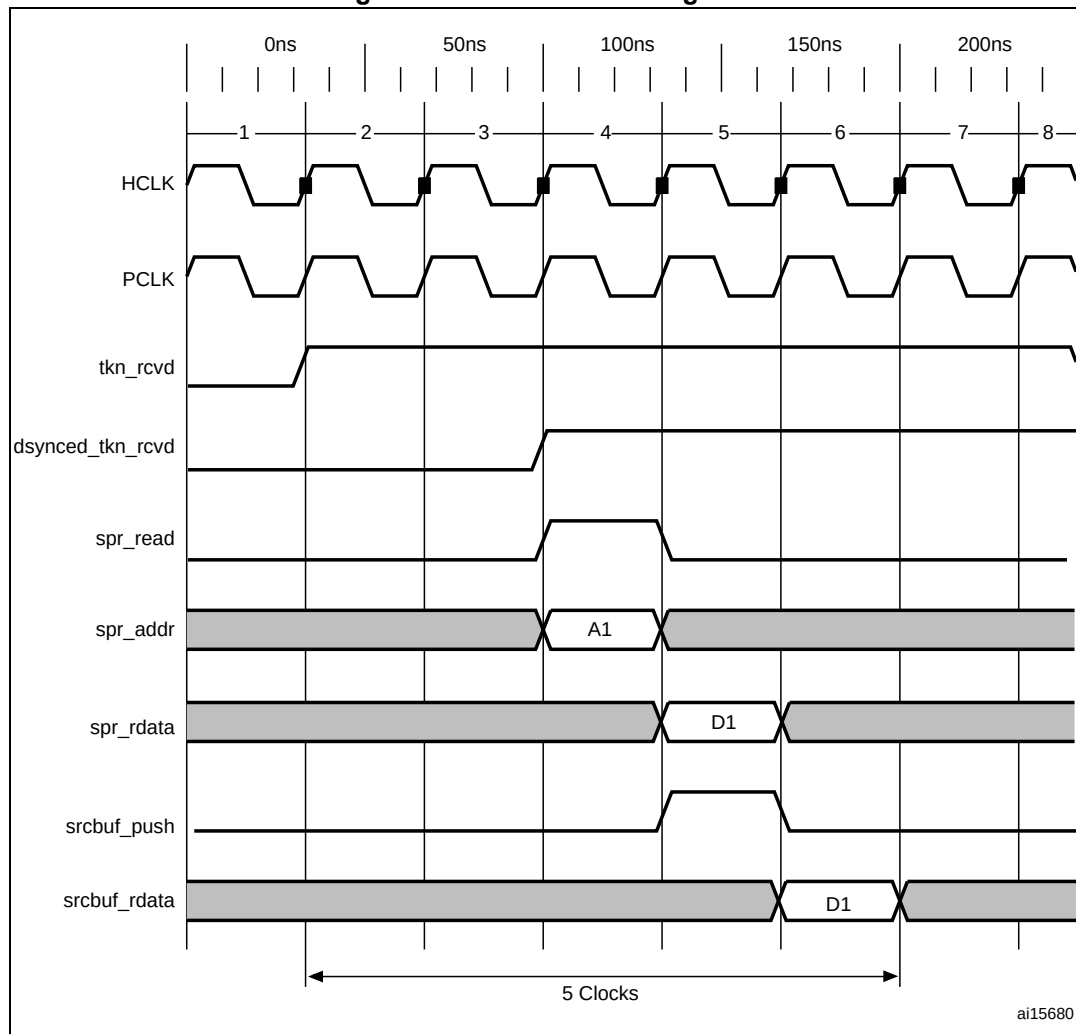
If the AHB is running at a higher frequency than the PHY, the application can use a smaller value for TRDT (in OTG\_HS\_GUSBCFG).

Figure 429 has the following signals:

- `tkn_rcvd`: Token received information from MAC to PFC
- `dynced_tkn_rcvd`: Doubled sync `tkn_rcvd`, from PCLK to HCLK domain
- `spr_read`: Read to SPRAM
- `spr_addr`: Address to SPRAM
- `spr_rdata`: Read data from SPRAM
- `srcbuf_push`: Push to the source buffer
- `srcbuf_rdata`: Read data from the source buffer. Data seen by MAC

Refer to [Table 213: TRDT values](#) for the values of TRDT versus AHB clock frequency.

**Figure 429. TRDT max timing case**



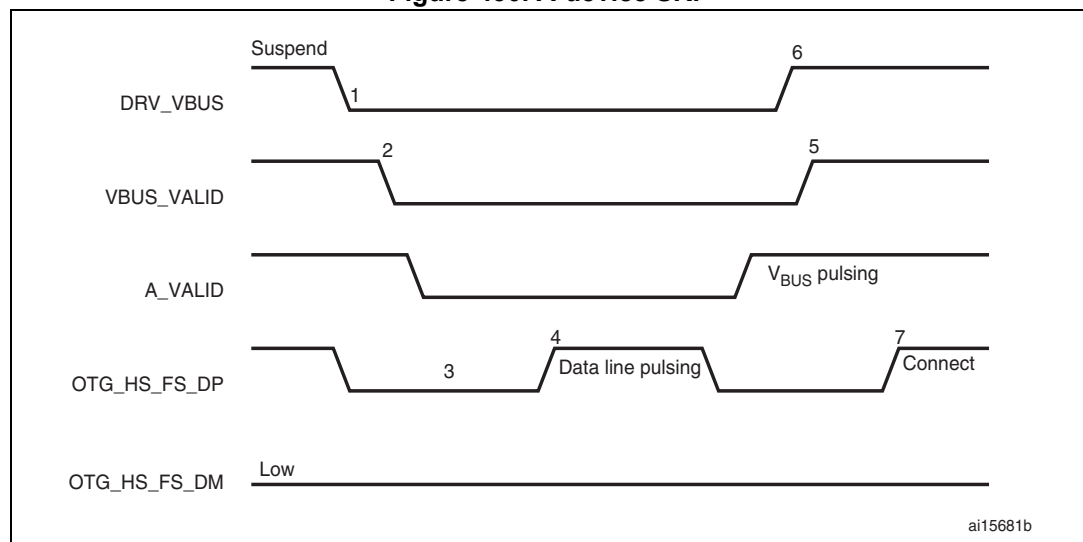
### 35.13.9 OTG programming model

The OTG\_HS controller is an OTG device supporting HNP and SRP. When the core is connected to an “A” plug, it is referred to as an A-device. When the core is connected to a “B” plug it is referred to as a B-device. In host mode, the OTG\_HS controller turns off  $V_{BUS}$  to conserve power. SRP is a method by which the B-device signals the A-device to turn on  $V_{BUS}$  power. A device must perform both data-line pulsing and  $V_{BUS}$  pulsing, but a host can detect either data-line pulsing or  $V_{BUS}$  pulsing for SRP. HNP is a method by which the B-device negotiates and switches to host role. In Negotiated mode after HNP, the B-device suspends the bus and reverts to the device role.

#### A-device session request protocol

The application must set the SRP-capable bit in the Core USB configuration register. This enables the OTG\_HS controller to detect SRP as an A-device.

Figure 430. A-device SRP



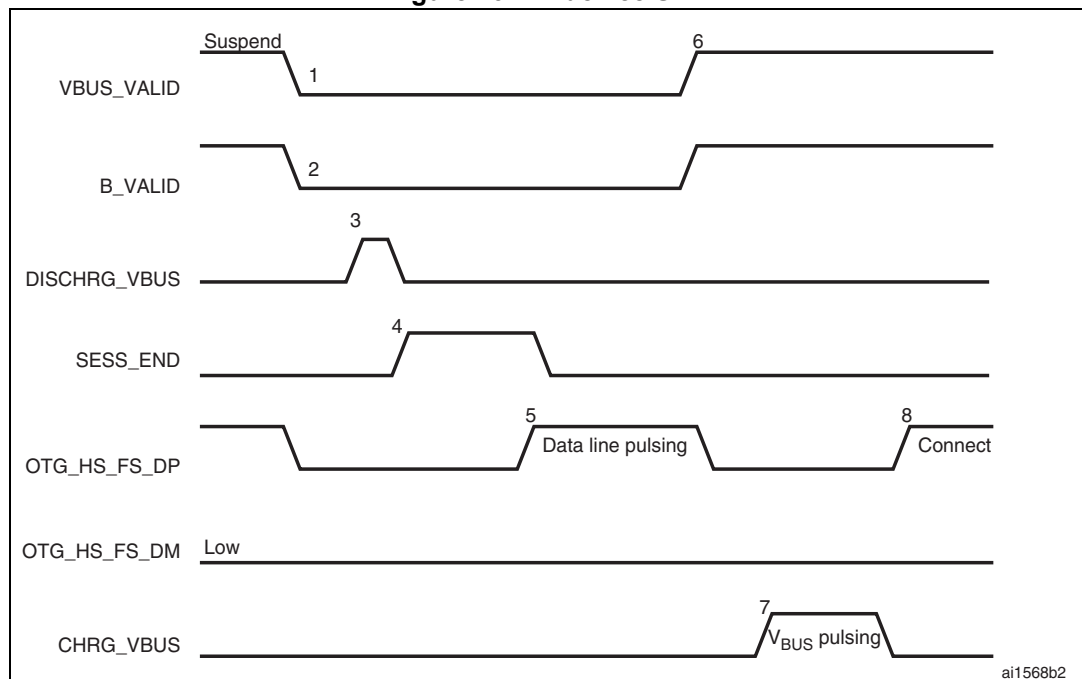
1. DRV\_VBUS =  $V_{BUS}$  drive signal to the PHY  
 VBUS\_VALID =  $V_{BUS}$  valid signal from PHY  
 A\_VALID = A-device  $V_{BUS}$  level signal to PHY  
 DP = Data plus line  
 DM = Data minus line

1. To save power, the application suspends and turns off port power when the bus is idle by writing the port suspend and port power bits in the host port control and status register.
2. PHY indicates port power off by deasserting the VBUS\_VALID signal.
3. The device must detect SE0 for at least 2 ms to start SRP when V<sub>BUS</sub> power is off.
4. To initiate SRP, the device turns on its data line pull-up resistor for 5 to 10 ms. The OTG\_HS controller detects data-line pulsing.
5. The device drives V<sub>BUS</sub> above the A-device session valid (2.0 V minimum) for V<sub>BUS</sub> pulsing.  
The OTG\_HS controller interrupts the application on detecting SRP. The Session request detected bit is set in Global interrupt status register (SRQINT set in OTG\_HS\_GINTSTS).
6. The application must service the Session request detected interrupt and turn on the port power bit by writing the port power bit in the host port control and status register. The PHY indicates port power-on by asserting the VBUS\_VALID signal.
7. When the USB is powered, the device connects, completing the SRP process.

### B-device session request protocol

The application must set the SRP-capable bit in the Core USB configuration register. This enables the OTG\_HS controller to initiate SRP as a B-device. SRP is a means by which the OTG\_HS controller can request a new session from the host.

Figure 431. B-device SRP



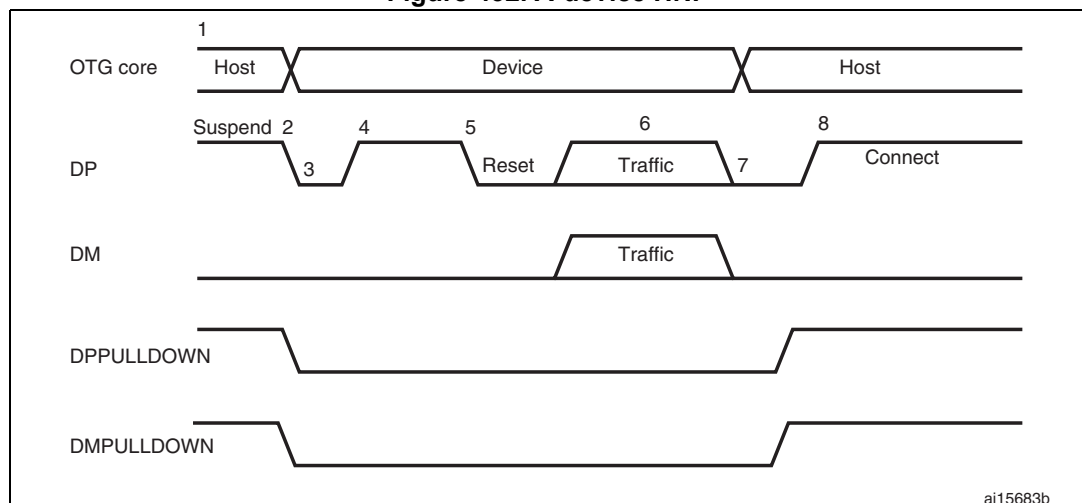
1. VBUS\_VALID = V<sub>BUS</sub> valid signal from PHY  
B\_VALID = B-device valid session to PHY  
DISCHRG\_VBUS = discharge signal to PHY  
SESS\_END = session end signal to PHY  
CHRG\_VBUS = charge V<sub>BUS</sub> signal to PHY  
DP = Data plus line  
DM = Data minus line

1. To save power, the host suspends and turns off port power when the bus is idle.  
The OTG\_HS controller sets the early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_HS controller sets the USB suspend bit in the Core interrupt register.  
The OTG\_HS controller informs the PHY to discharge  $V_{BUS}$ .
2. The PHY indicates the session's end to the device. This is the initial condition for SRP.  
The OTG\_HS controller requires 2 ms of SE0 before initiating SRP.  
For a USB 1.1 full-speed serial transceiver, the application must wait until  $V_{BUS}$  discharges to 0.2 V after BSVLD (in OTG\_HS\_GOTGCTL) is deasserted. This discharge time can be obtained from the transceiver vendor and varies from one transceiver to another.
3. The USB OTG core informs the PHY to speed up  $V_{BUS}$  discharge.
4. The application initiates SRP by writing the session request bit in the OTG Control and status register. The OTG\_HS controller perform data-line pulsing followed by  $V_{BUS}$  pulsing.
5. The host detects SRP from either the data-line or  $V_{BUS}$  pulsing, and turns on  $V_{BUS}$ .  
The PHY indicates  $V_{BUS}$  power-on to the device.
6. The OTG\_HS controller performs  $V_{BUS}$  pulsing.  
The host starts a new session by turning on  $V_{BUS}$ , indicating SRP success. The OTG\_HS controller interrupts the application by setting the session request success status change bit in the OTG interrupt status register. The application reads the session request success bit in the OTG control and status register.
7. When the USB is powered, the OTG\_HS controller connects, completing the SRP process.

### A-device host negotiation protocol

HNP switches the USB host role from the A-device to the B-device. The application must set the HNP-capable bit in the Core USB configuration register to enable the OTG\_HS controller to perform HNP as an A-device.

**Figure 432. A-device HNP**



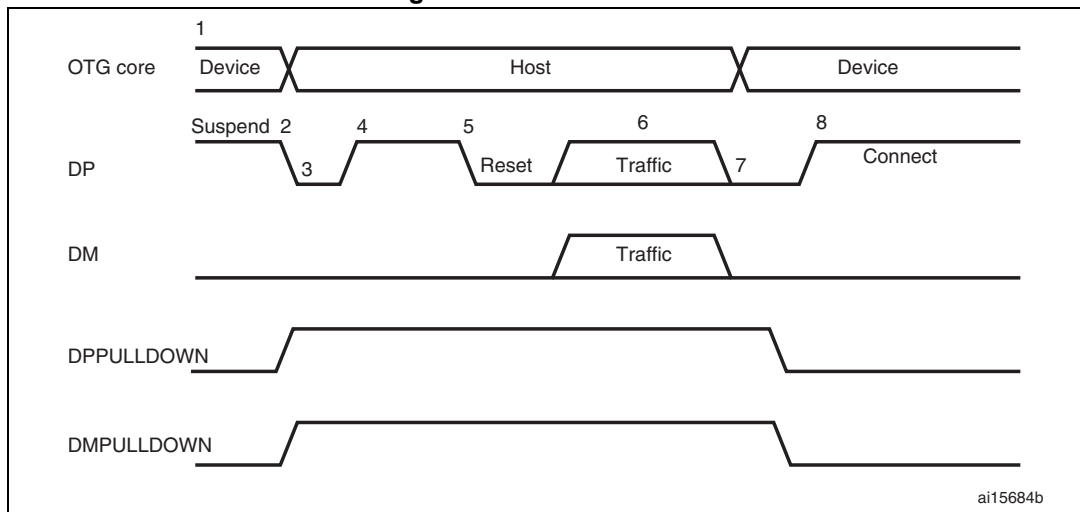
1. DPPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DP line inside the PHY.  
DMPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DM line inside the PHY.

1. The OTG\_HS controller sends the B-device a SetFeature b\_hnp\_enable descriptor to enable HNP support. The B-device's ACK response indicates that the B-device supports HNP. The application must set host Set HNP Enable bit in the OTG Control and status register to indicate to the OTG\_HS controller that the B-device supports HNP.
2. When it has finished using the bus, the application suspends by writing the Port suspend bit in the host port control and status register.
3. When the B-device observes a USB suspend, it disconnects, indicating the initial condition for HNP. The B-device initiates HNP only when it must switch to the host role; otherwise, the bus continues to be suspended.  
The OTG\_HS controller sets the host negotiation detected interrupt in the OTG interrupt status register, indicating the start of HNP.  
The OTG\_HS controller deasserts the DM pull down and DM pull down in the PHY to indicate a device role. The PHY enables the OTG\_HS\_DP pull-up resistor to indicate a connect for B-device.  
The application must read the current mode bit in the OTG Control and status register to determine peripheral mode operation.
4. The B-device detects the connection, issues a USB reset, and enumerates the OTG\_HS controller for data traffic.
5. The B-device continues the host role, initiating traffic, and suspends the bus when done.  
The OTG\_HS controller sets the early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_HS controller sets the USB Suspend bit in the Core interrupt register.
6. In Negotiated mode, the OTG\_HS controller detects the suspend, disconnects, and switches back to the host role. The OTG\_HS controller asserts the DM pull down and DM pull down in the PHY to indicate its assumption of the host role.
7. The OTG\_HS controller sets the Connector ID status change interrupt in the OTG Interrupt Status register. The application must read the connector ID status in the OTG Control and Status register to determine the OTG\_HS controller operation as an A-device. This indicates the completion of HNP to the application. The application must read the Current mode bit in the OTG control and status register to determine host mode operation.
8. The B-device connects, completing the HNP process.

### **B-device host negotiation protocol**

HNP switches the USB host role from B-device to A-device. The application must set the HNP-capable bit in the Core USB configuration register to enable the OTG\_HS controller to perform HNP as a B-device.

Figure 433. B-device HNP



1. DPPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DP line inside the PHY.  
DMPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DM line inside the PHY.
1. The A-device sends the SetFeature b\_hnp\_enable descriptor to enable HNP support. The OTG\_HS controller's ACK response indicates that it supports HNP. The application must set the Device HNP enable bit in the OTG Control and status register to indicate HNP support.  
The application sets the HNP request bit in the OTG Control and status register to indicate to the OTG\_HS controller to initiate HNP.
2. When it has finished using the bus, the A-device suspends by writing the Port suspend bit in the host port control and status register.  
The OTG\_HS controller sets the Early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_HS controller sets the USB suspend bit in the Core interrupt register.  
The OTG\_HS controller disconnects and the A-device detects SE0 on the bus, indicating HNP. The OTG\_HS controller asserts the DP pull down and DM pull down in the PHY to indicate its assumption of the host role.  
The A-device responds by activating its OTG\_HS\_DP pull-up resistor within 3 ms of detecting SE0. The OTG\_HS controller detects this as a connect.  
The OTG\_HS controller sets the host negotiation success status change interrupt in the OTG Interrupt status register, indicating the HNP status. The application must read the host negotiation success bit in the OTG Control and status register to determine

host negotiation success. The application must read the current Mode bit in the Core interrupt register (OTG\_HS\_GINTSTS) to determine host mode operation.

3. The application sets the reset bit (PRST in OTG\_HS\_HPRT) and the OTG\_HS controller issues a USB reset and enumerates the A-device for data traffic.
4. The OTG\_HS controller continues the host role of initiating traffic, and when done, suspends the bus by writing the Port suspend bit in the host port control and status register.
5. In Negotiated mode, when the A-device detects a suspend, it disconnects and switches back to the host role. The OTG\_HS controller deasserts the DP pull down and DM pull down in the PHY to indicate the assumption of the device role.
6. The application must read the current mode bit in the Core interrupt (OTG\_HS\_GINTSTS) register to determine the host mode operation.
7. The OTG\_HS controller connects, completing the HNP process.